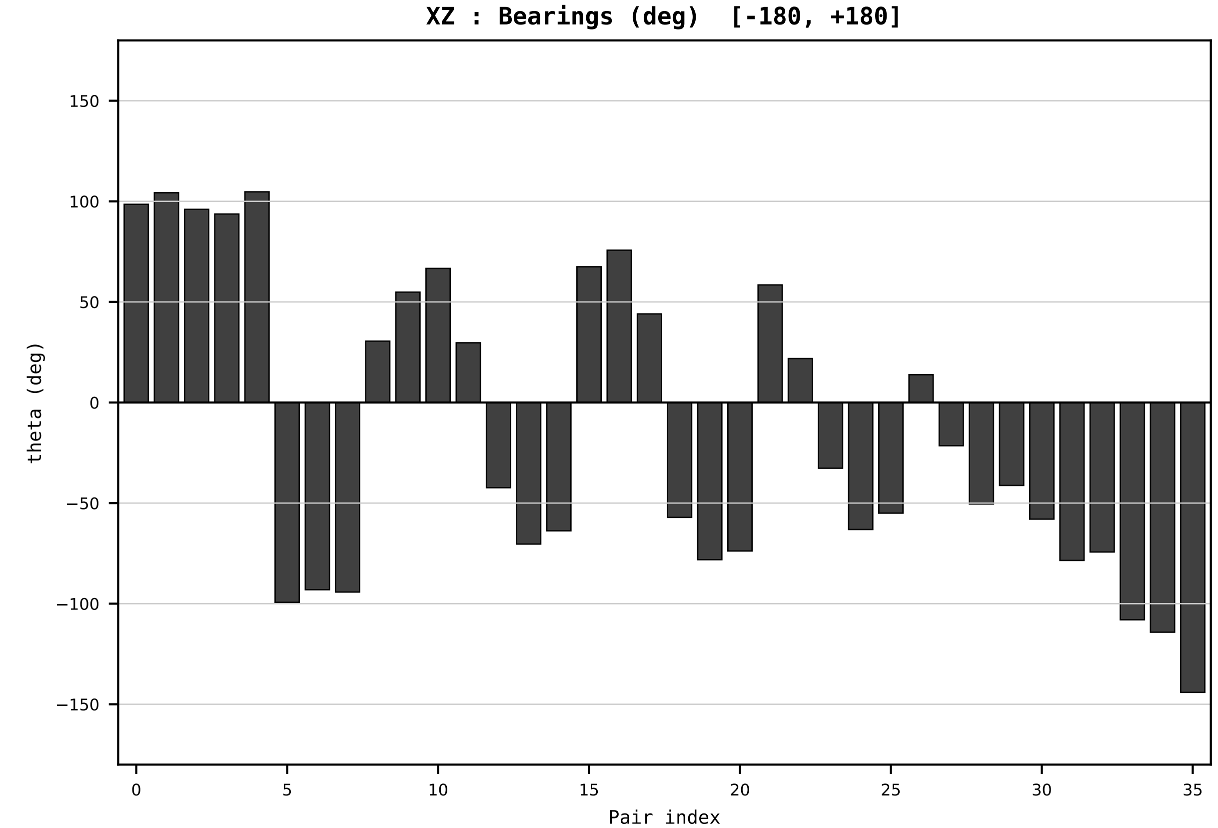
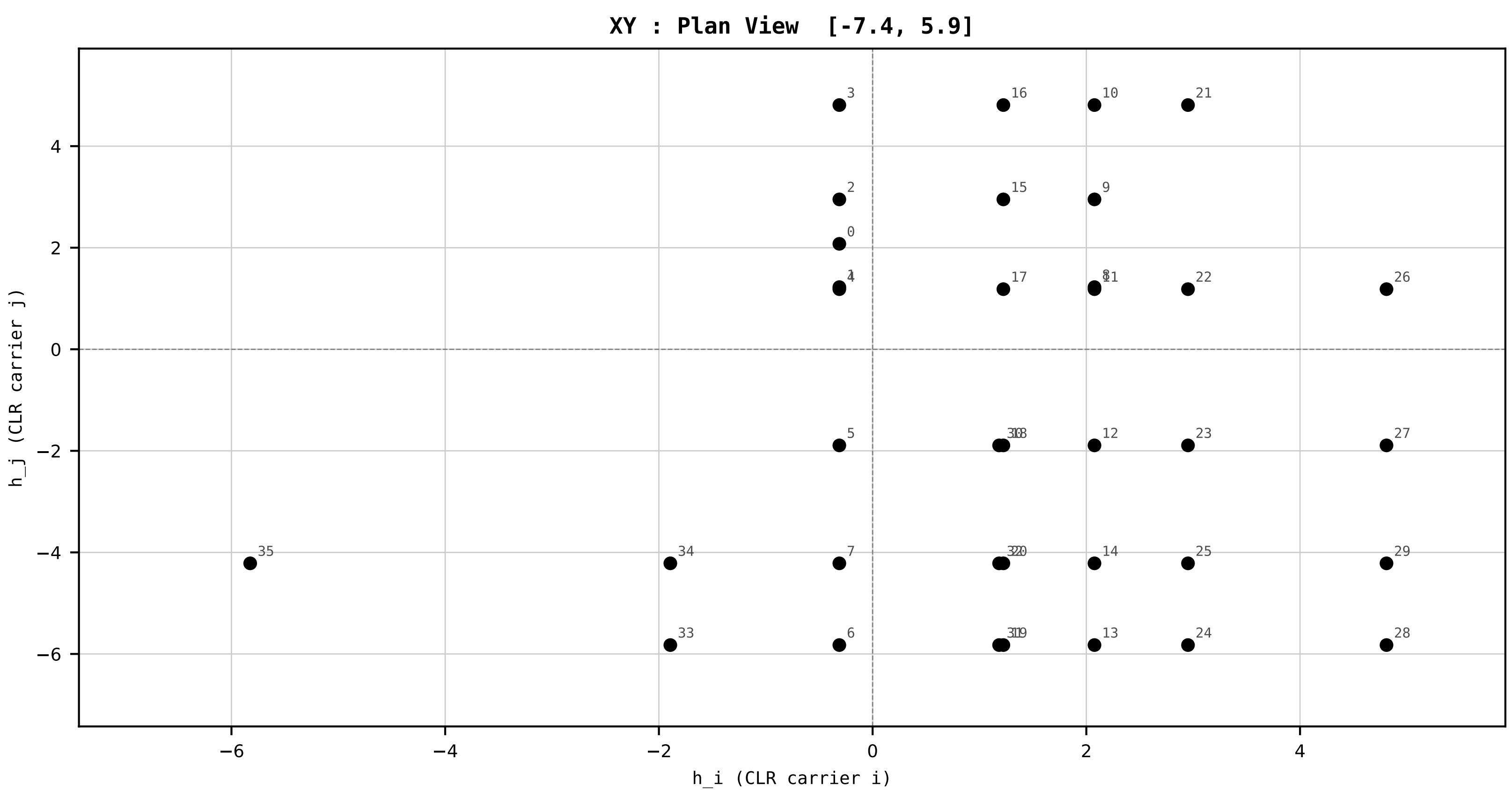
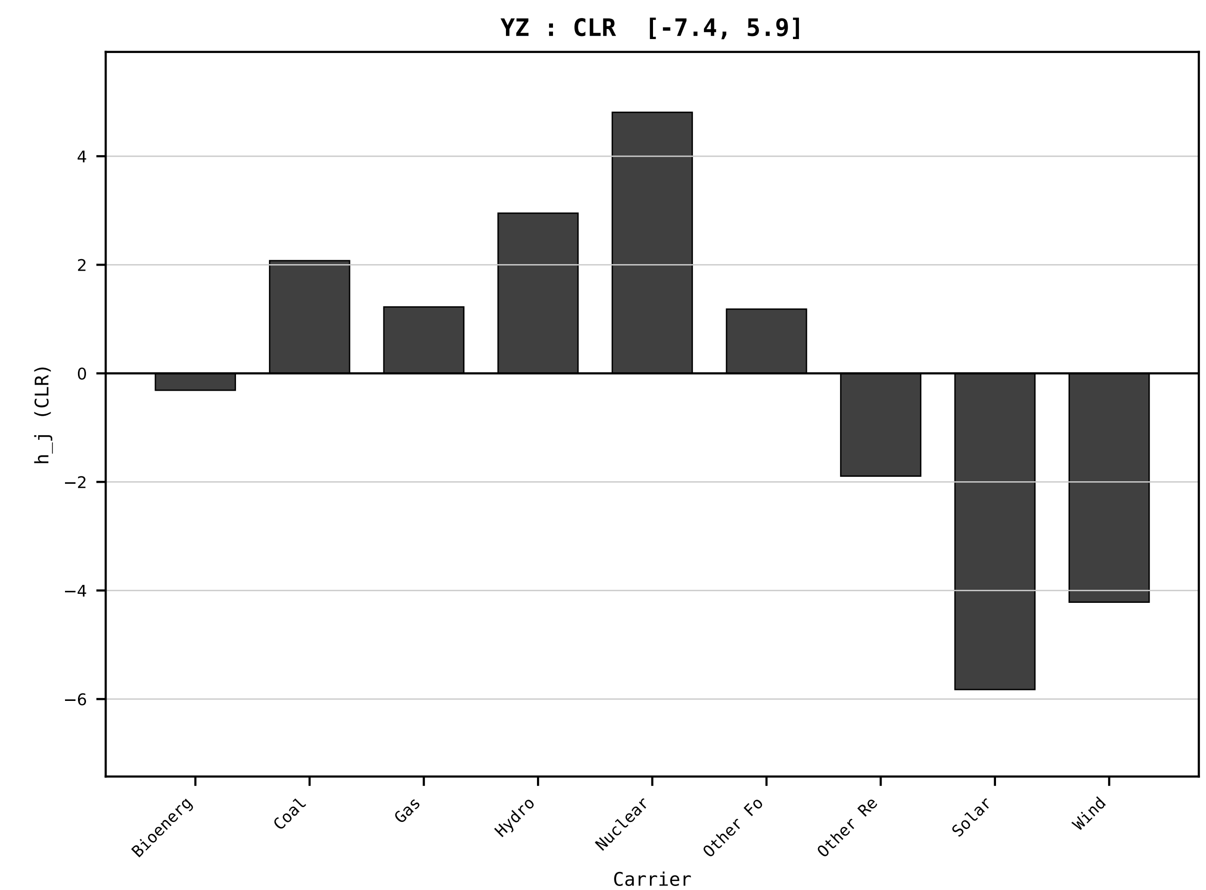
SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2000
t       : 0 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.636980
Ring    = Hs-4
E_metric= 94.4242
kappa_HS= 41516.00
omega   = 0.0000 deg
d_A     = 0.000000
Helm    = ---
Helm d  = ---
DR      = 10.634 HLR
DR ratio= 41516.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



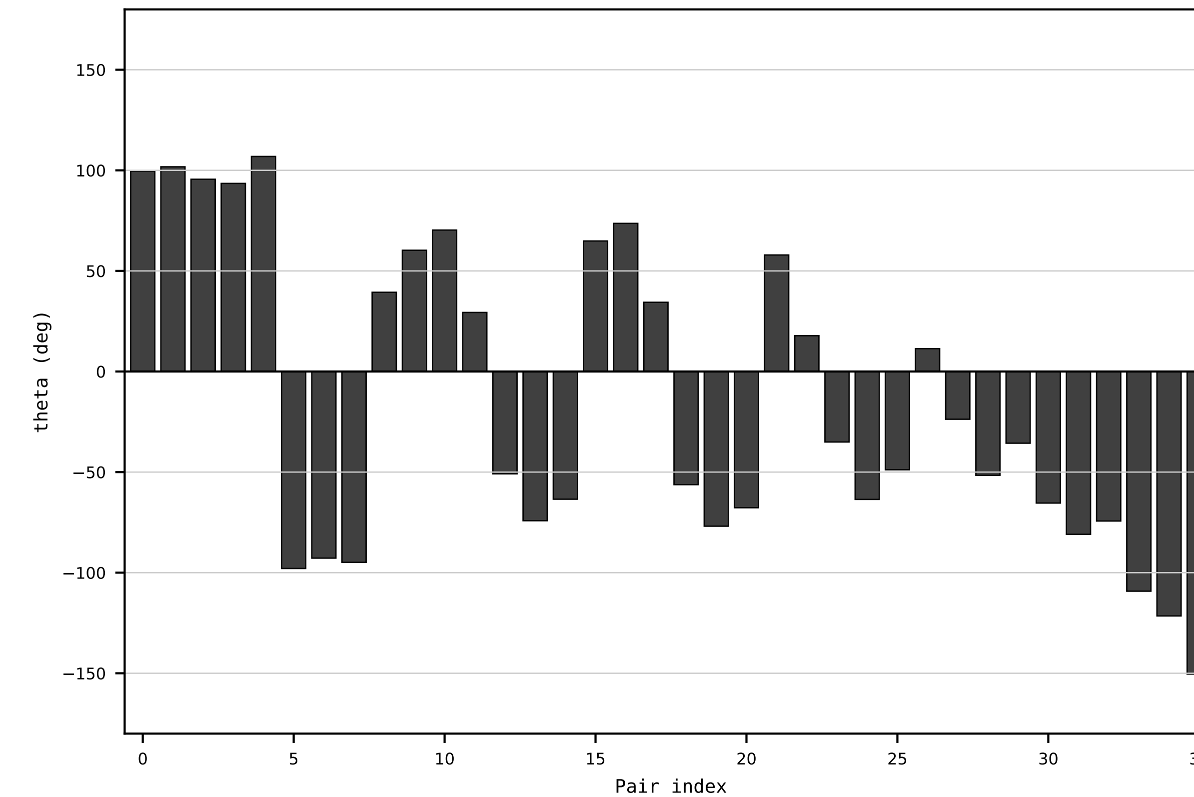


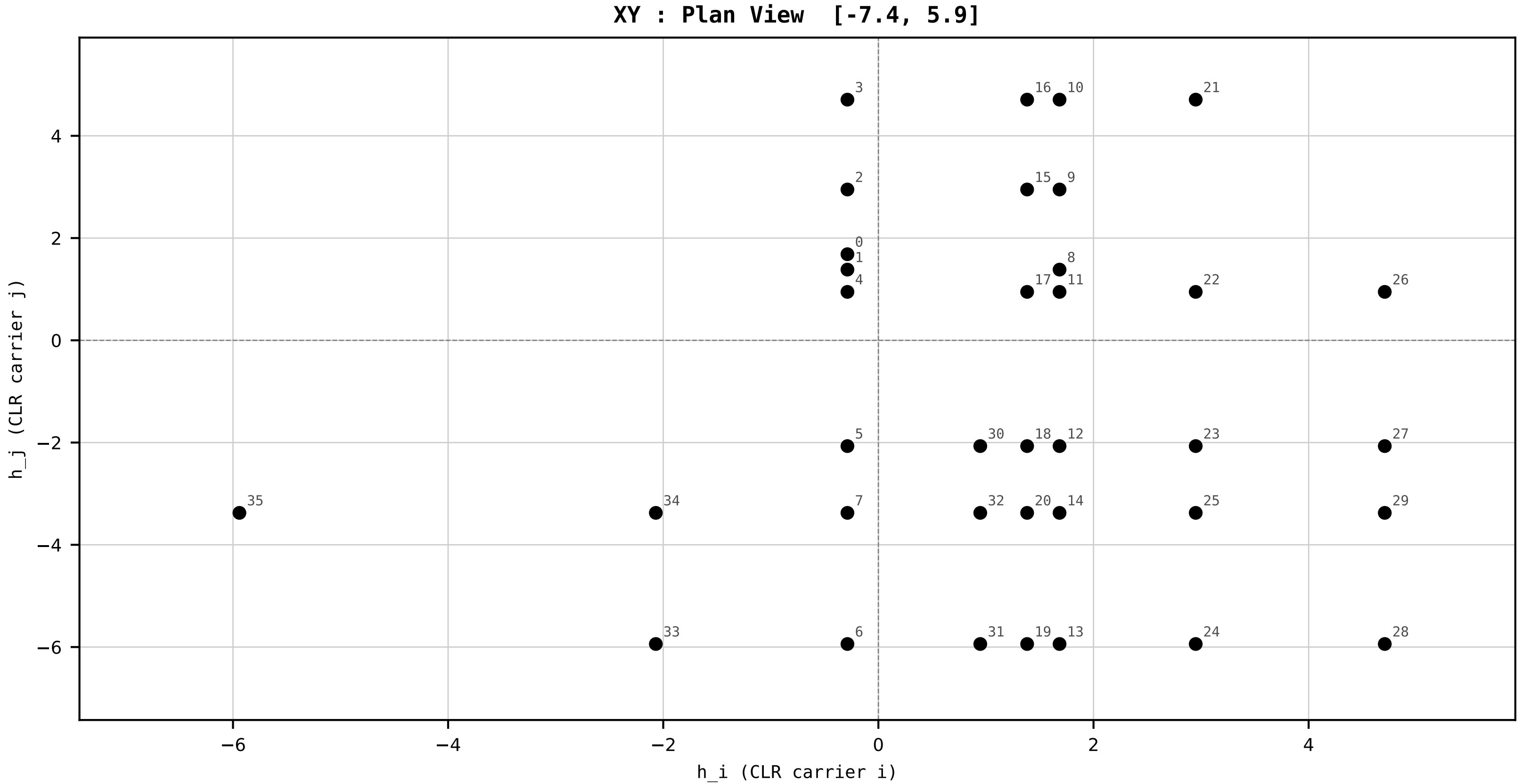


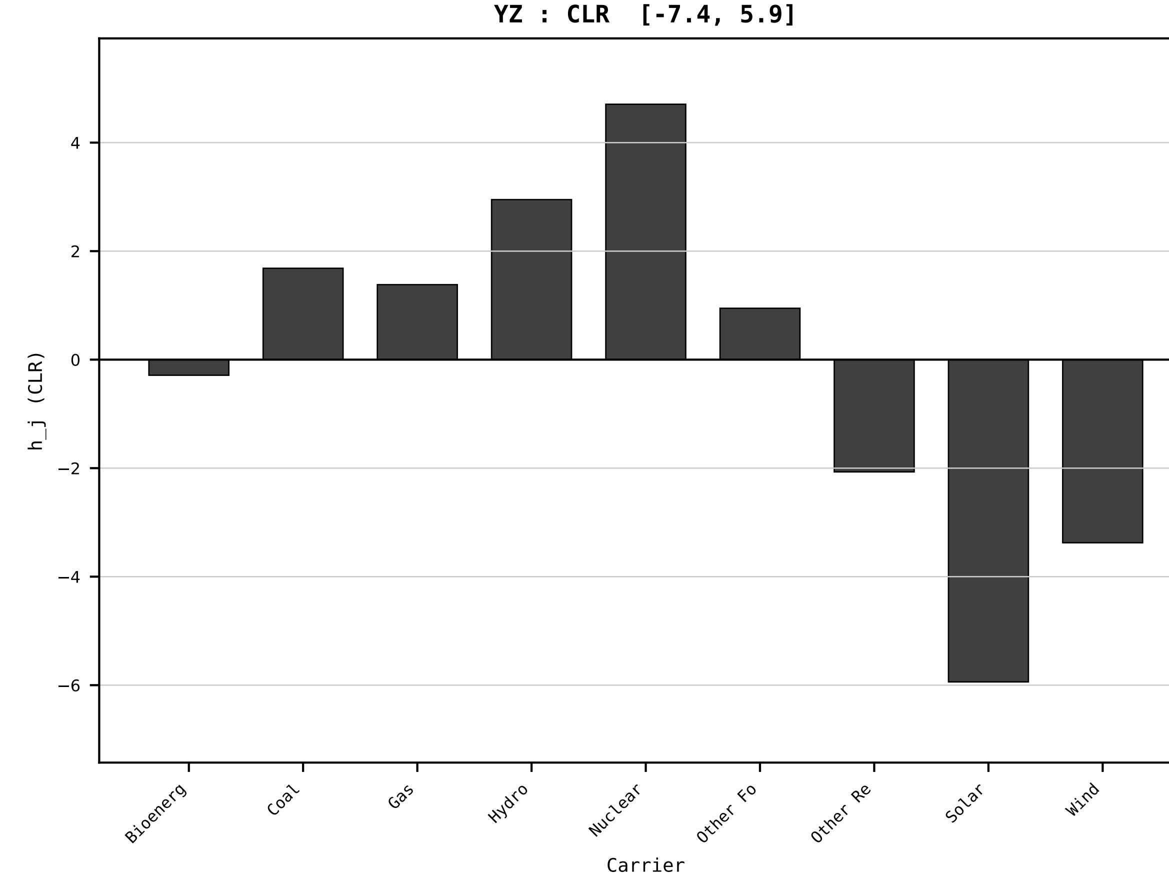
SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2001
t       : 1 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.636809
Ring    = Hs-4
E_metric= 87.5562
kappa_HS= 42108.00
omega   = 5.5939 deg
d_A     = 0.997829
Helm    = Wind
Helm d  = +0.839979
DR      = 10.648 HLR
DR ratio= 42108.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



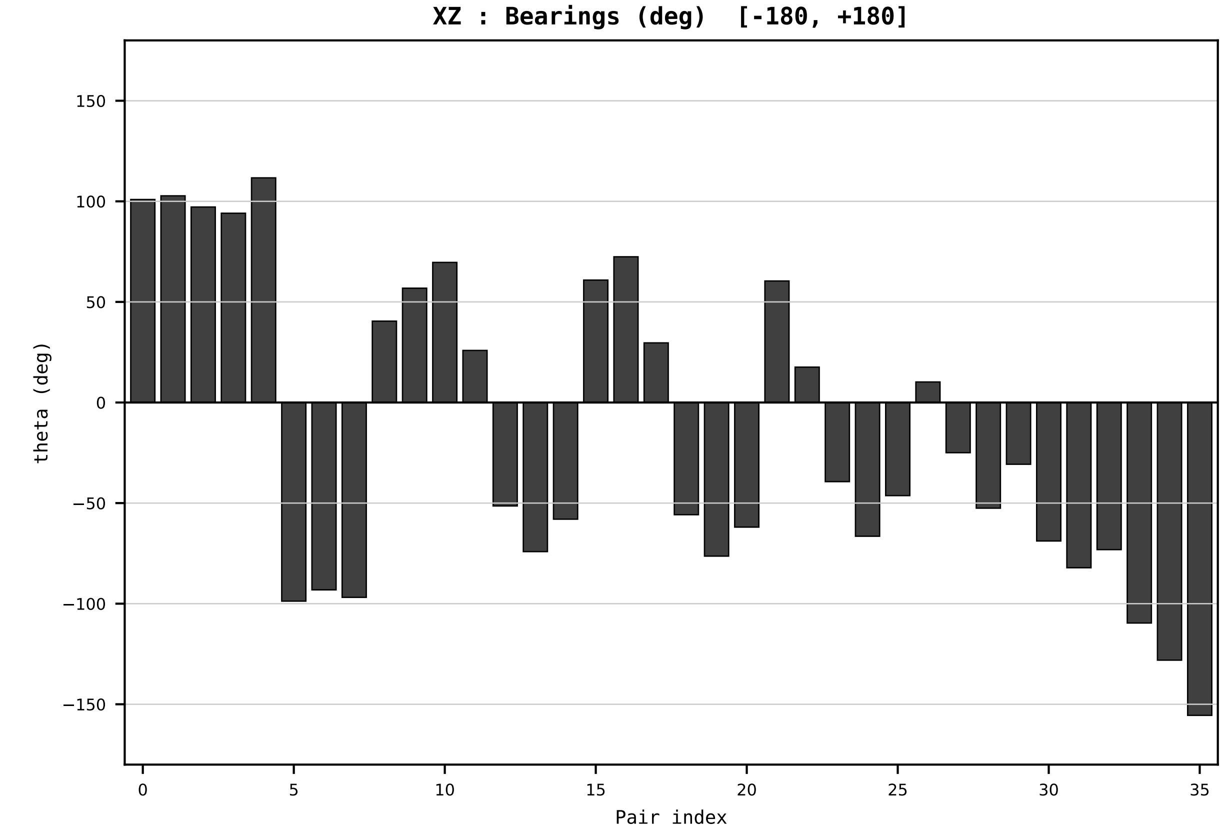


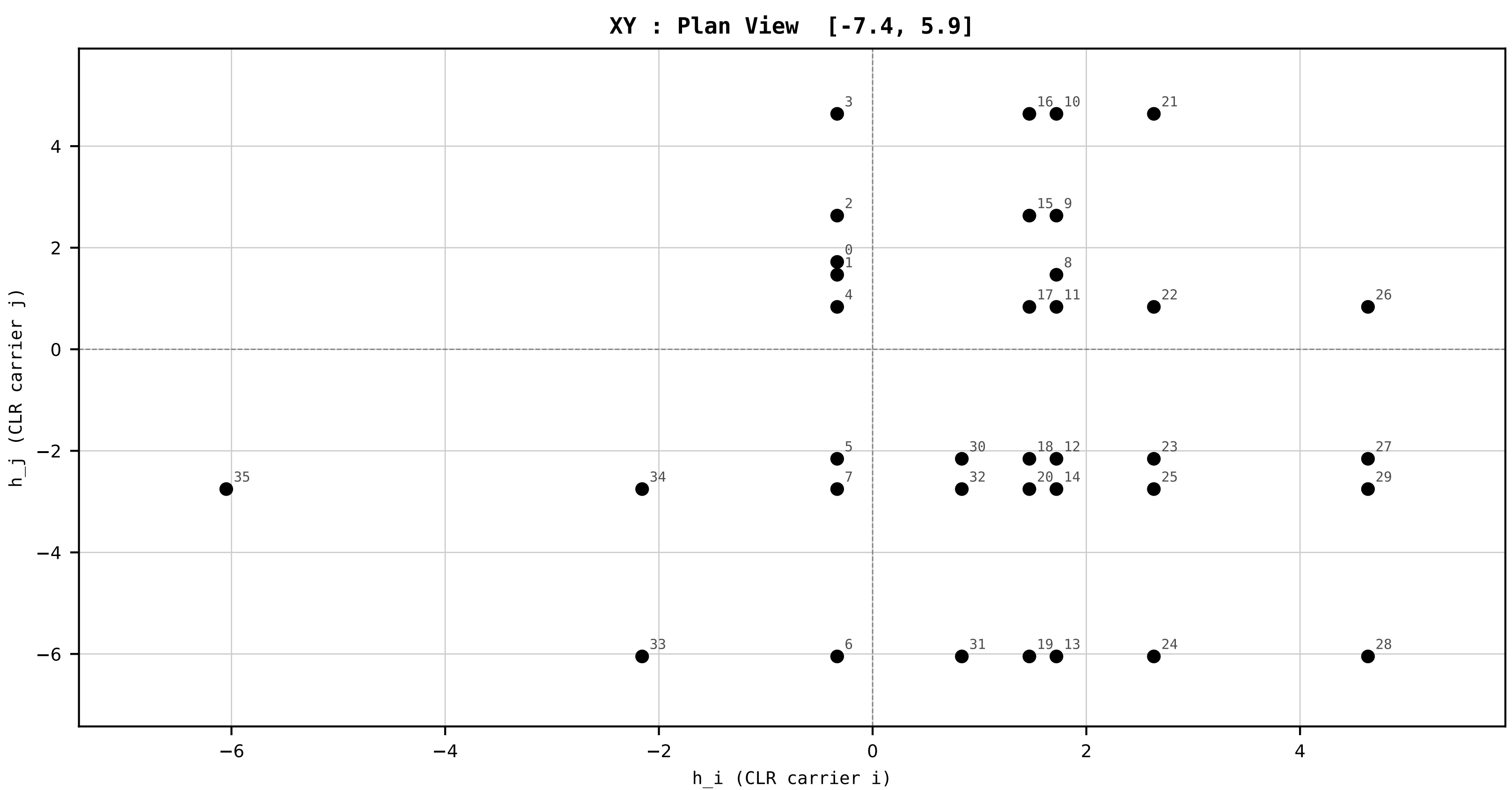


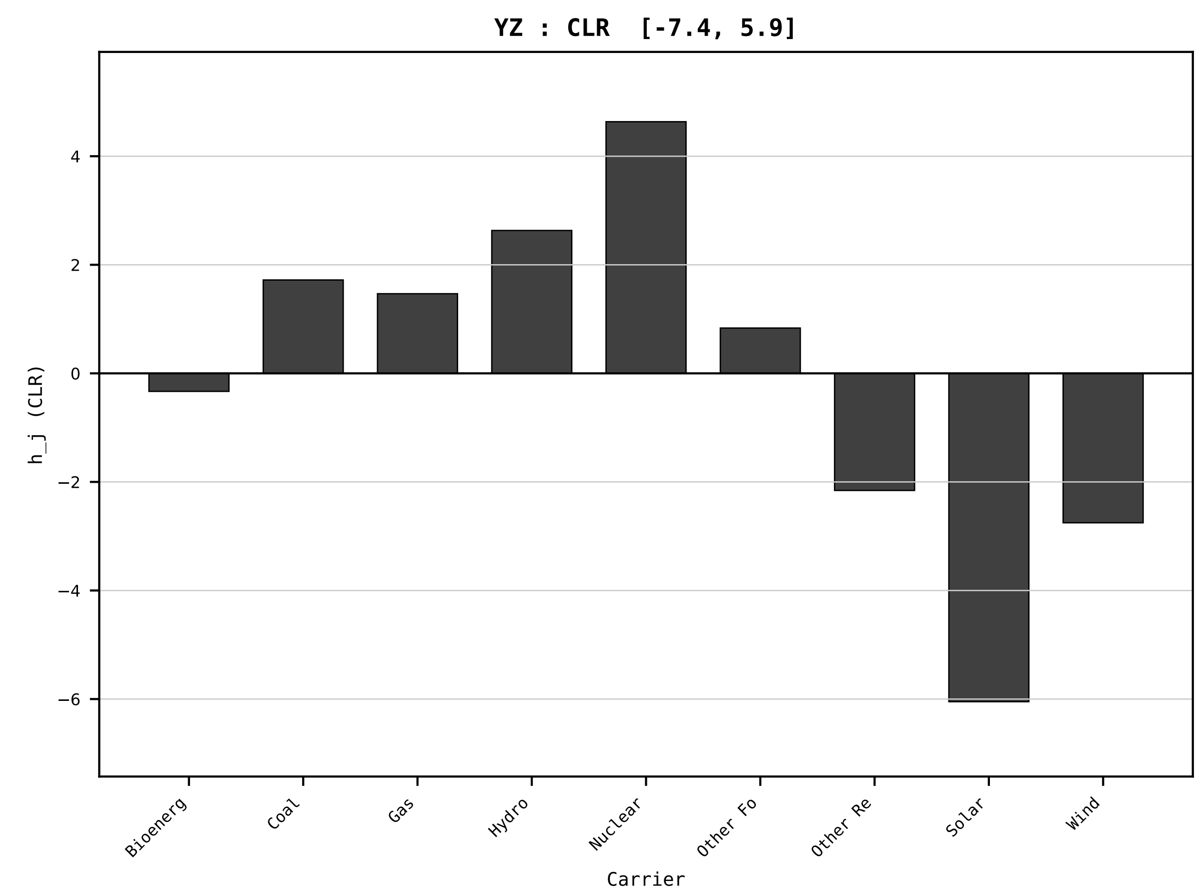
SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2002
t       : 2 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.644046
Ring    = Hs-4
E_metric= 83.1543
kappa_HS= 43676.00
omega   = 4.2963 deg
d_A     = 0.732324
Helm    = Wind
Helm d  = +0.622462
DR      = 10.685 HLR
DR ratio= 43676.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



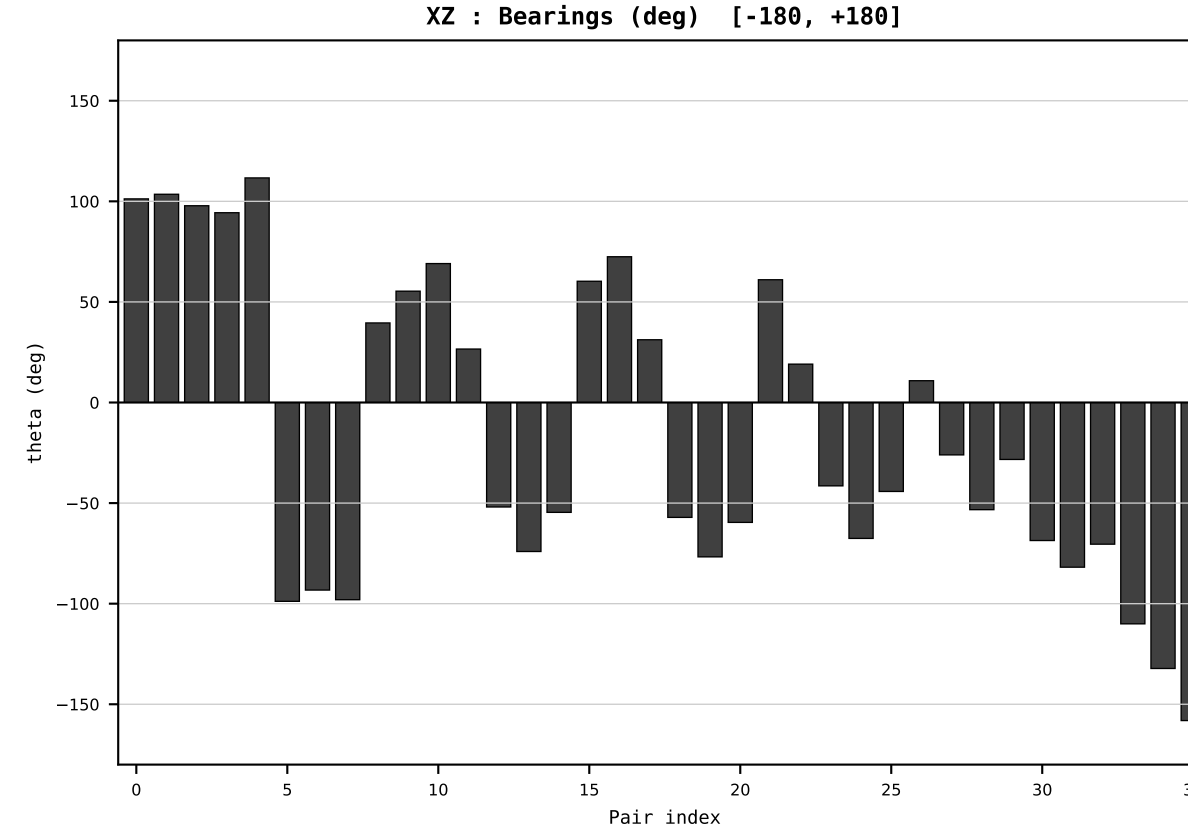


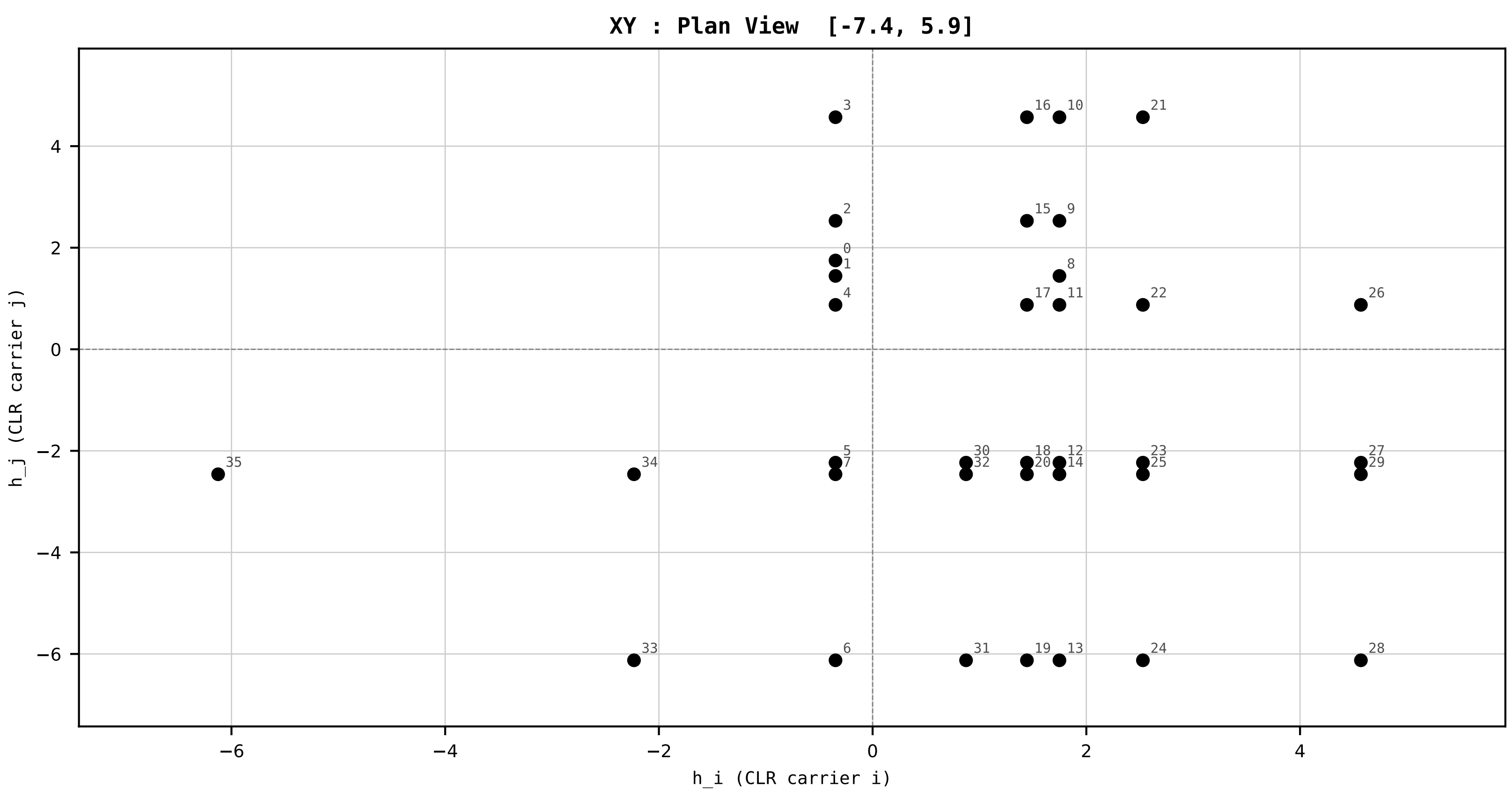


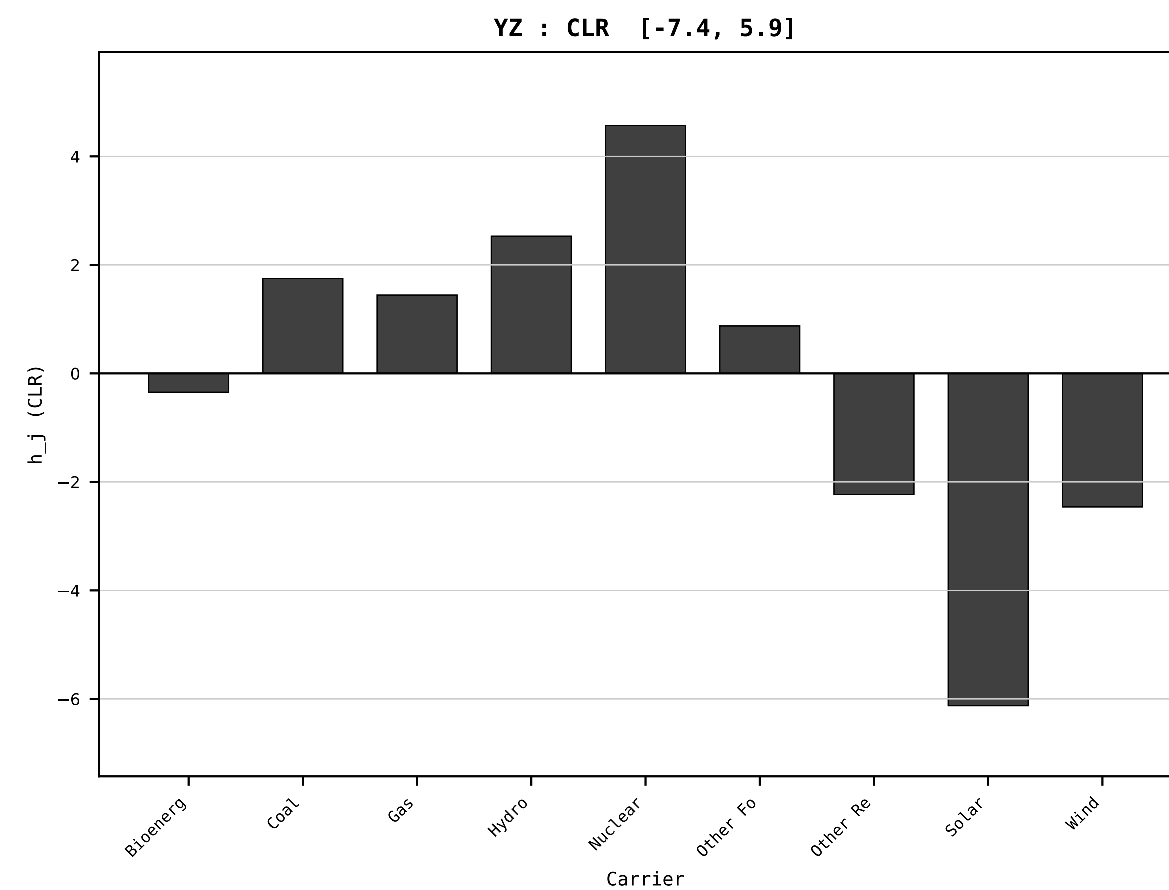
SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2003
t       : 3 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.636320
Ring    = Hs-4
E_metric= 81.8645
kappa_HS= 44107.00
omega   = 2.0886 deg
d_A     = 0.338625
Helm    = Wind
Helm d  = +0.291606
DR      = 10.694 HLR
DR ratio= 44107.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



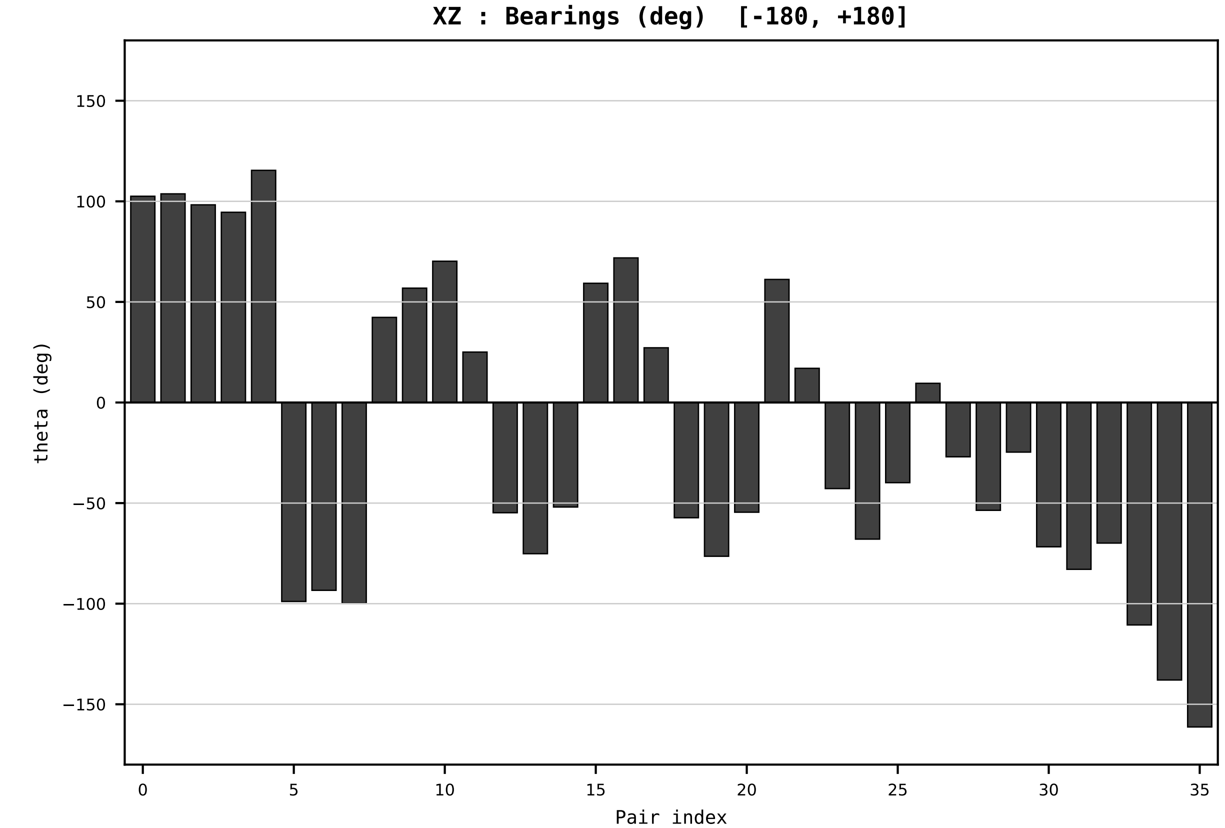


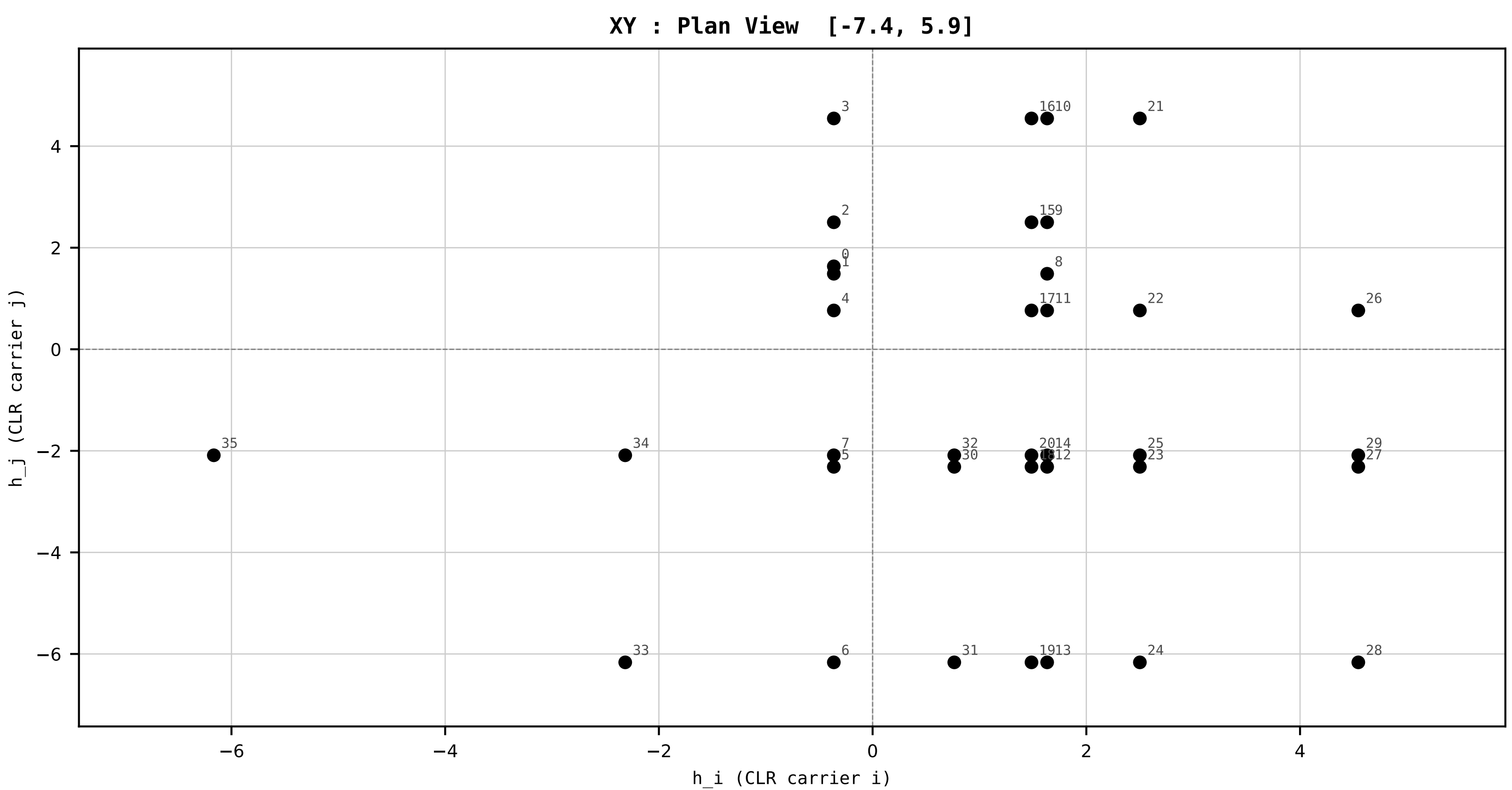


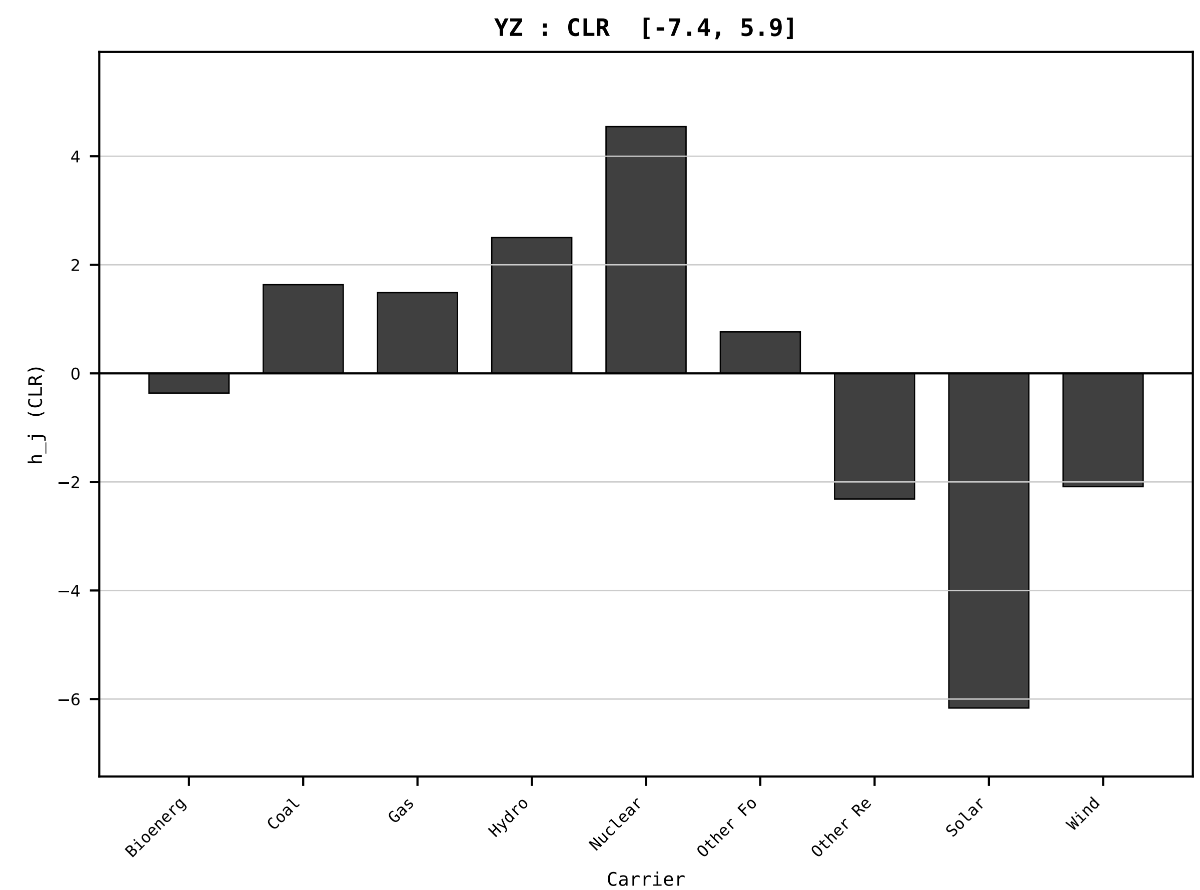
SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2004
t       : 4 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.639536
Ring    = Hs-4
E_metric= 80.2373
kappa_HS= 44824.00
omega   = 2.6139 deg
d_A     = 0.420503
Helm    = Wind
Helm d  = +0.373595
DR      = 10.710 HLR
DR ratio= 44824.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```





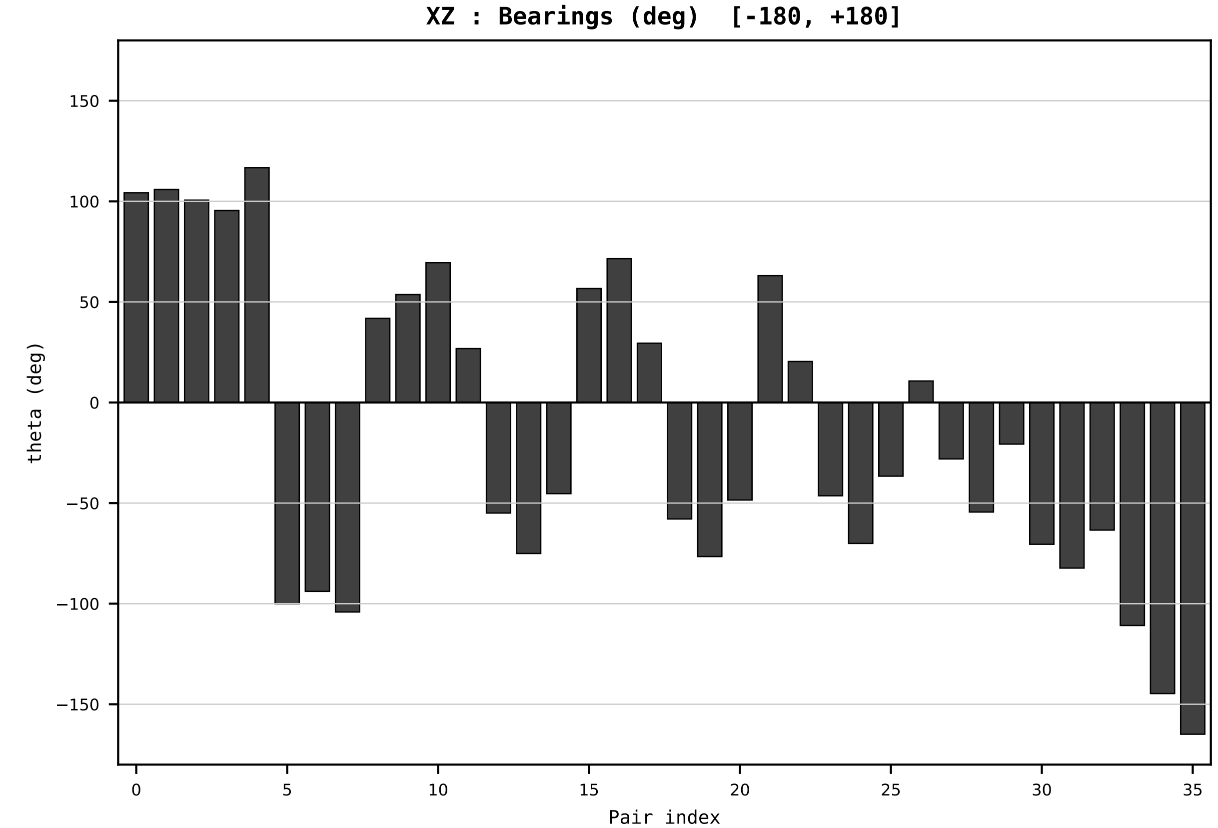


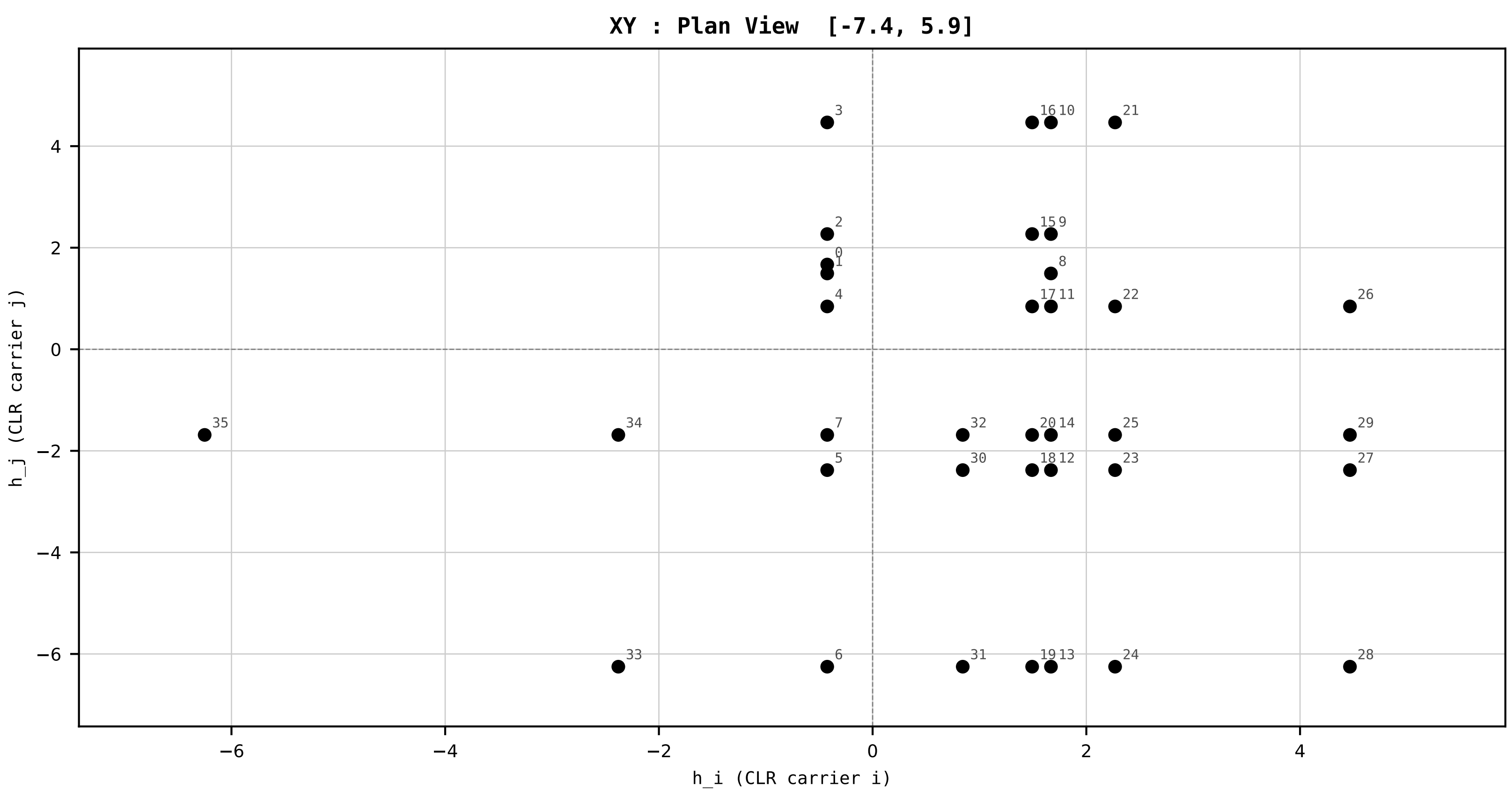
SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

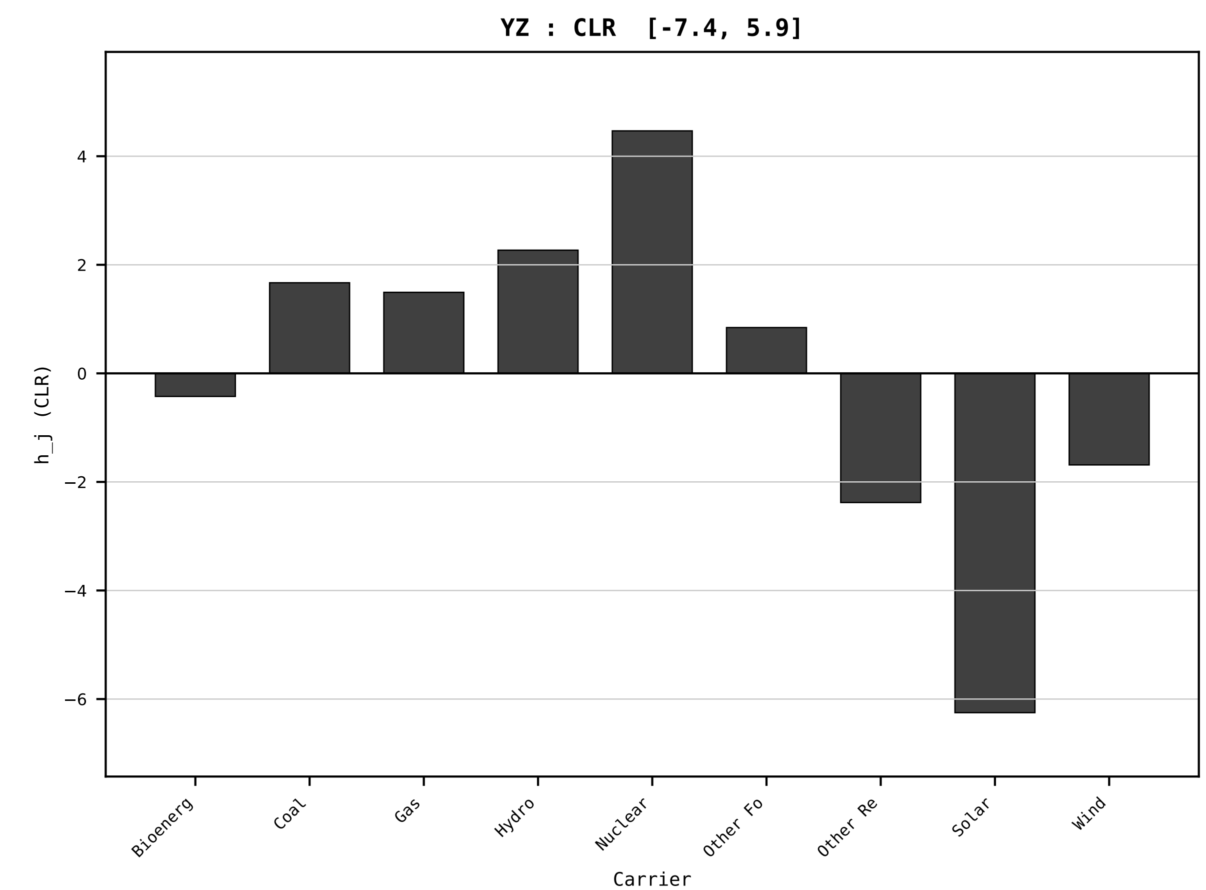
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2005
t       : 5 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.634489
Ring    = Hs-4
E_metric= 78.5884
kappa_HS= 45153.00
omega   = 3.1185 deg
d_A     = 0.493709
Helm    = Wind
Helm d  = +0.401223
DR      = 10.718 HLR
DR ratio= 45153.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





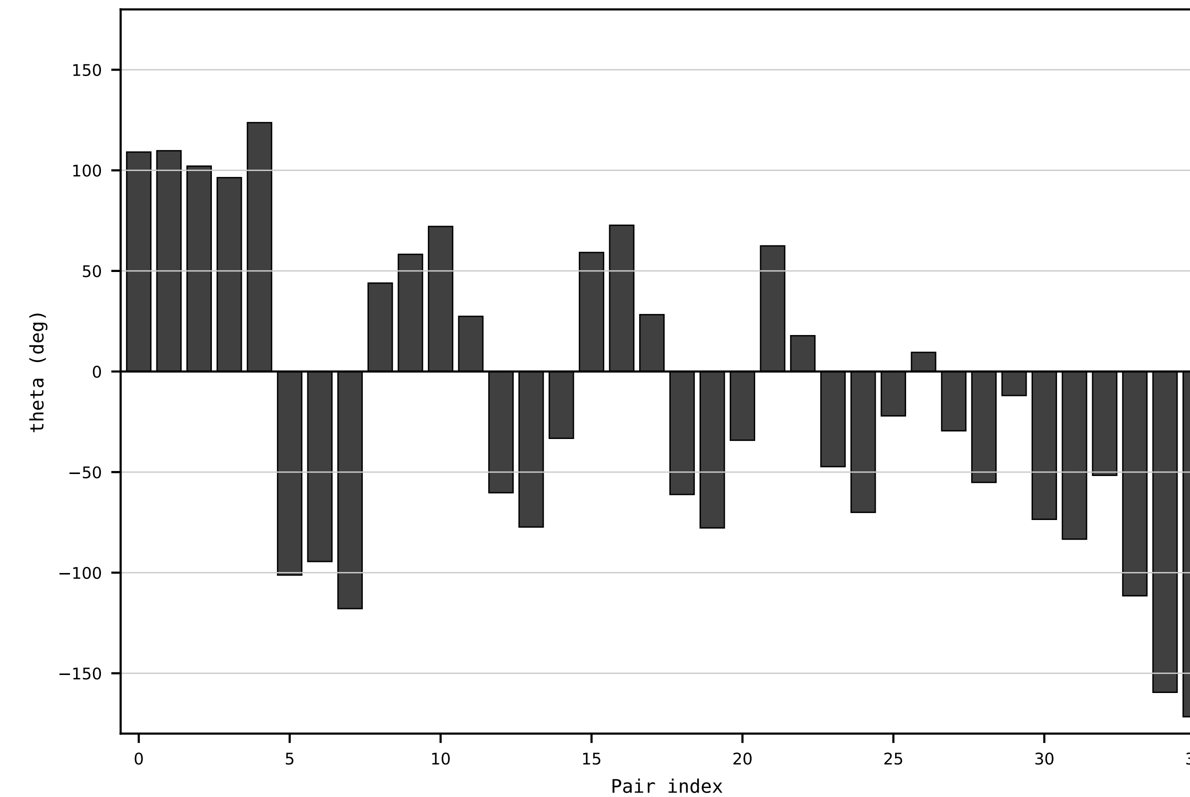


SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

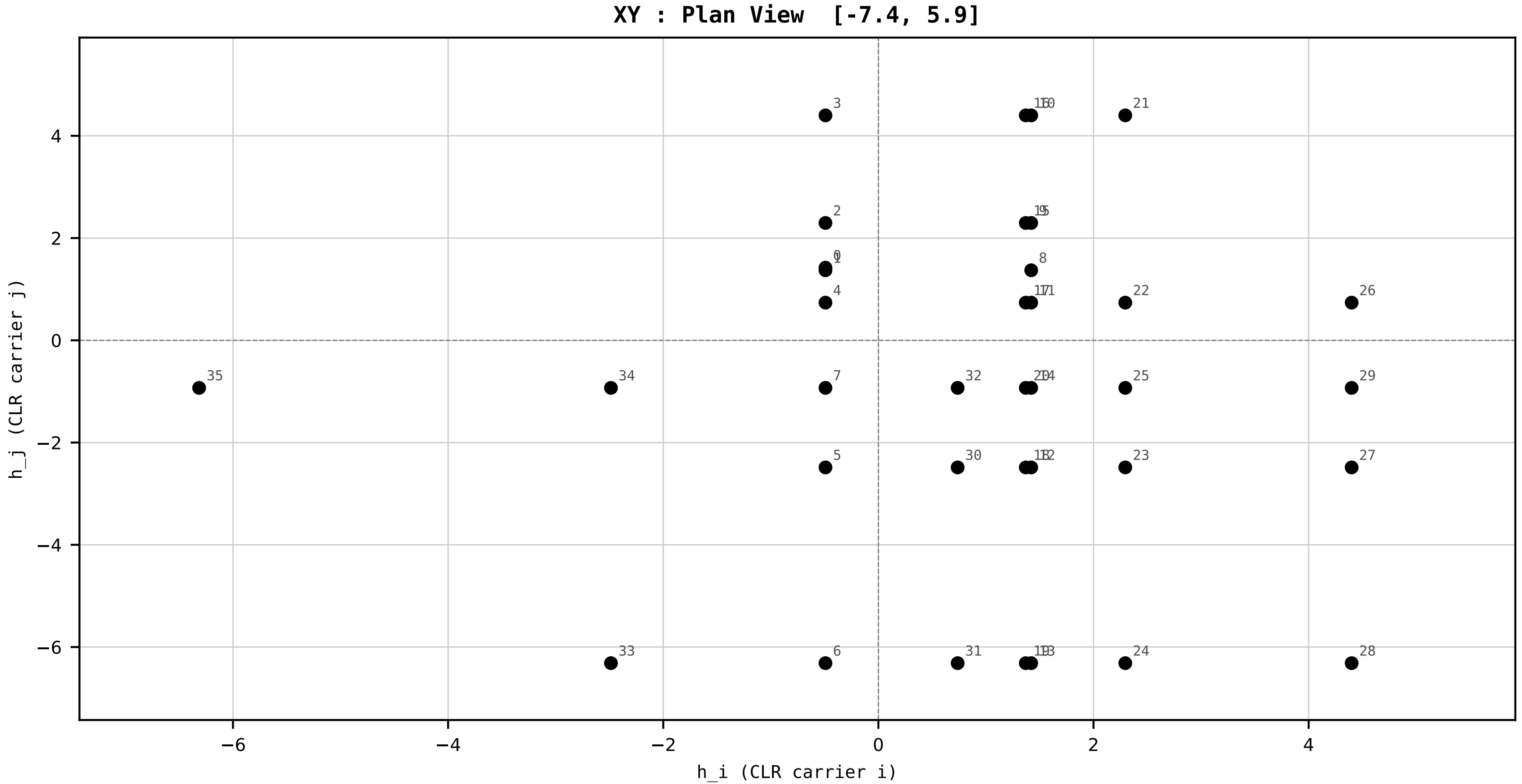
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2006
t       : 6 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.635362
Ring    = Hs-4
E_metric= 76.2355
kappa_HS= 45019.00
omega   = 5.3196 deg
d_A     = 0.827413
Helm    = Wind
Helm d  = +0.756002
DR      = 10.715 HLR
DR ratio= 45019.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```

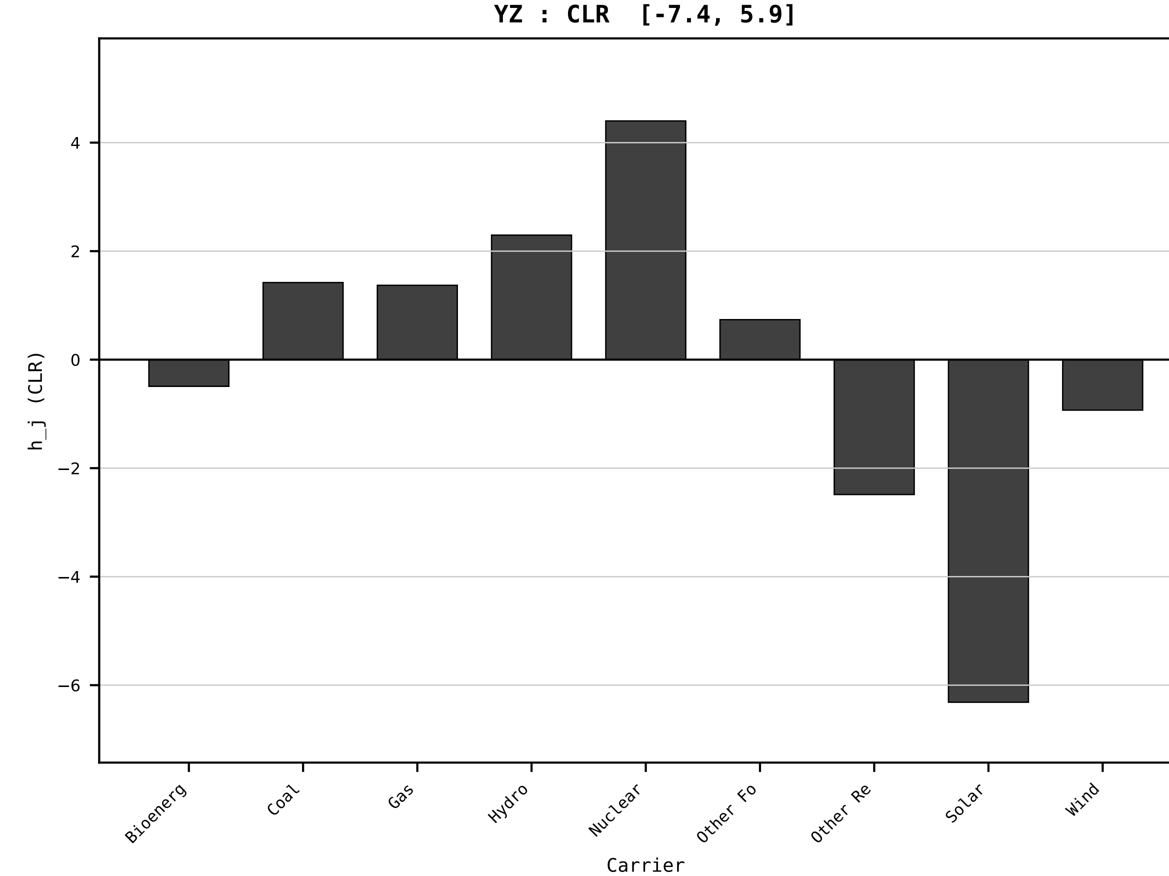
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-7.4, 5.9]



YZ : CLR [-7.4, 5.9]

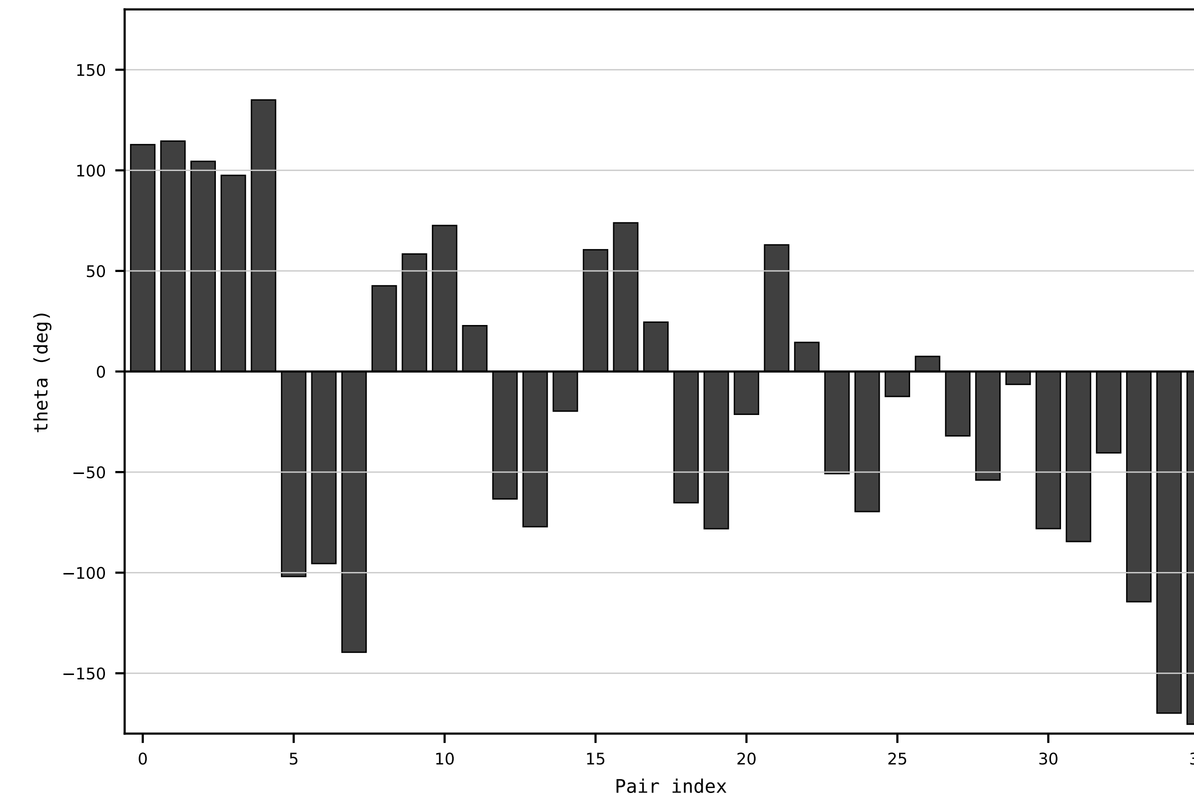


SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

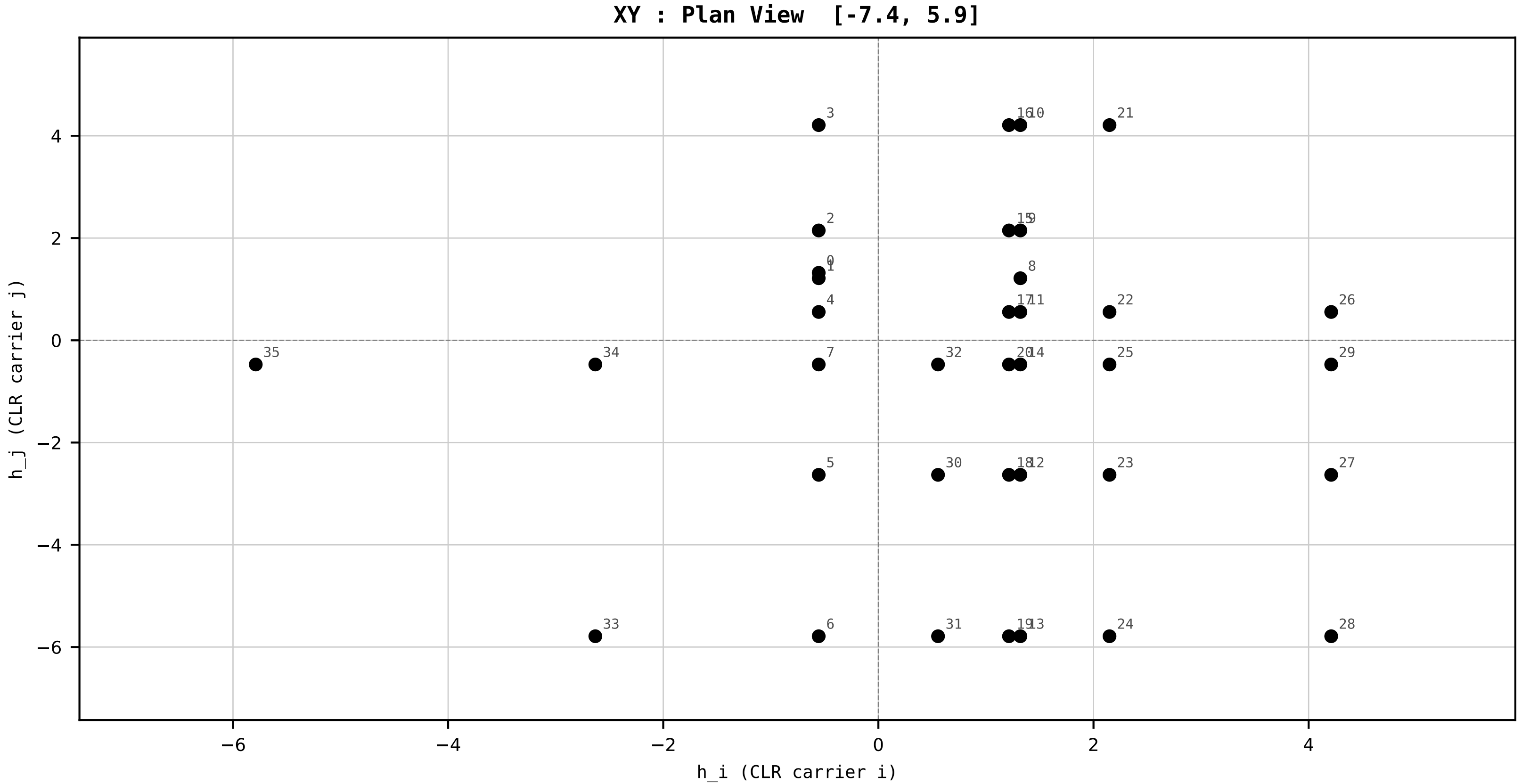
```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2007
t       : 7 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.618062
Ring    = Hs-4
E_metric= 66.8255
kappa_HS= 21986.50
omega   = 3.8806 deg
d_A     = 0.798187
Helm    = Solar
Helm d  = +0.526469
DR      = 9.998 HLR
DR ratio= 21986.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```

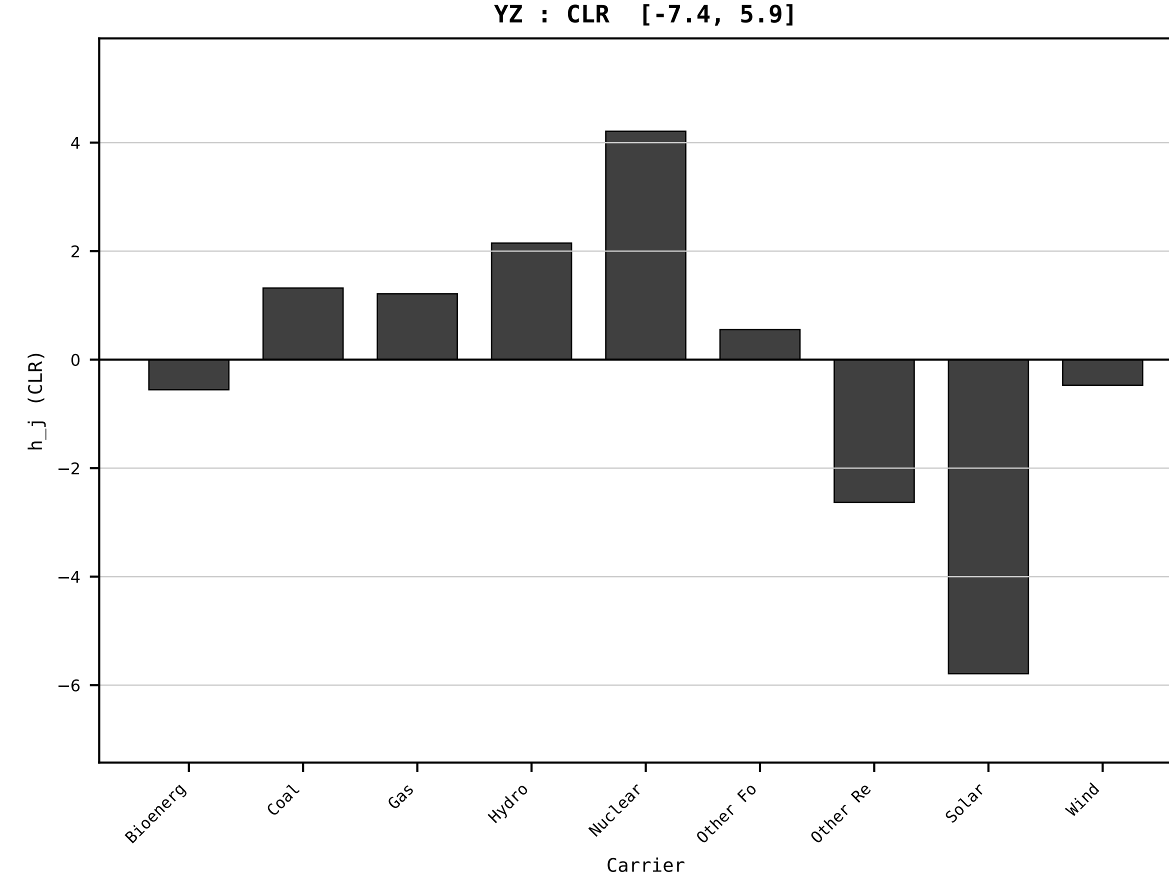
XZ : Bearings (deg) [-180, +180]



XY : Plan View [-7.4, 5.9]



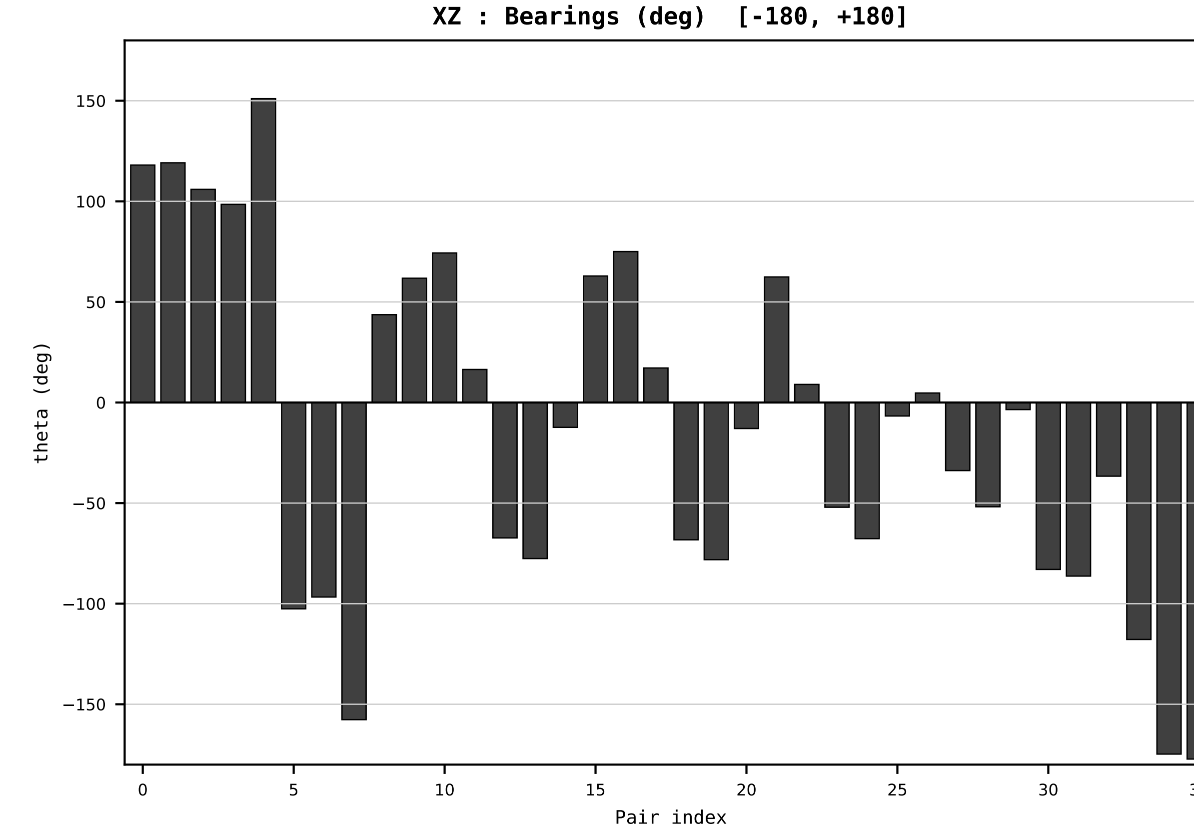
YZ : CLR [-7.4, 5.9]

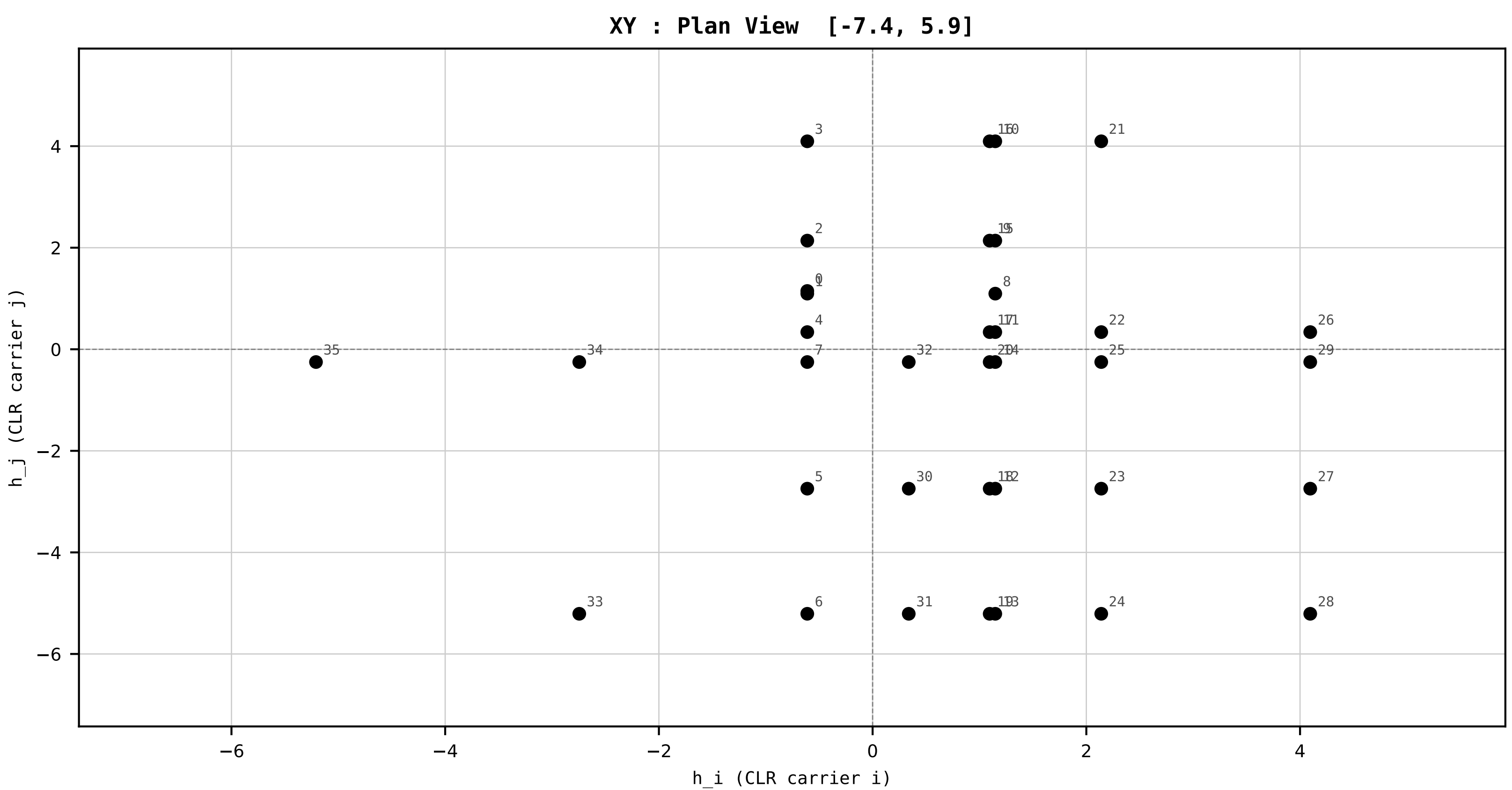


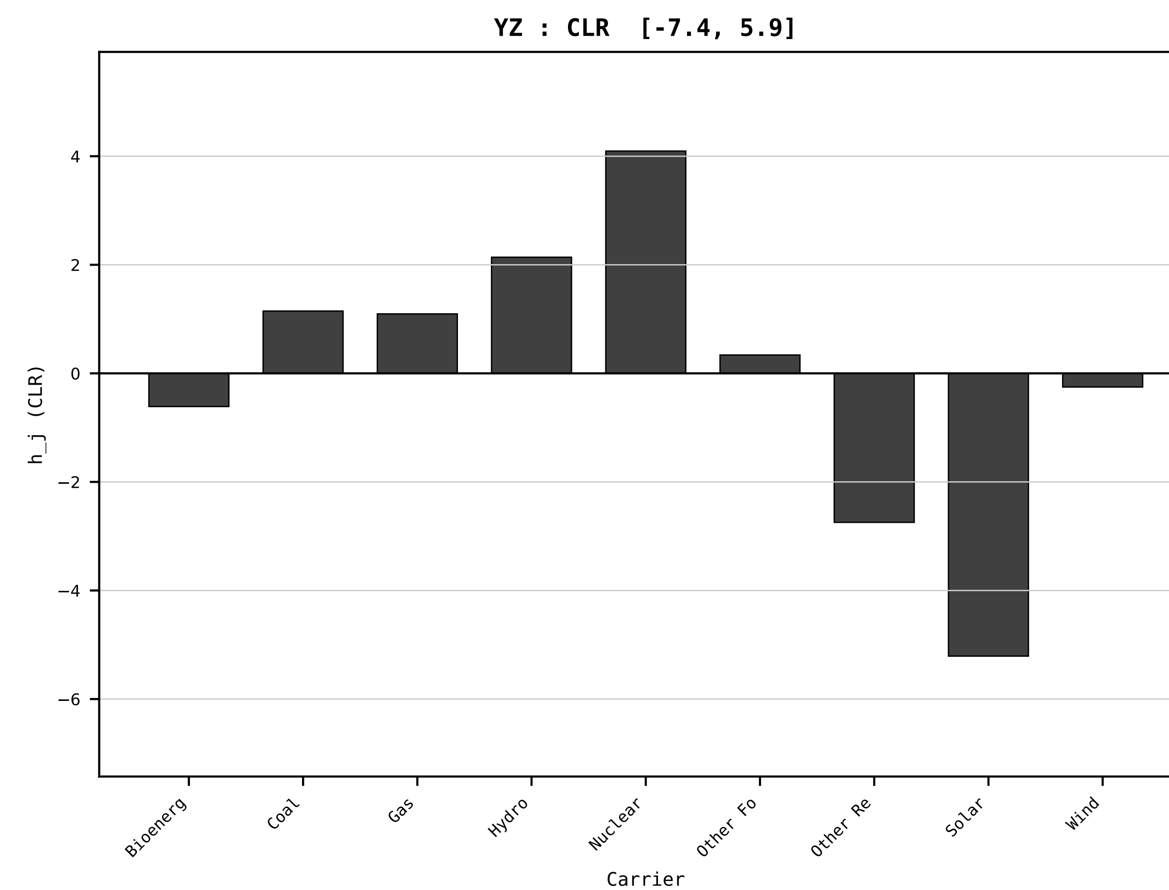
SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2008
t       : 8 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.610524
Ring    = Hs-4
E_metric= 59.0888
kappa_HS= 10986.25
omega   = 3.7337 deg
d_A     = 0.710392
Helm    = Solar
Helm d  = +0.579497
DR      = 9.304 HLR
DR ratio= 10986.3x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



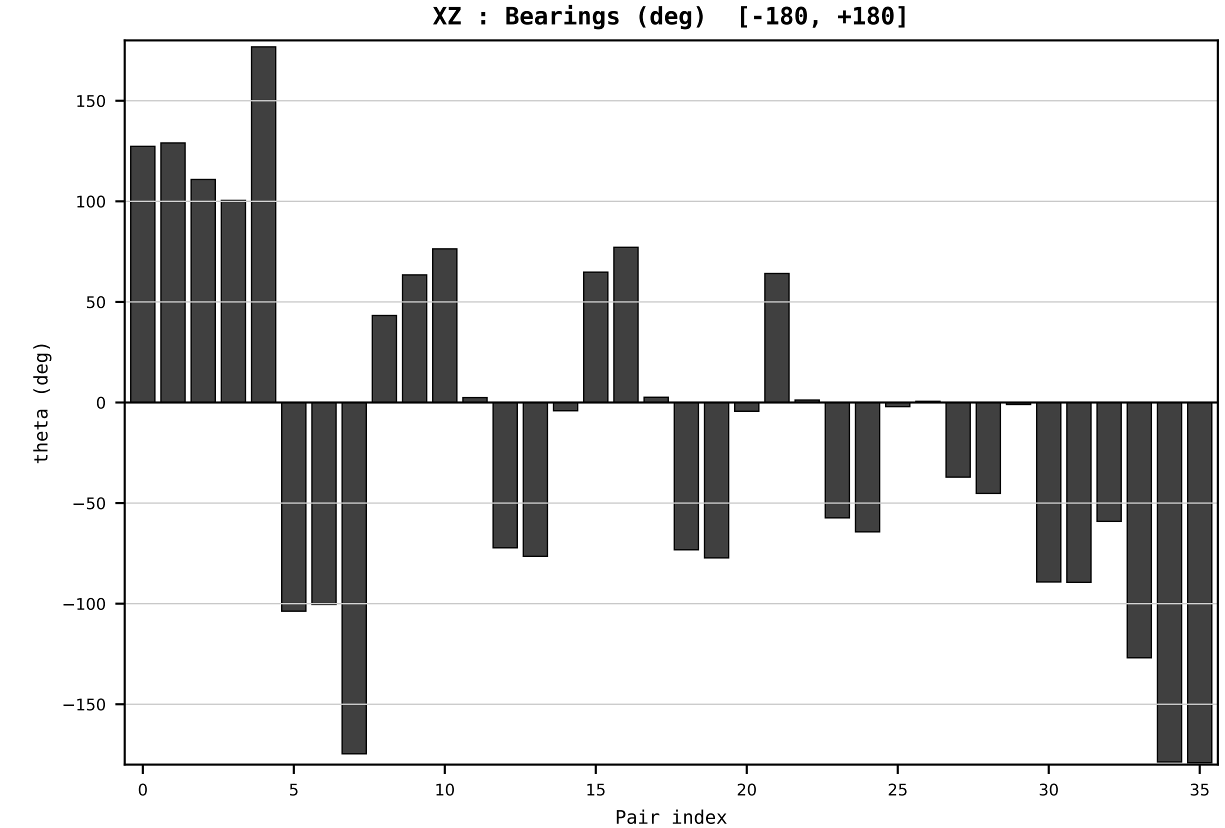


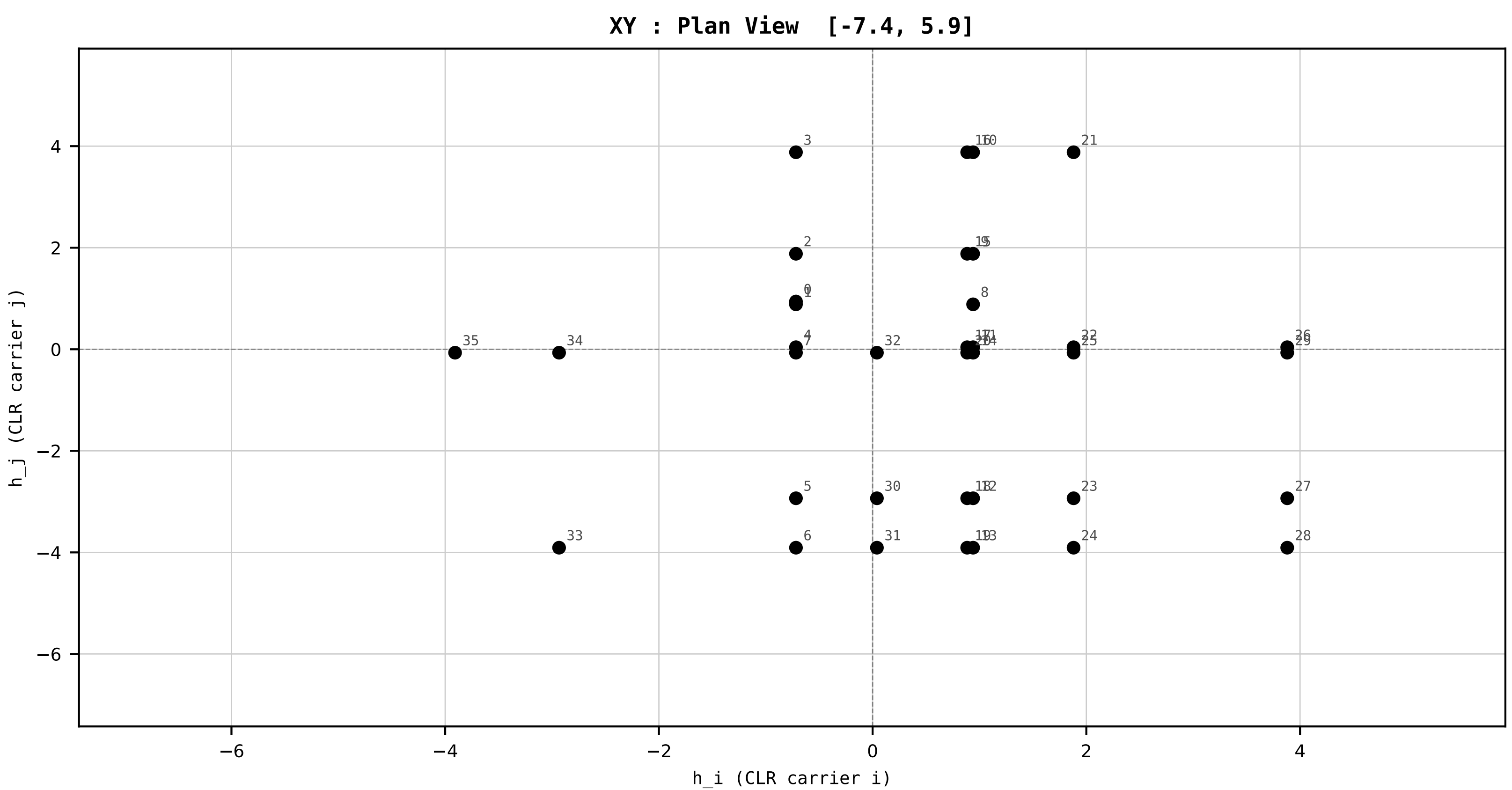


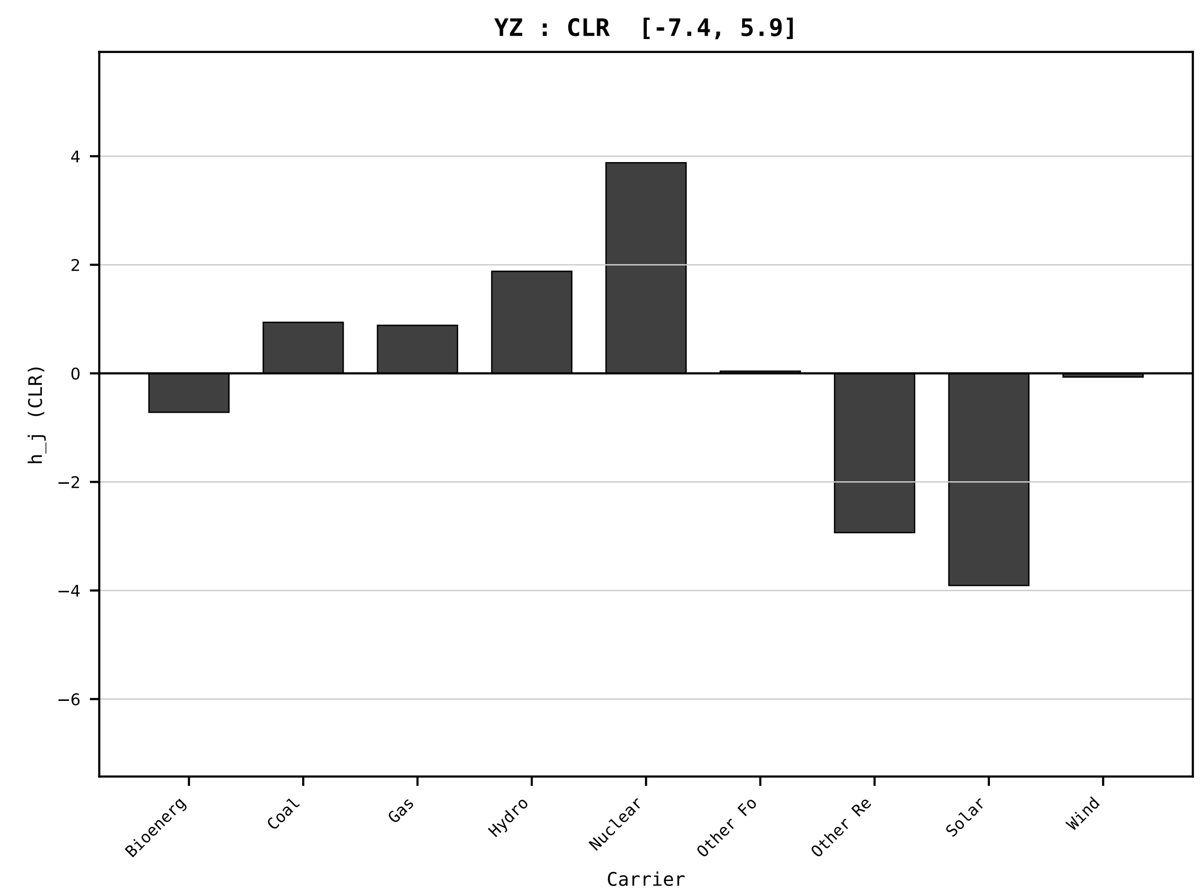
SECTION PLATE t=9 | Year 2009 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2009
t       : 9 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.604621
Ring    = Hs-4
E_metric= 44.6563
kappa_HS= 2410.24
omega   = 8.2198 deg
d_A     = 1.436738
Helm    = Solar
Helm d  = +1.301456
DR      = 7.787 HLR
DR ratio= 2410.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



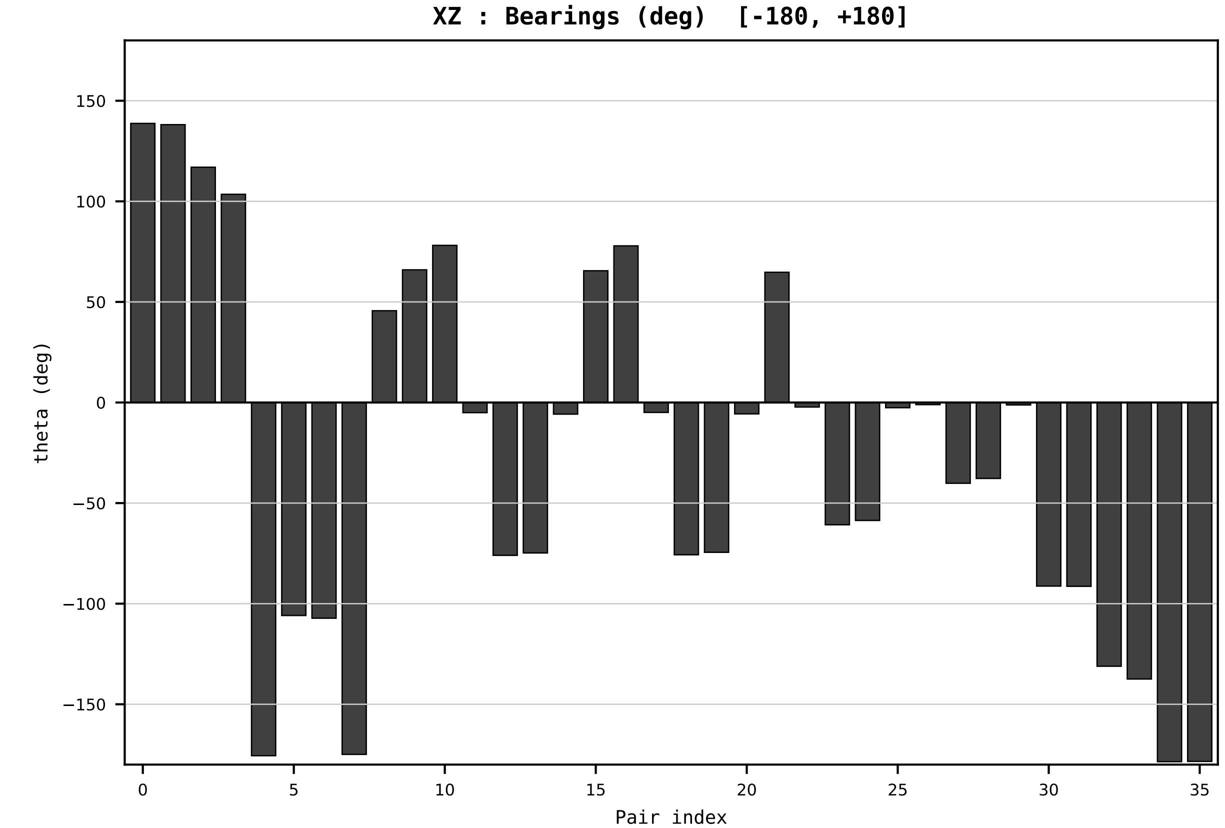


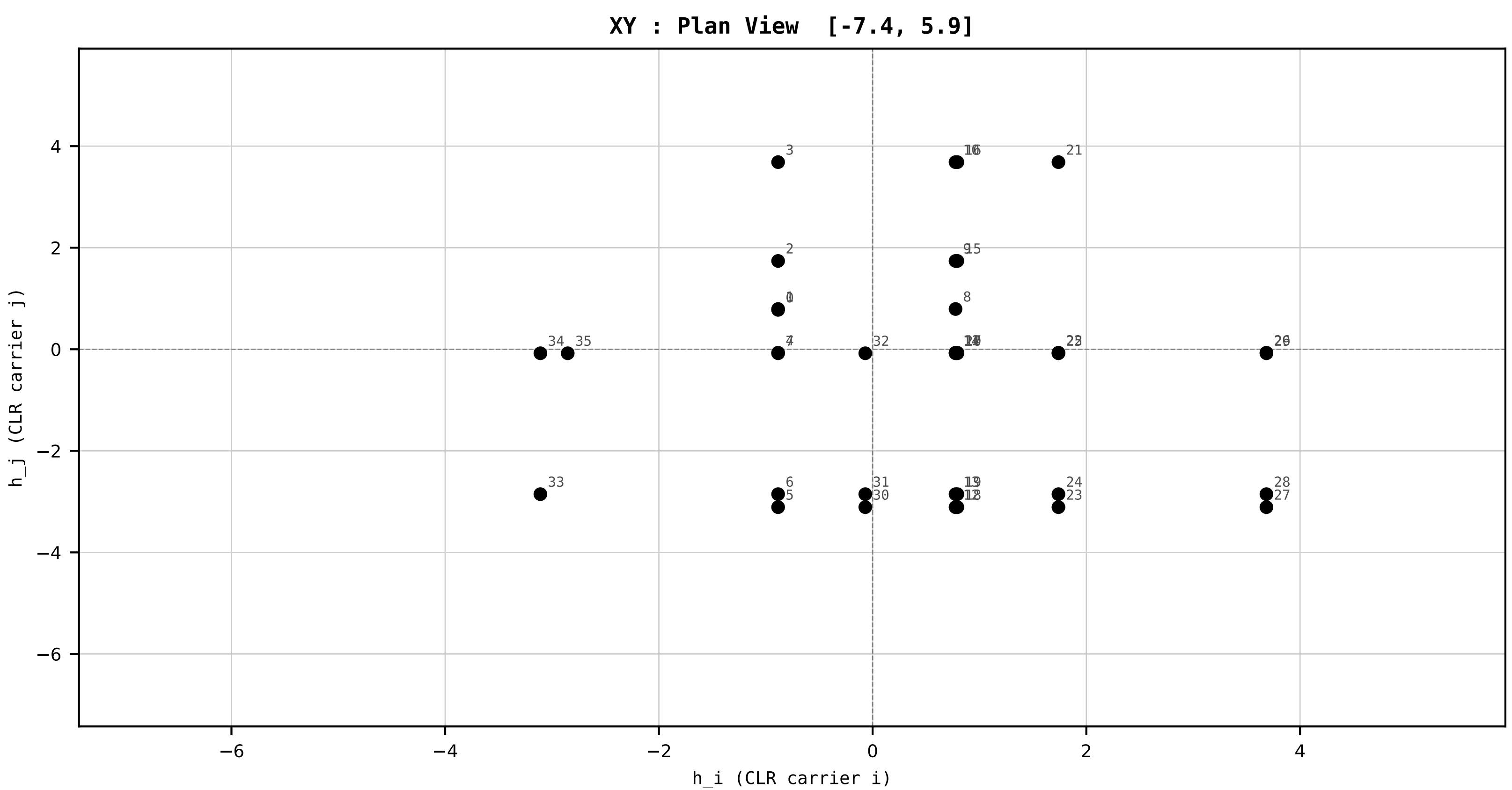


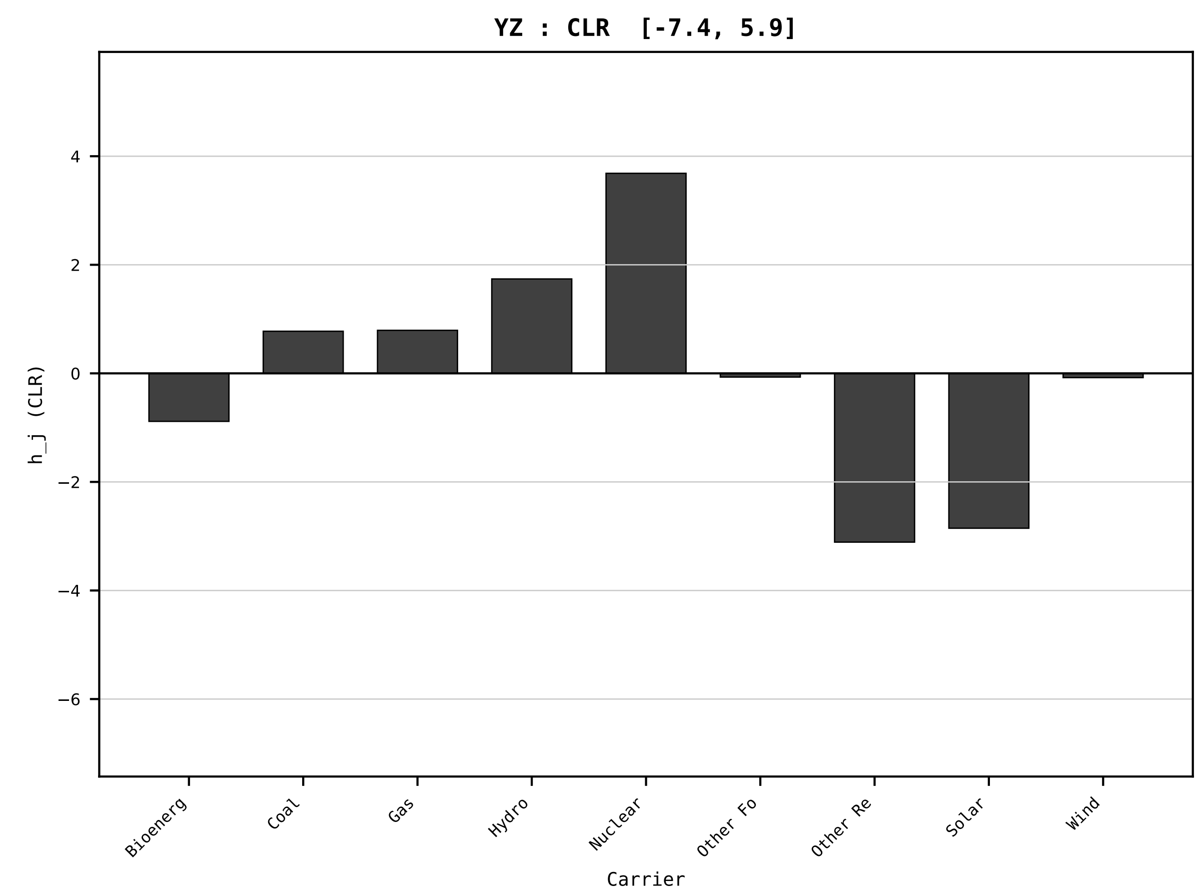
SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2010
t       : 10 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.586166
Ring    = Hs-4
E_metric= 36.4358
kappa_HS= 892.75
omega   = 8.3636 deg
d_A     = 1.129480
Helm    = Solar
Helm d  = +1.054432
DR      = 6.794 HLR
DR ratio= 892.7x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



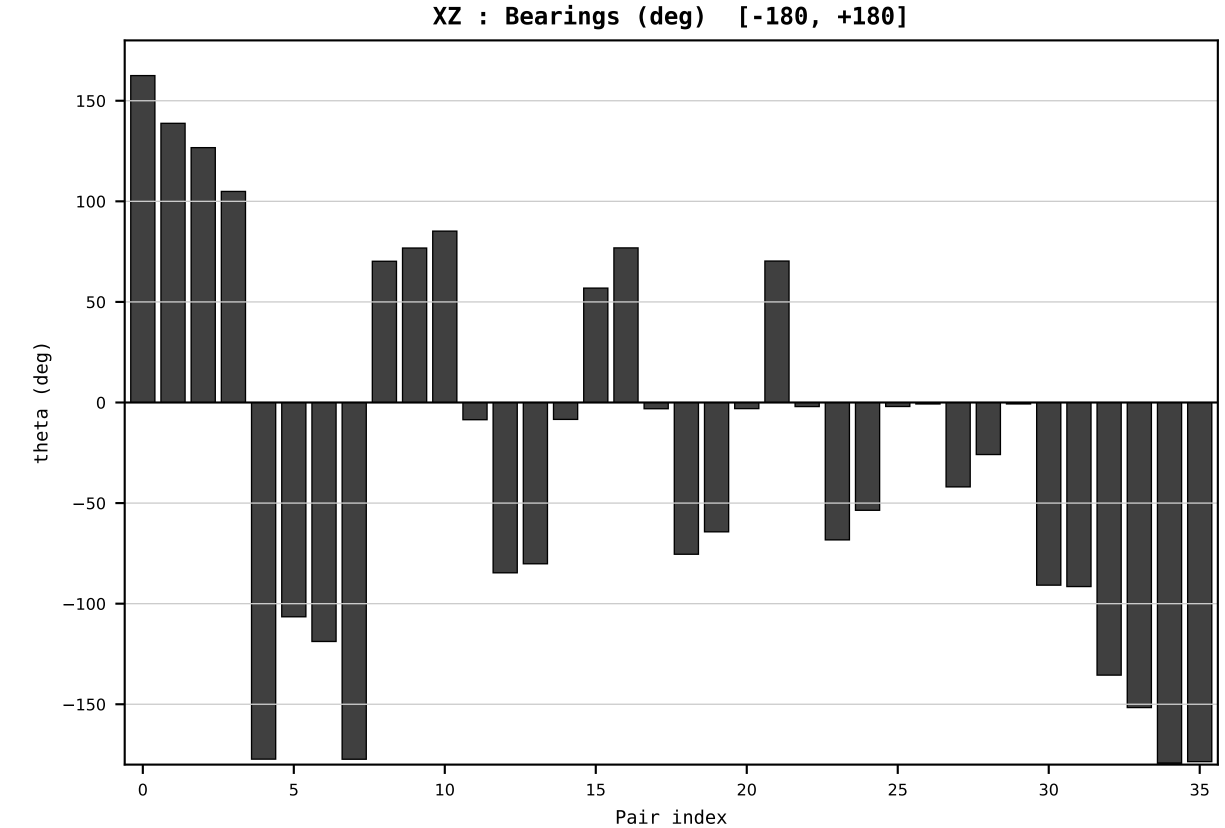


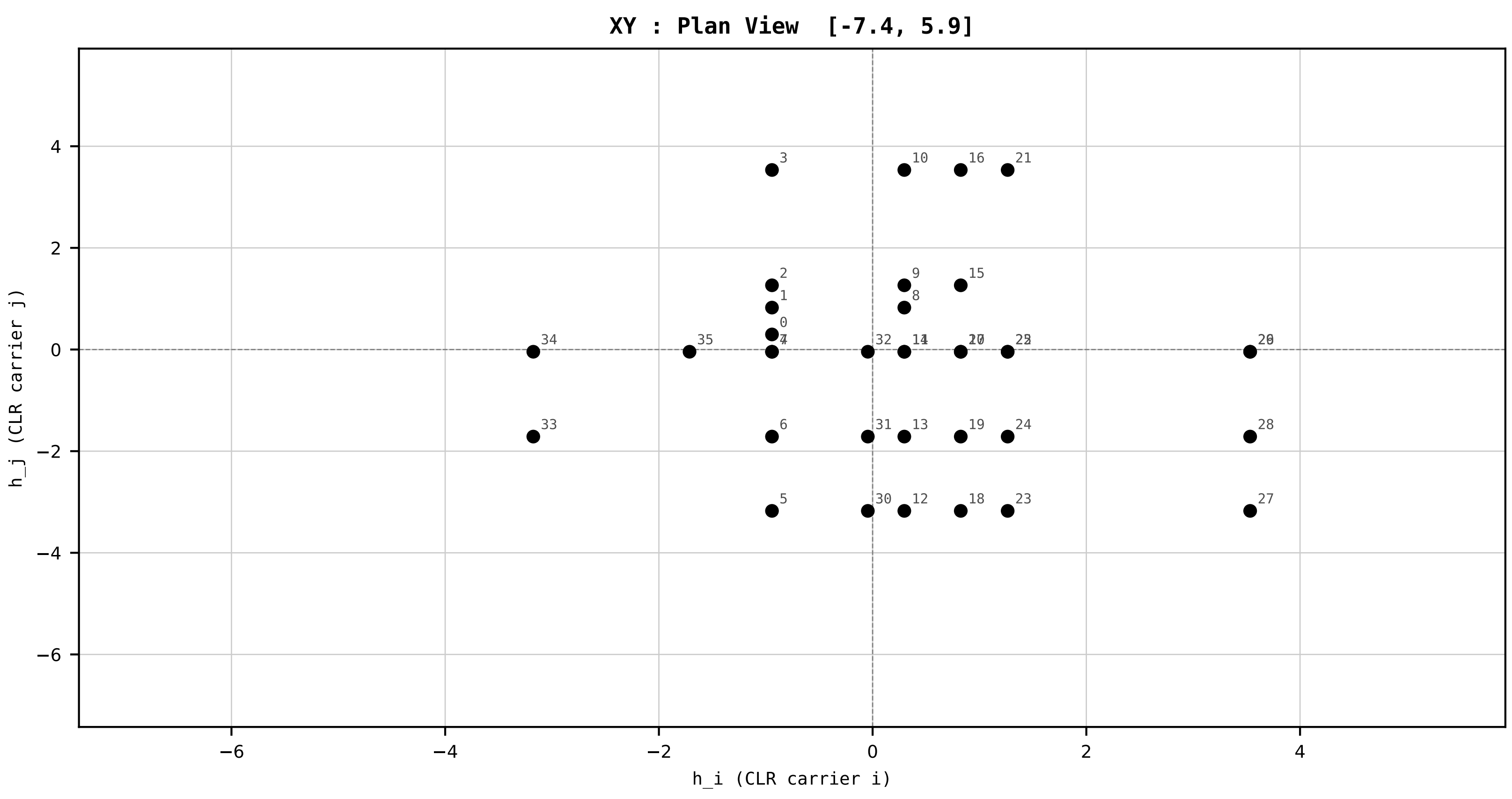


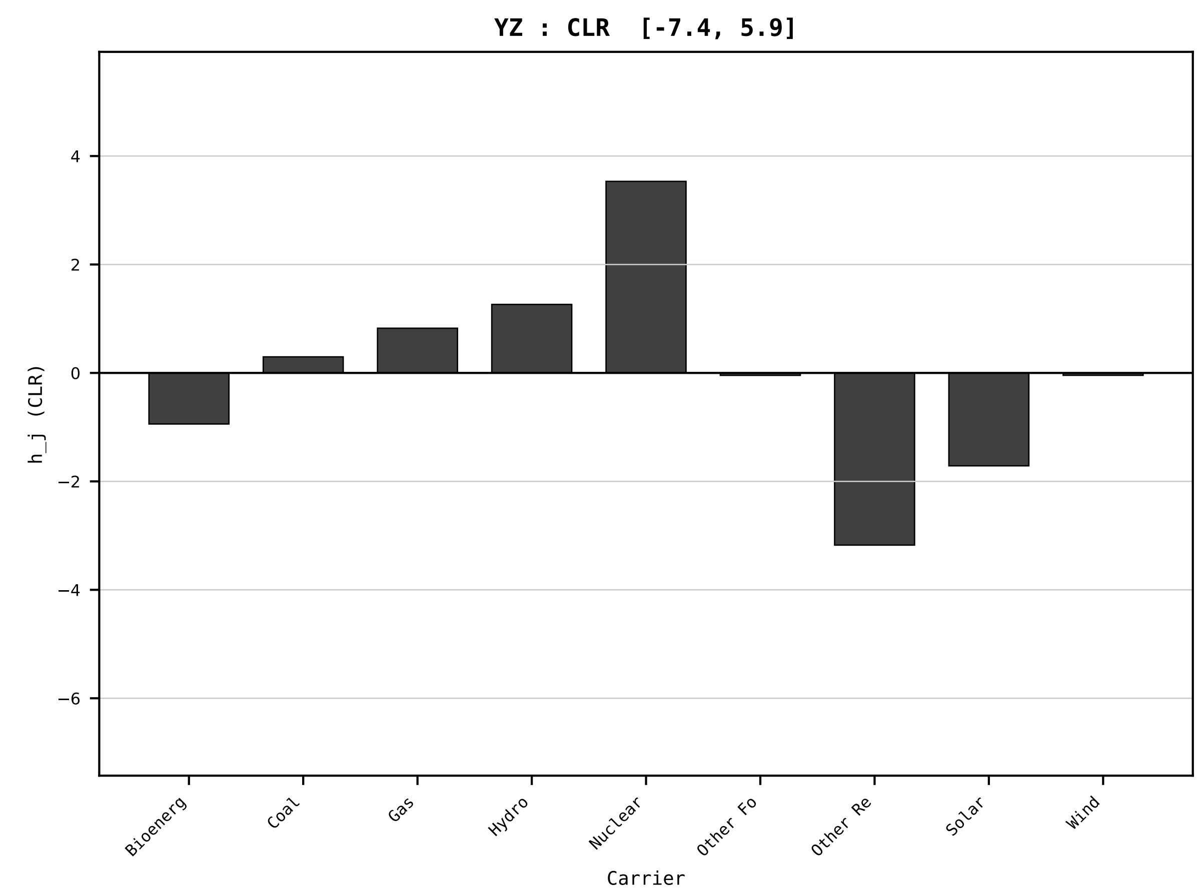
SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2011
t       : 11 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.592444
Ring    = Hs-4
E_metric= 28.7571
kappa_HS= 819.24
omega   = 11.6539 deg
d_A     = 1.337292
Helm    = Solar
Helm d  = +1.139834
DR      = 6.708 HLR
DR ratio= 819.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```







SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

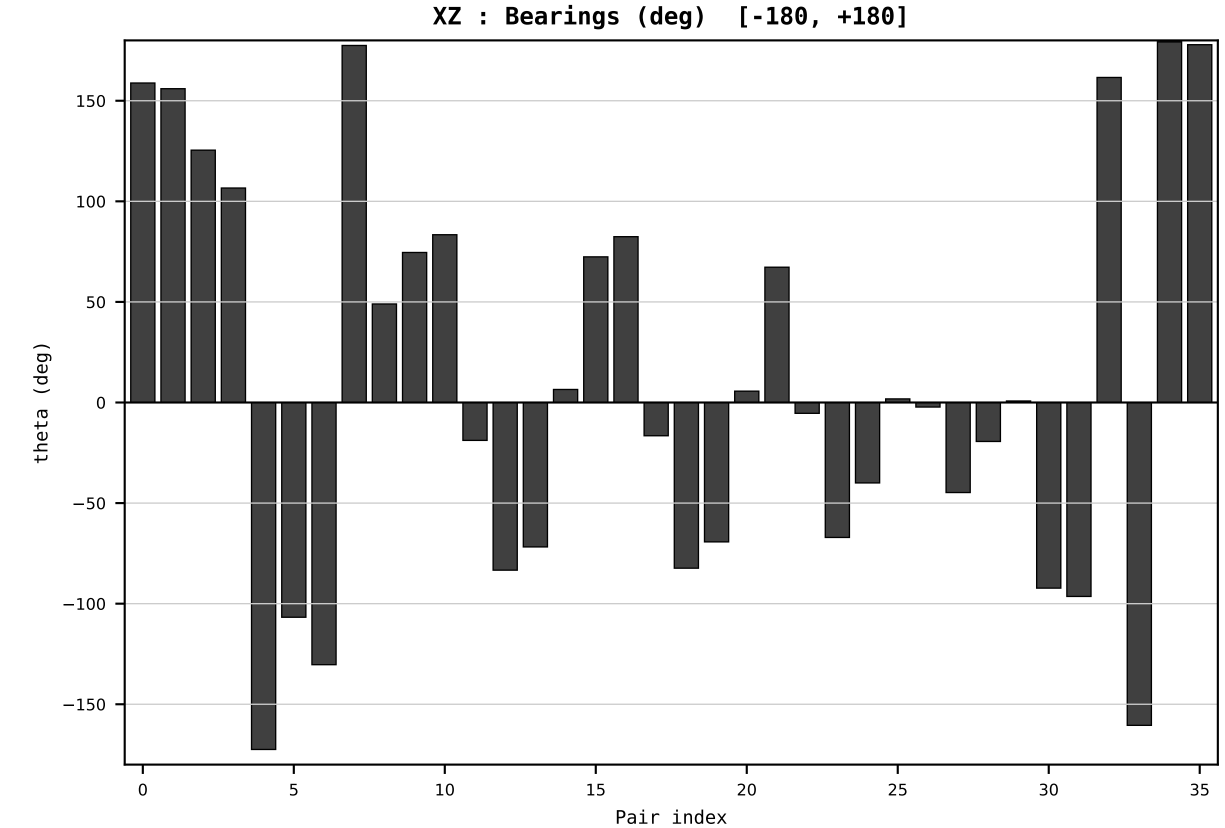
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

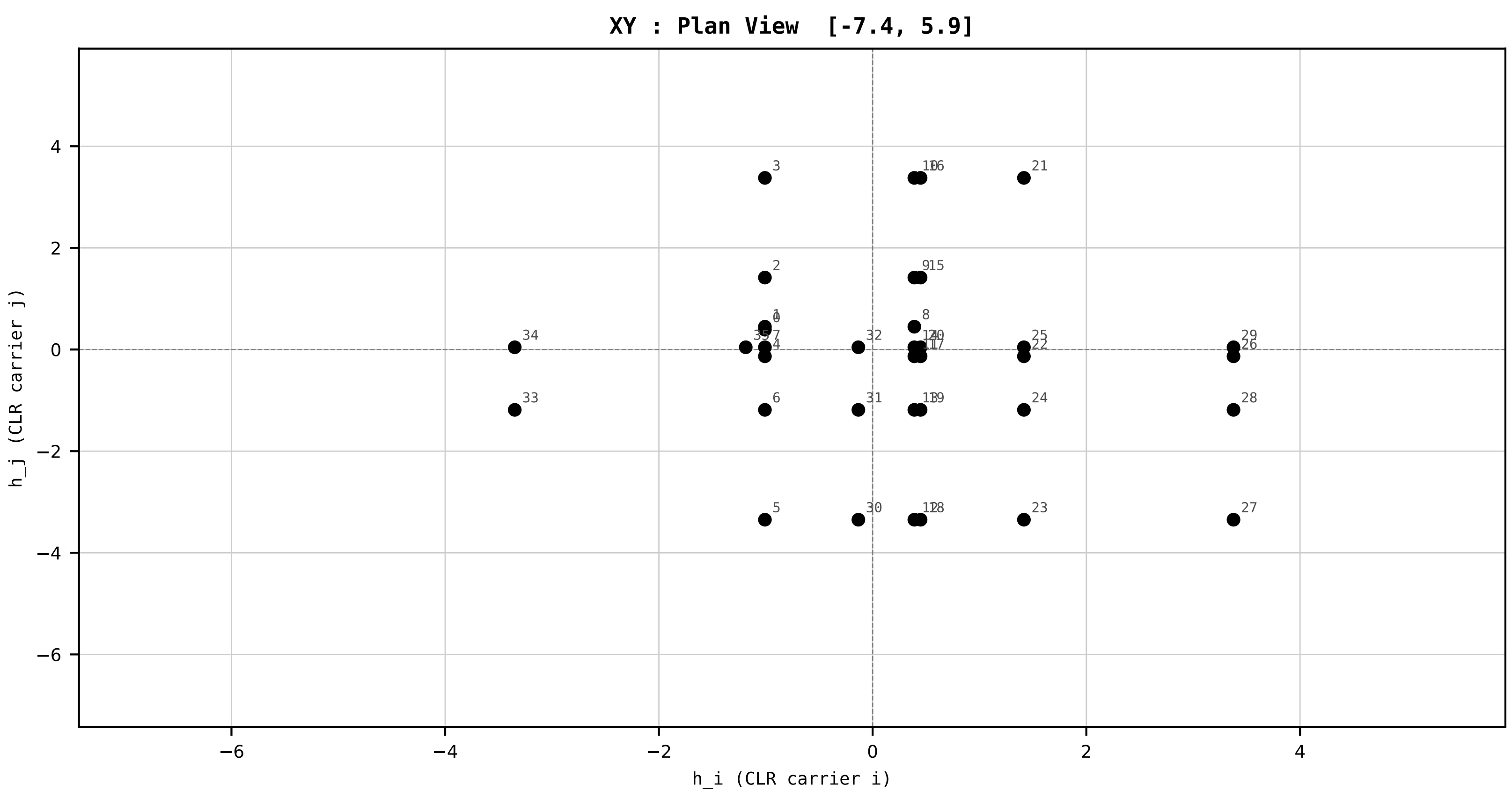
Hs = 0.555897
Ring = Hs-4
E_metric= 27.4266
kappa_HS= 834.14
omega = 7.7165 deg
d_A = 0.724138
Helm = Solar
Helm d = +0.526031
DR = 6.726 HLR
DR ratio= 834.1x

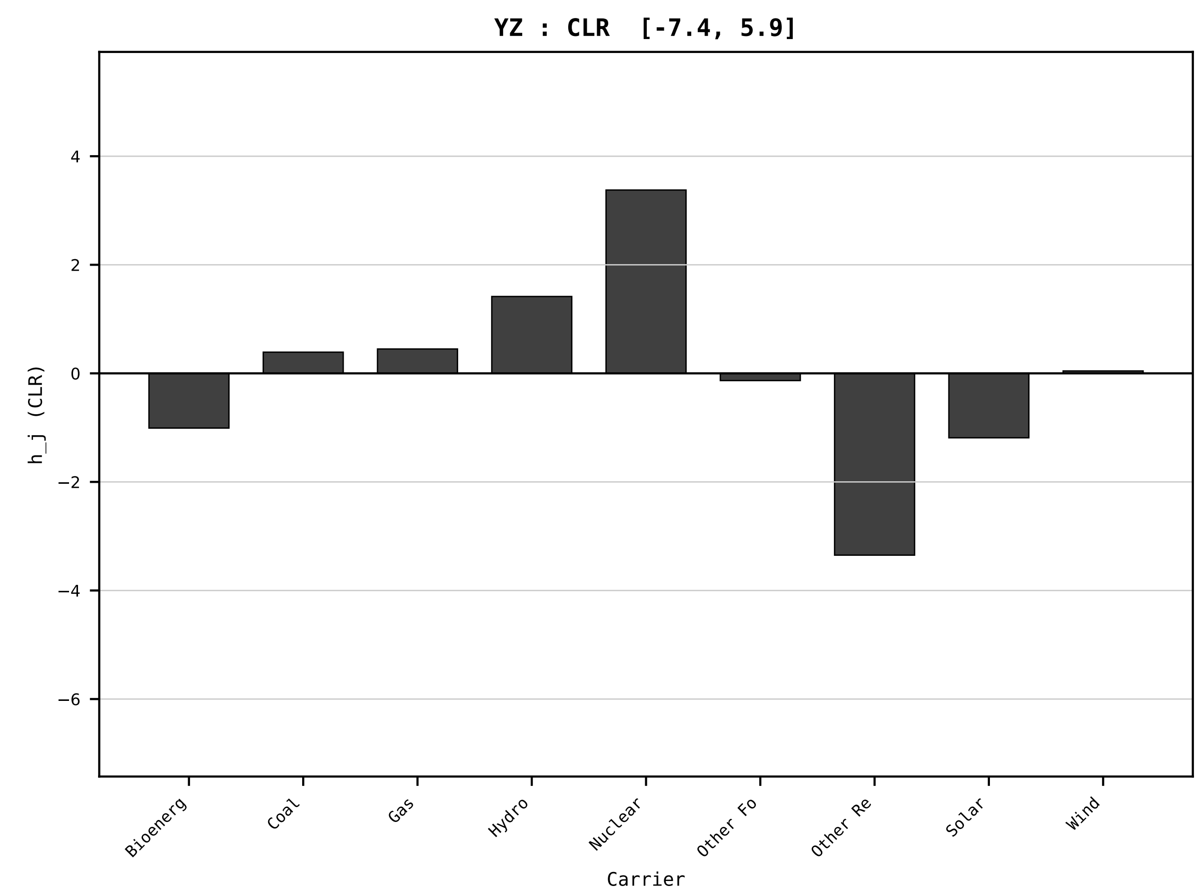
XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



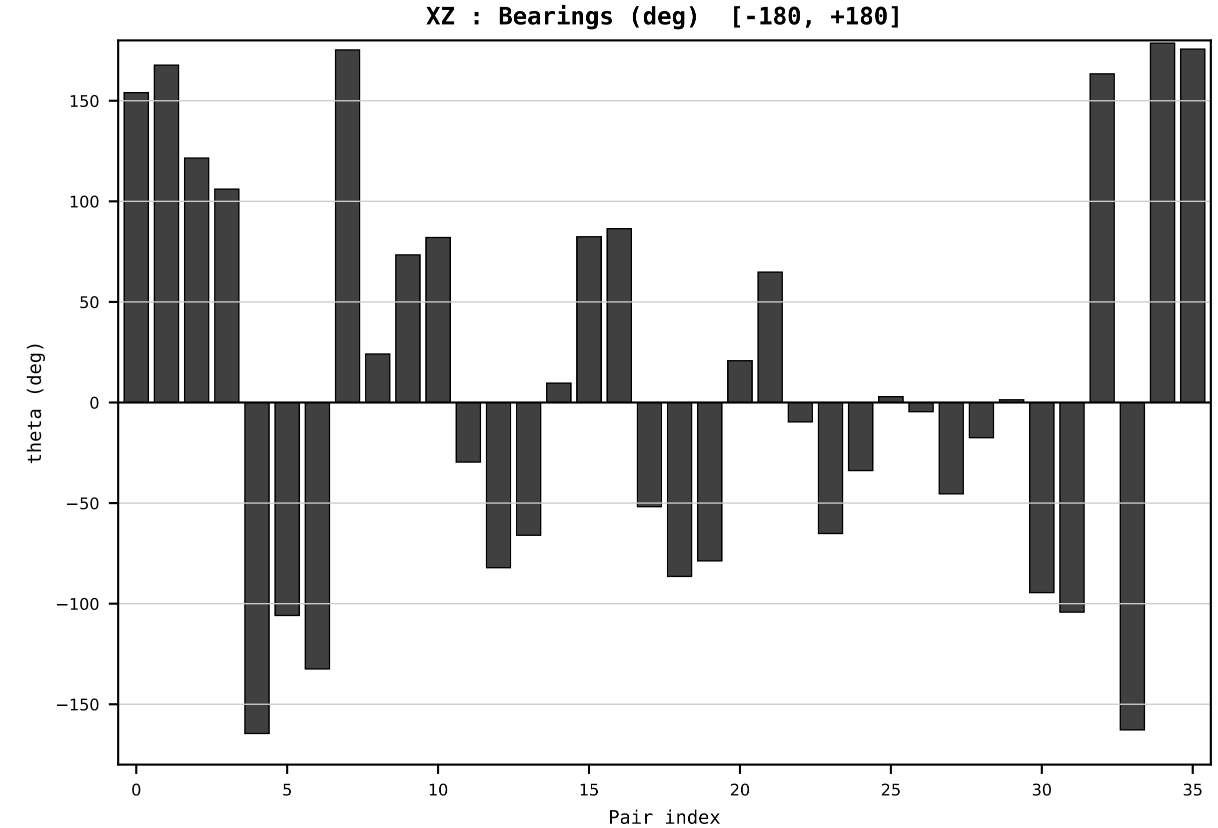


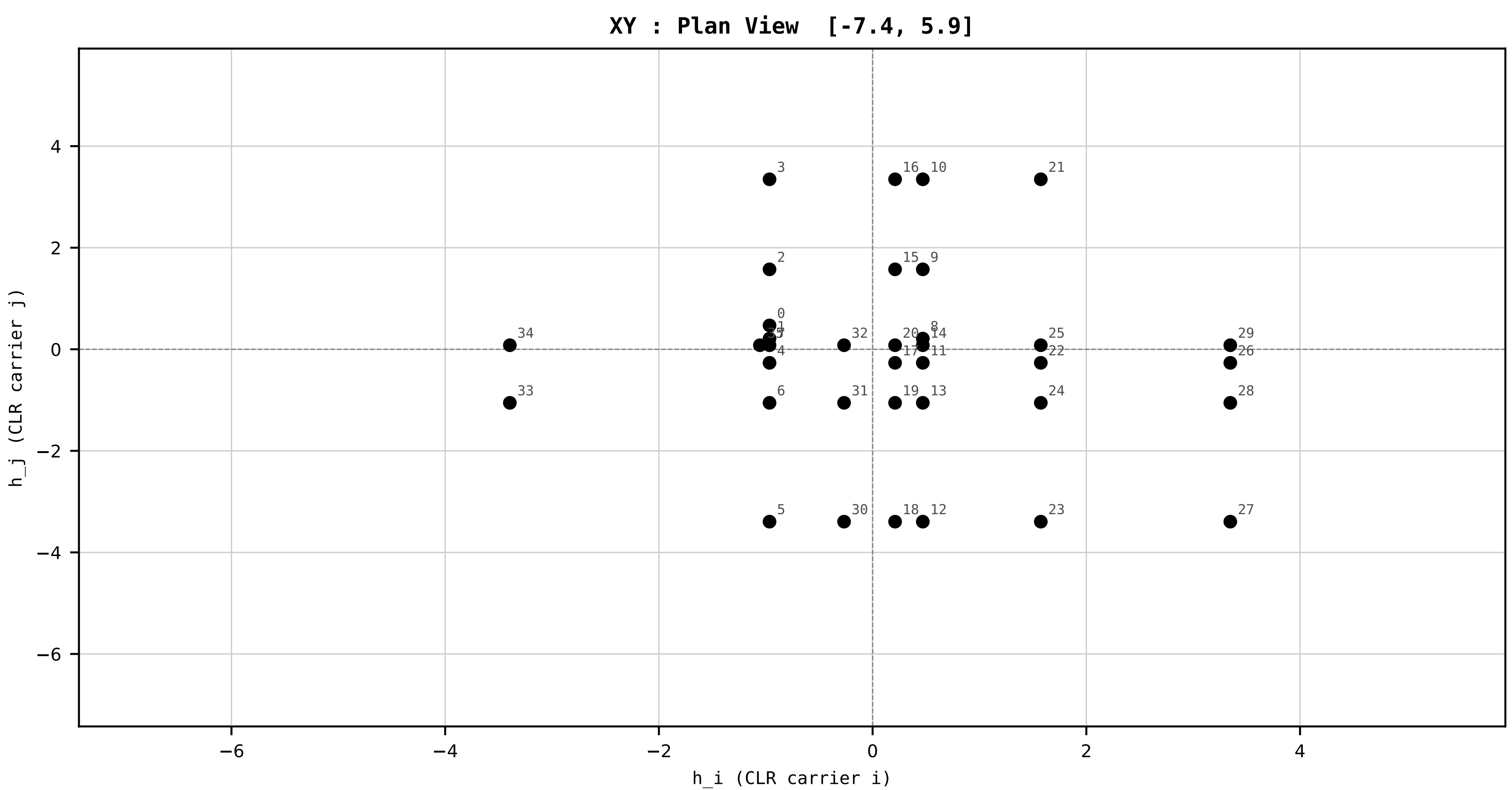


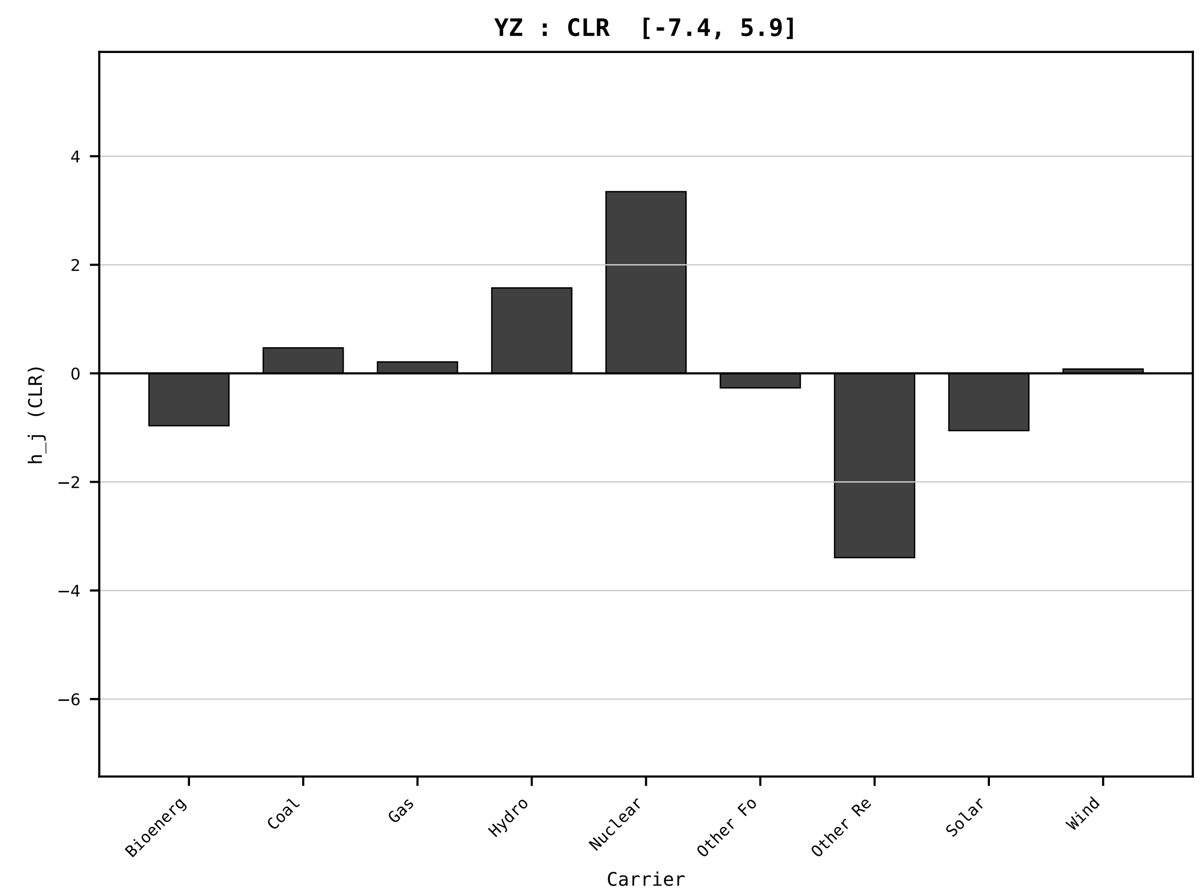
SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2013
t       : 13 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.545141
Ring    = Hs-4
E_metric= 27.5924
kappa_HS= 847.36
omega   = 3.9329 deg
d_A     = 0.360297
Helm    = Gas
Helm d  = -0.238712
DR      = 6.742 HLR
DR ratio= 847.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal        18=Gas-Other Renewables
 1=Bioenergy-Gas         19=Gas-Solar
 2=Bioenergy-Hydro       20=Gas-Wind
 3=Bioenergy-Nuclear     21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar       24=Hydro-Solar
 7=Bioenergy-Wind        25=Hydro-Wind
 8=Coal-Gas              26=Nuclear-Other Fossil
 9=Coal-Hydro            27=Nuclear-Other Renewables
10=Coal-Nuclear          28=Nuclear-Solar
11=Coal-Other Fossil     29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar            31=Other Fossil-Solar
14=Coal-Wind             32=Other Fossil-Wind
15=Gas-Hydro             33=Other Renewables-Solar
16=Gas-Nuclear           34=Other Renewables-Wind
17=Gas-Other Fossil      35=Solar-Wind
```



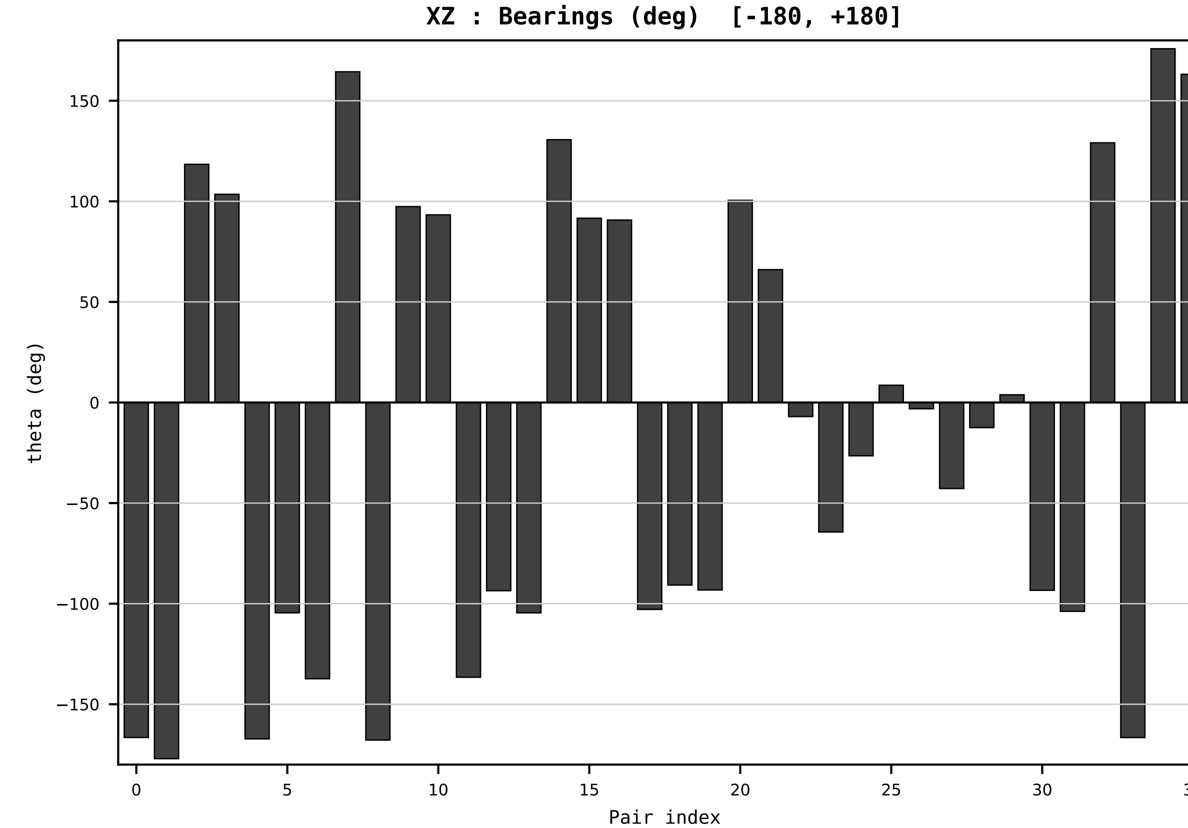


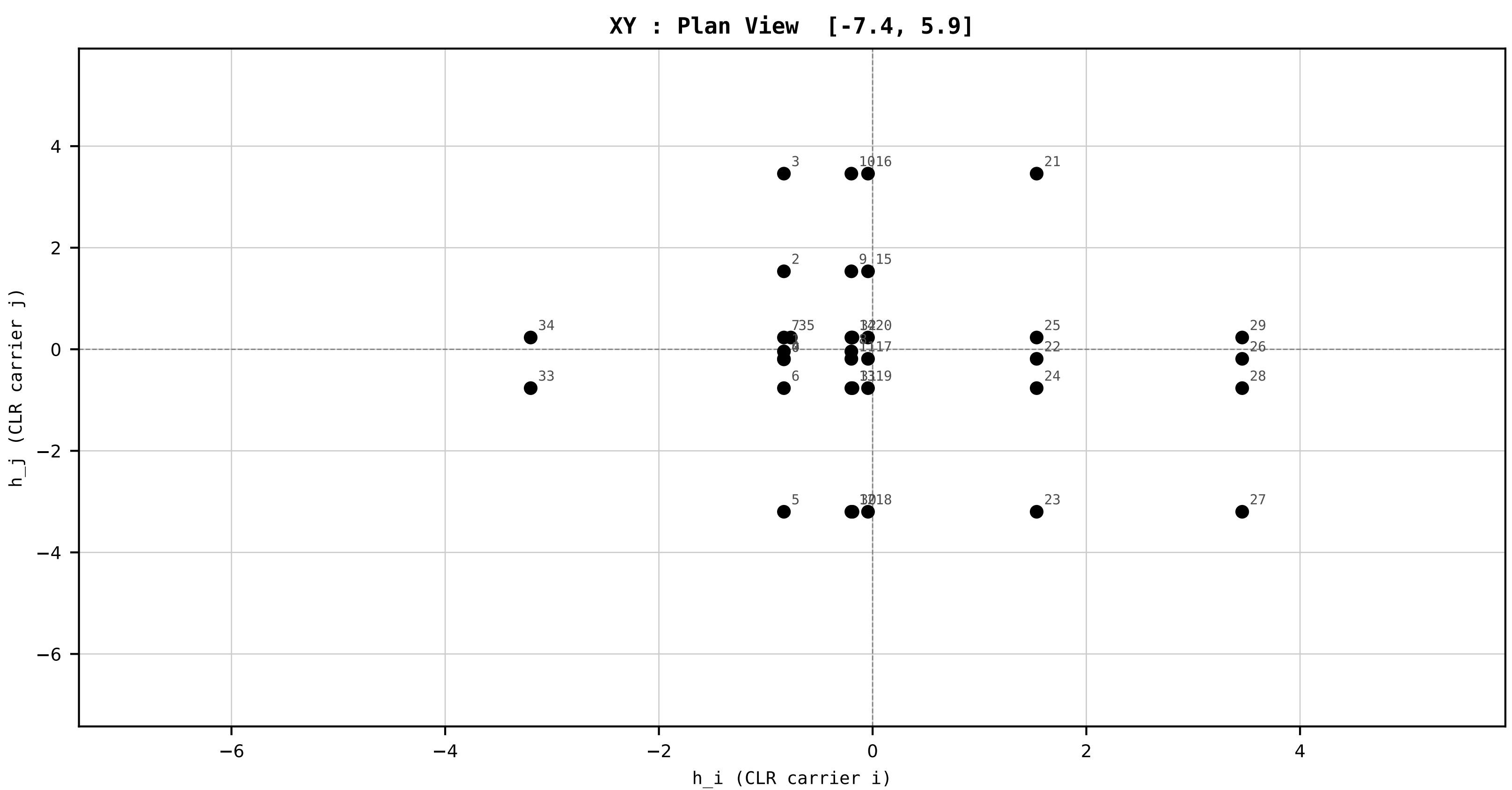


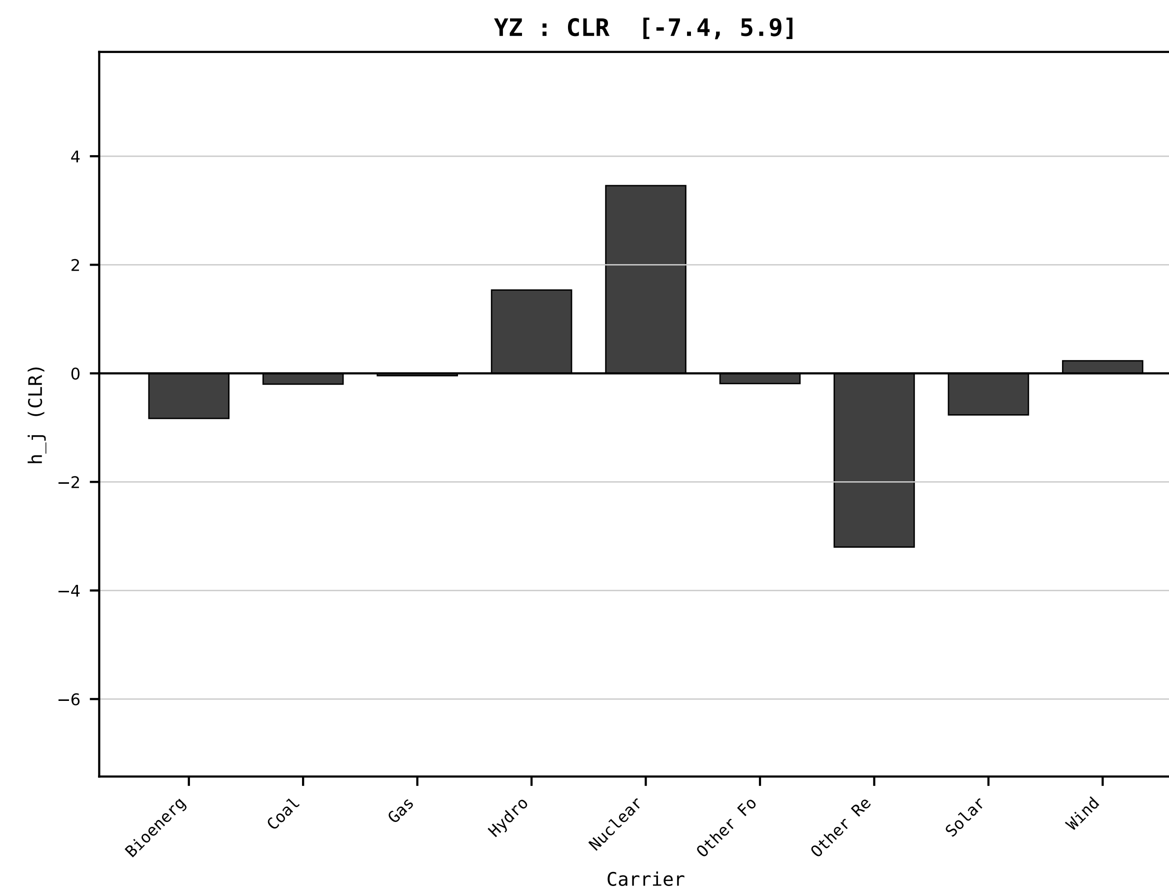
SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2014
t       : 14 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.589178
Ring    = Hs-4
E_metric= 25.9637
kappa_HS= 779.43
omega   = 9.0709 deg
d_A     = 0.833206
Helm    = Coal
Helm d  = -0.668681
DR      = 6.659 HLR
DR ratio= 779.4x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



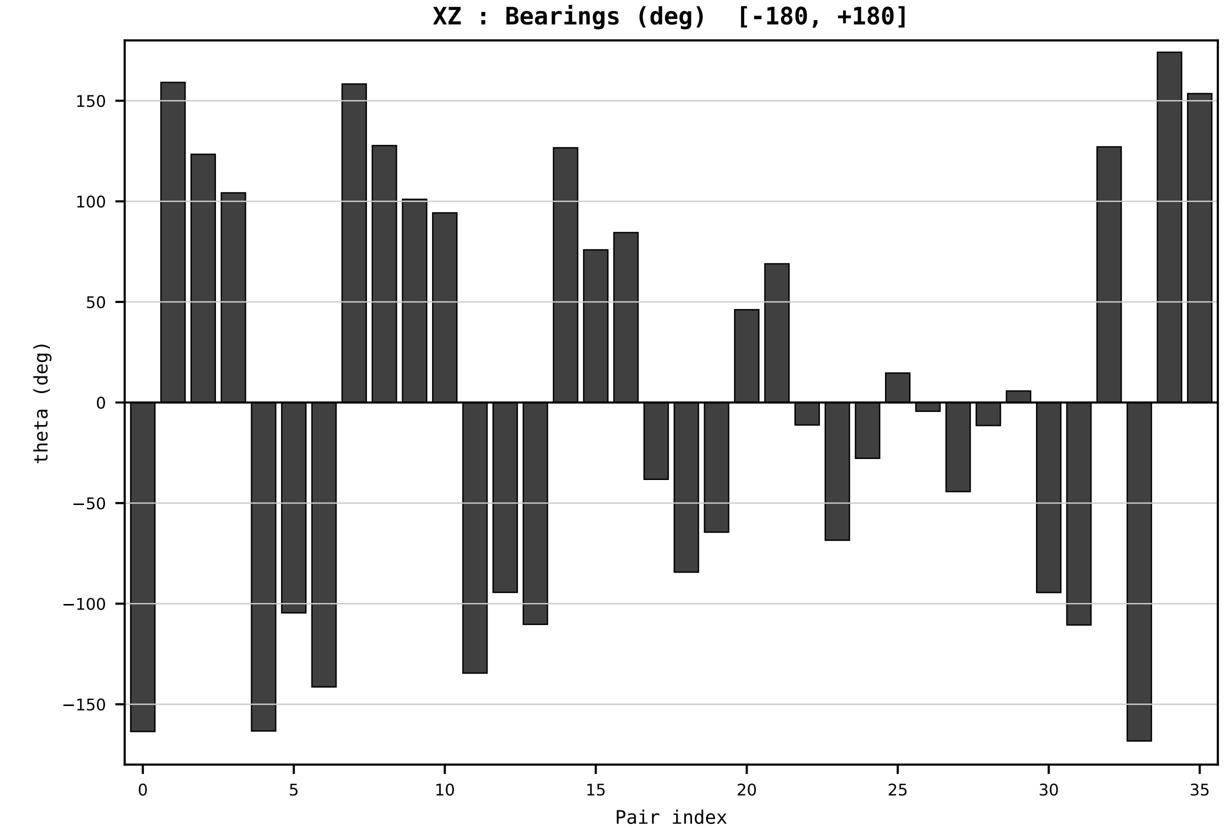


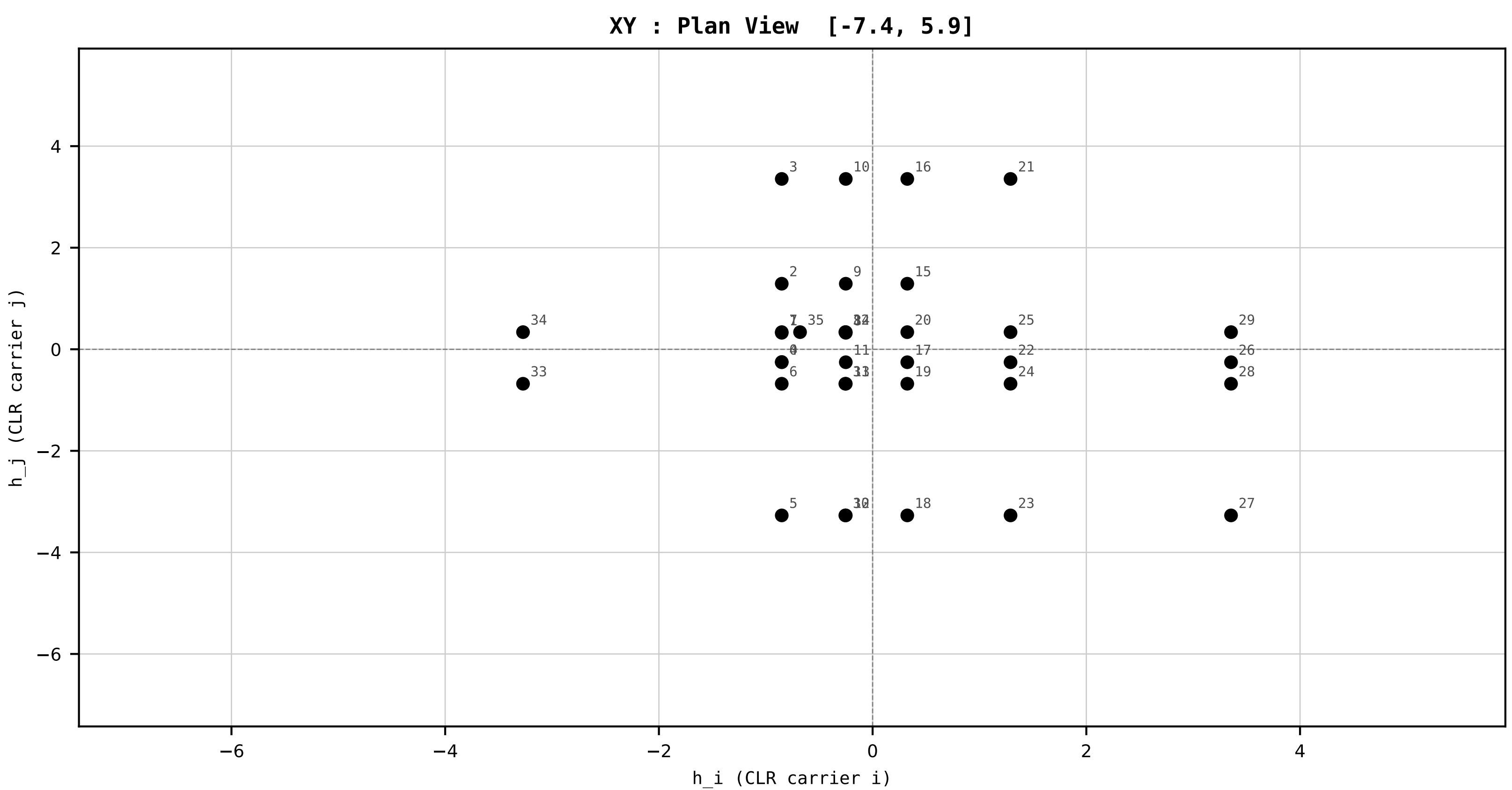


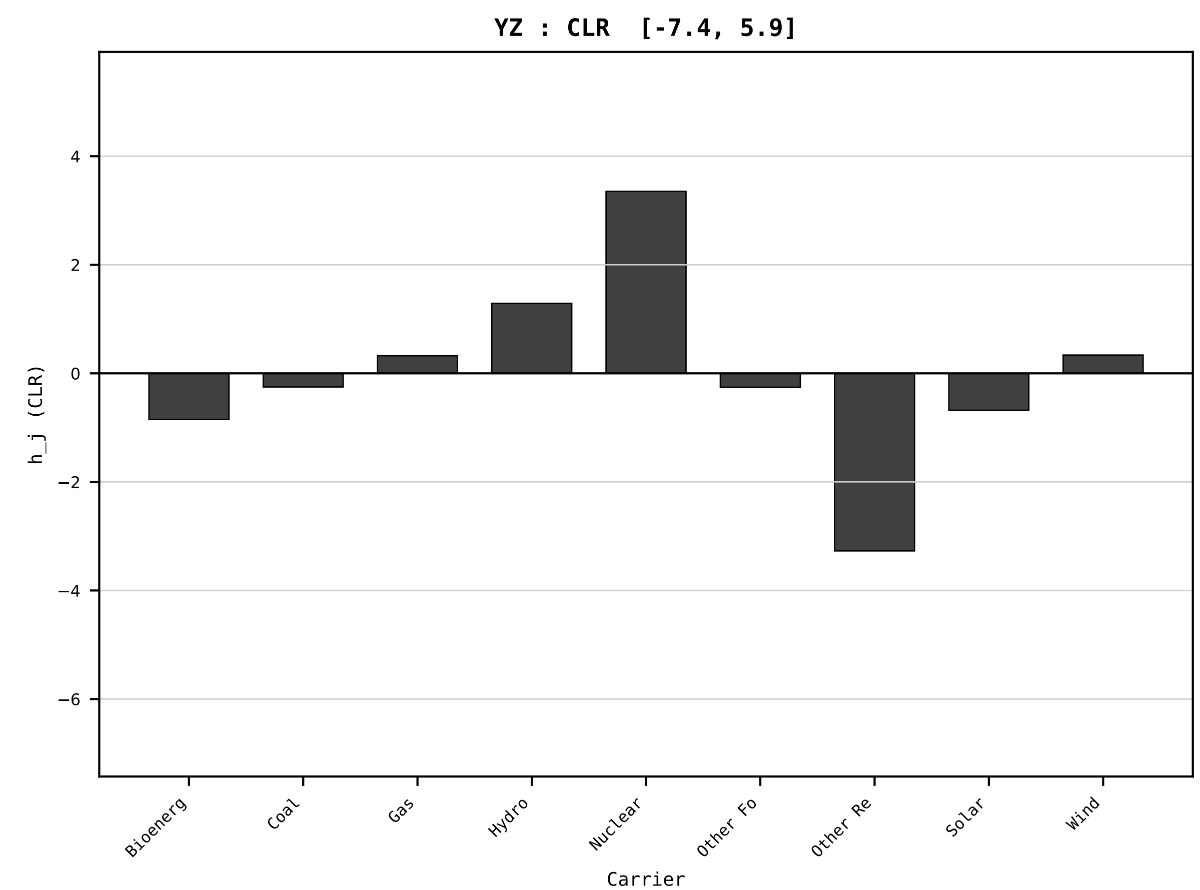
SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2015
t       : 15 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.565912
Ring    = Hs-4
E_metric= 25.1505
kappa_HS= 754.19
omega   = 5.4442 deg
d_A     = 0.486840
Helm    = Gas
Helm d  = +0.367403
DR      = 6.626 HLR
DR ratio= 754.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



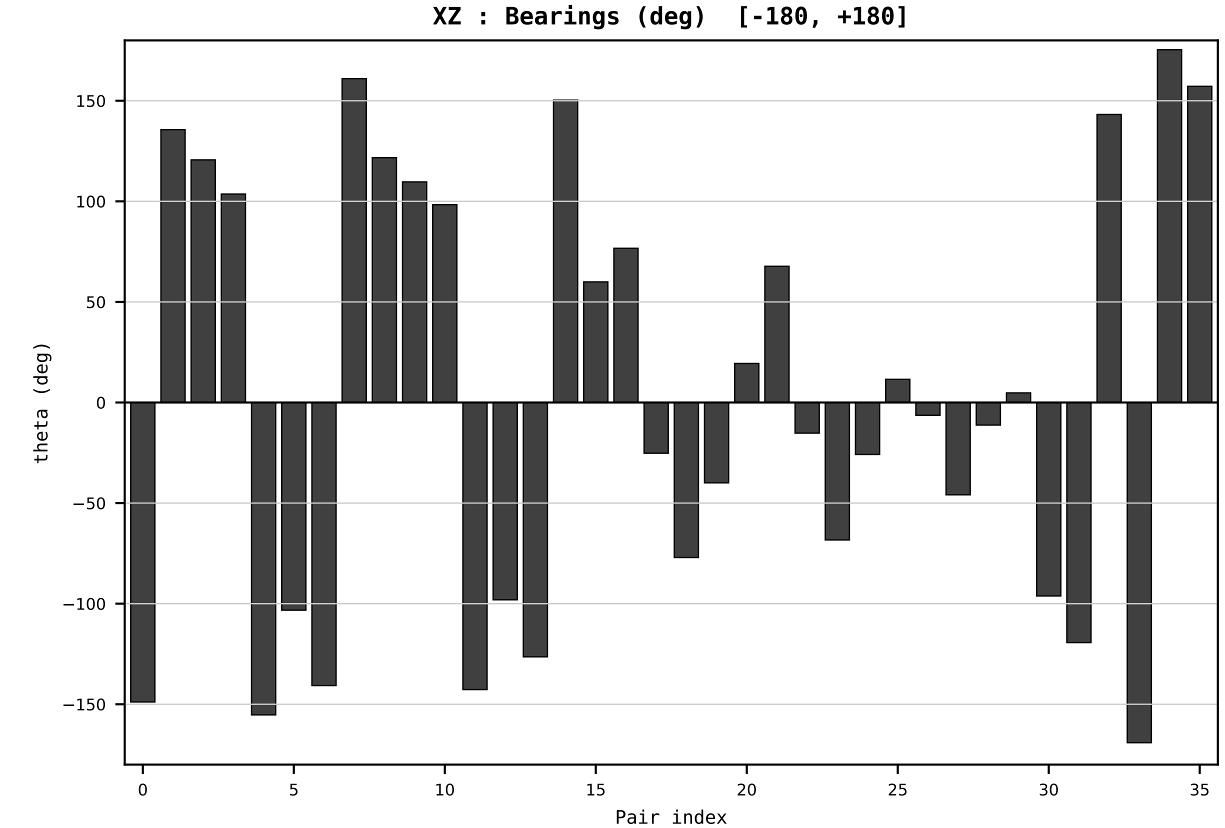


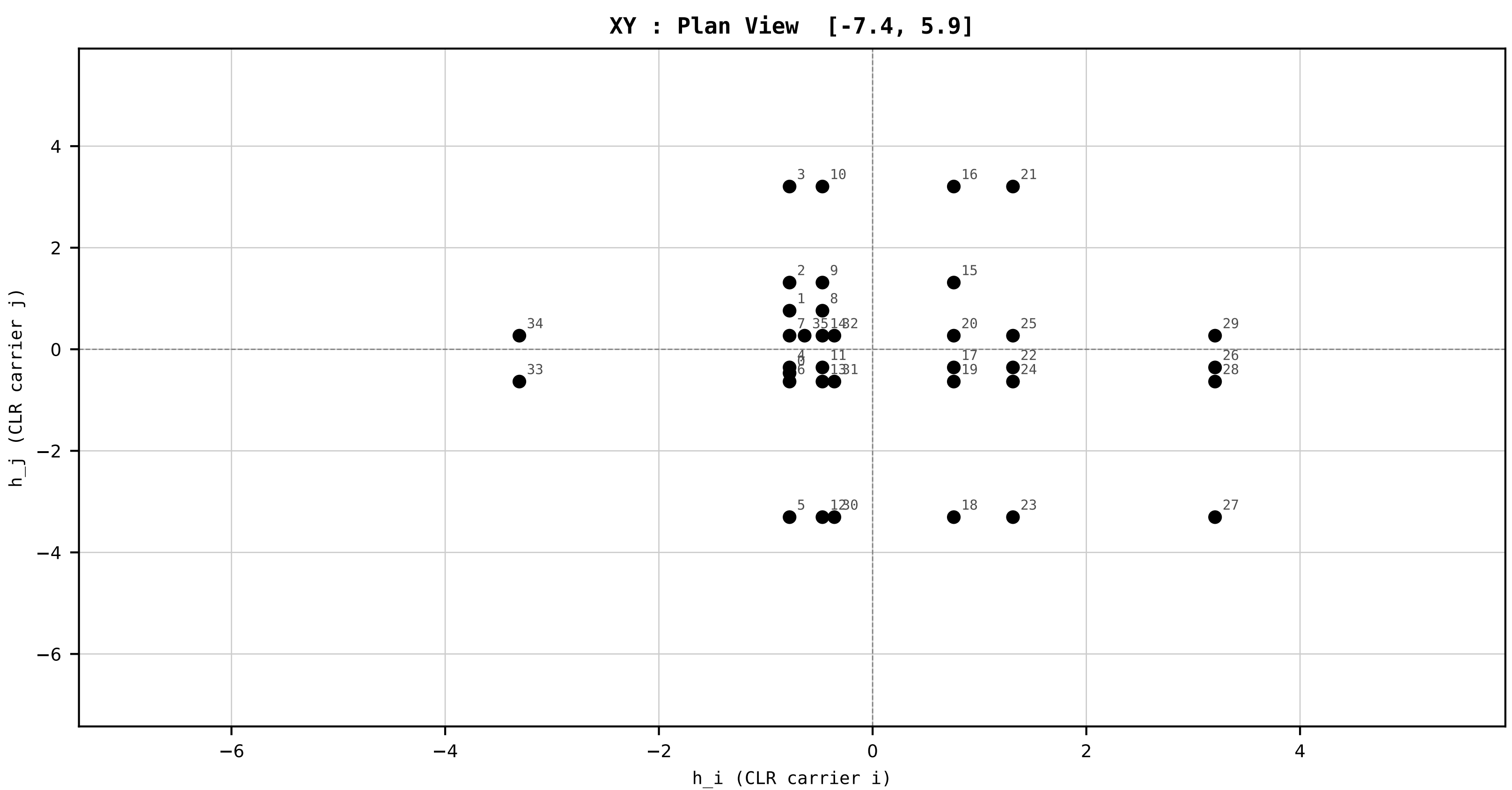


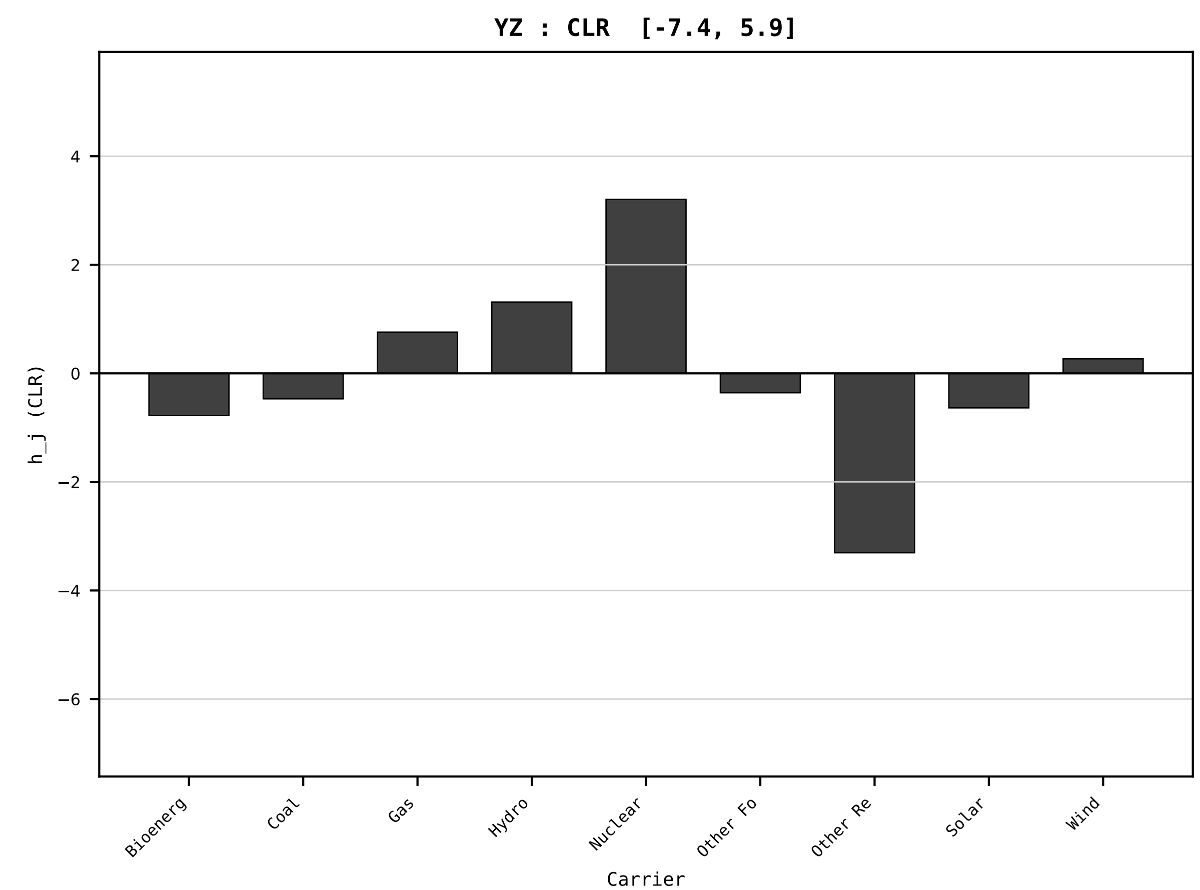
SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2016
t       : 16 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.518507
Ring    = Hs-4
E_metric= 24.9278
kappa_HS= 672.00
omega   = 6.0973 deg
d_A     = 0.532714
Helm    = Gas
Helm d  = +0.435313
DR      = 6.510 HLR
DR ratio= 672.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



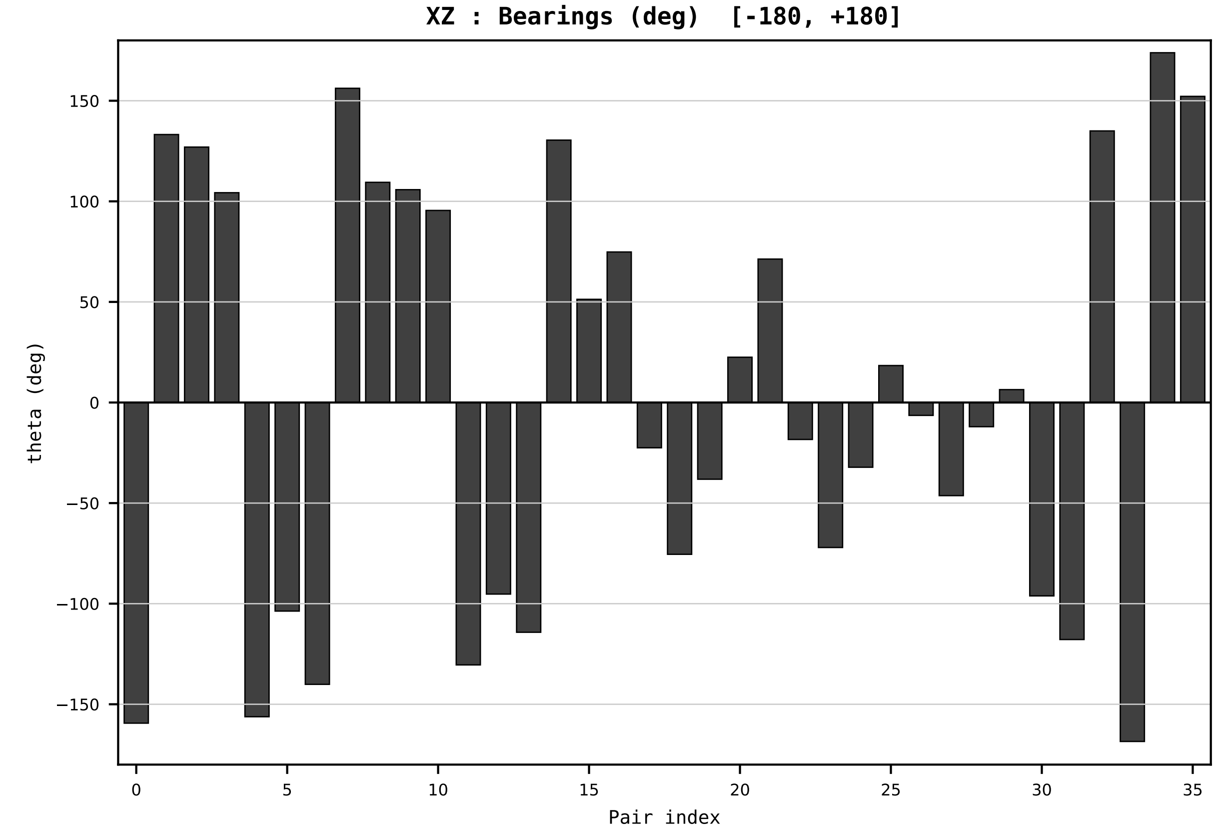


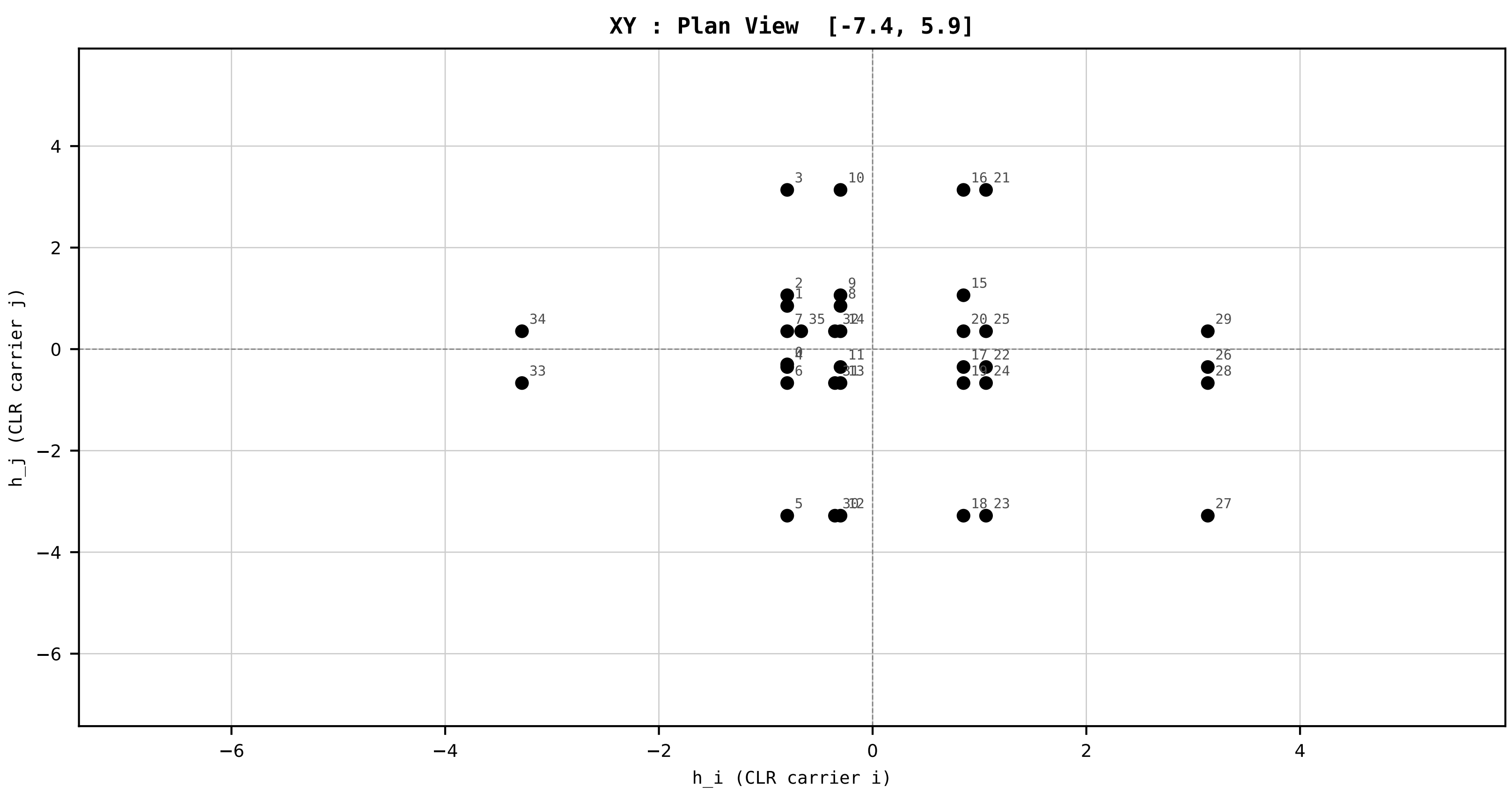


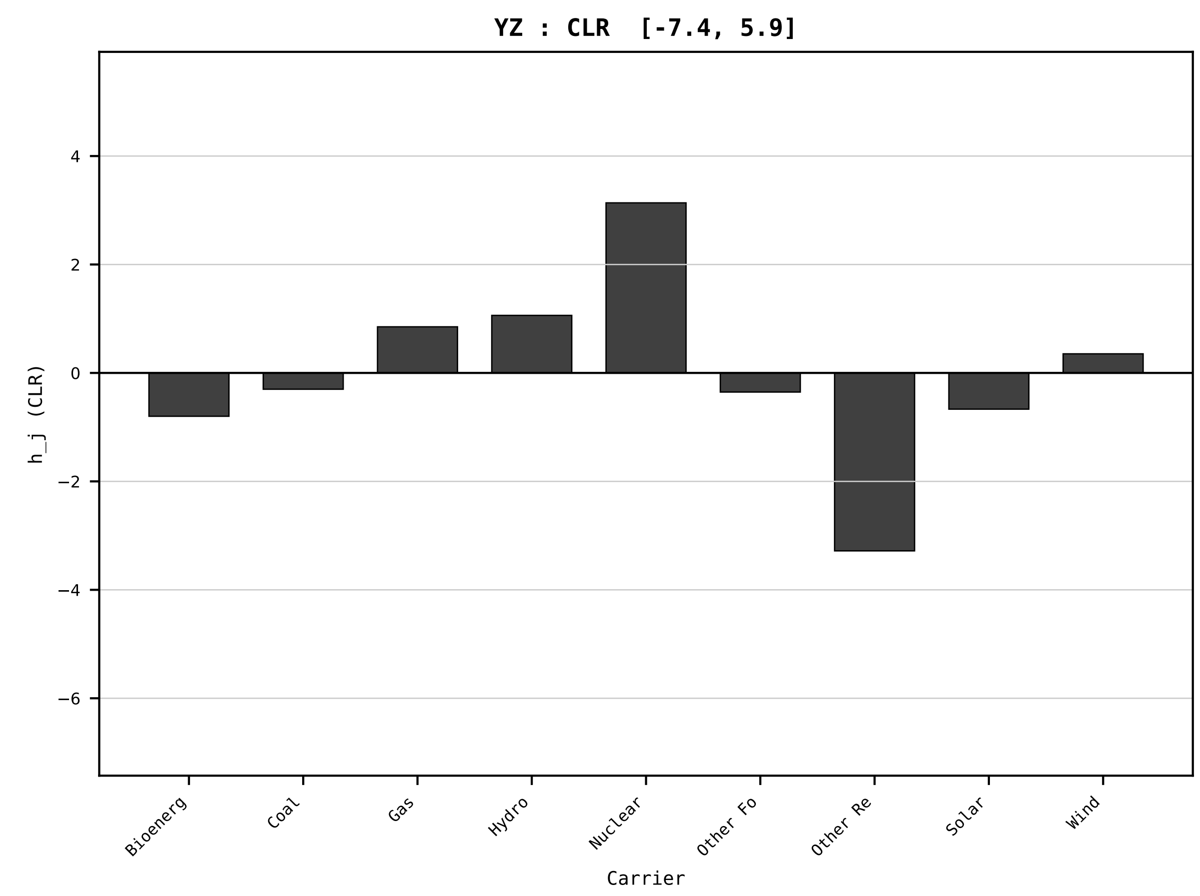
SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2017
t       : 17 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.504621
Ring    = Hs-4
E_metric= 23.8804
kappa_HS= 612.86
omega   = 3.7247 deg
d_A     = 0.338103
Helm    = Hydro
Helm d  = -0.252129
DR      = 6.418 HLR
DR ratio= 612.9x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



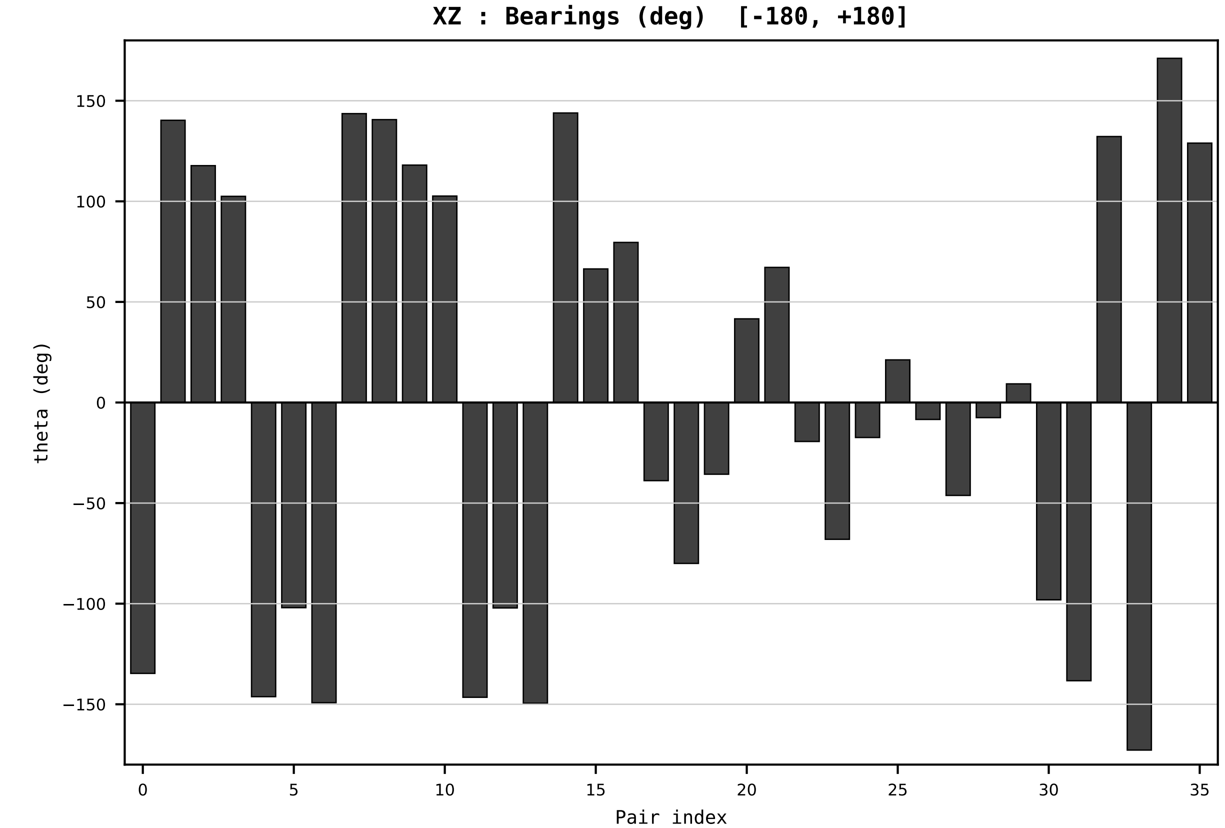


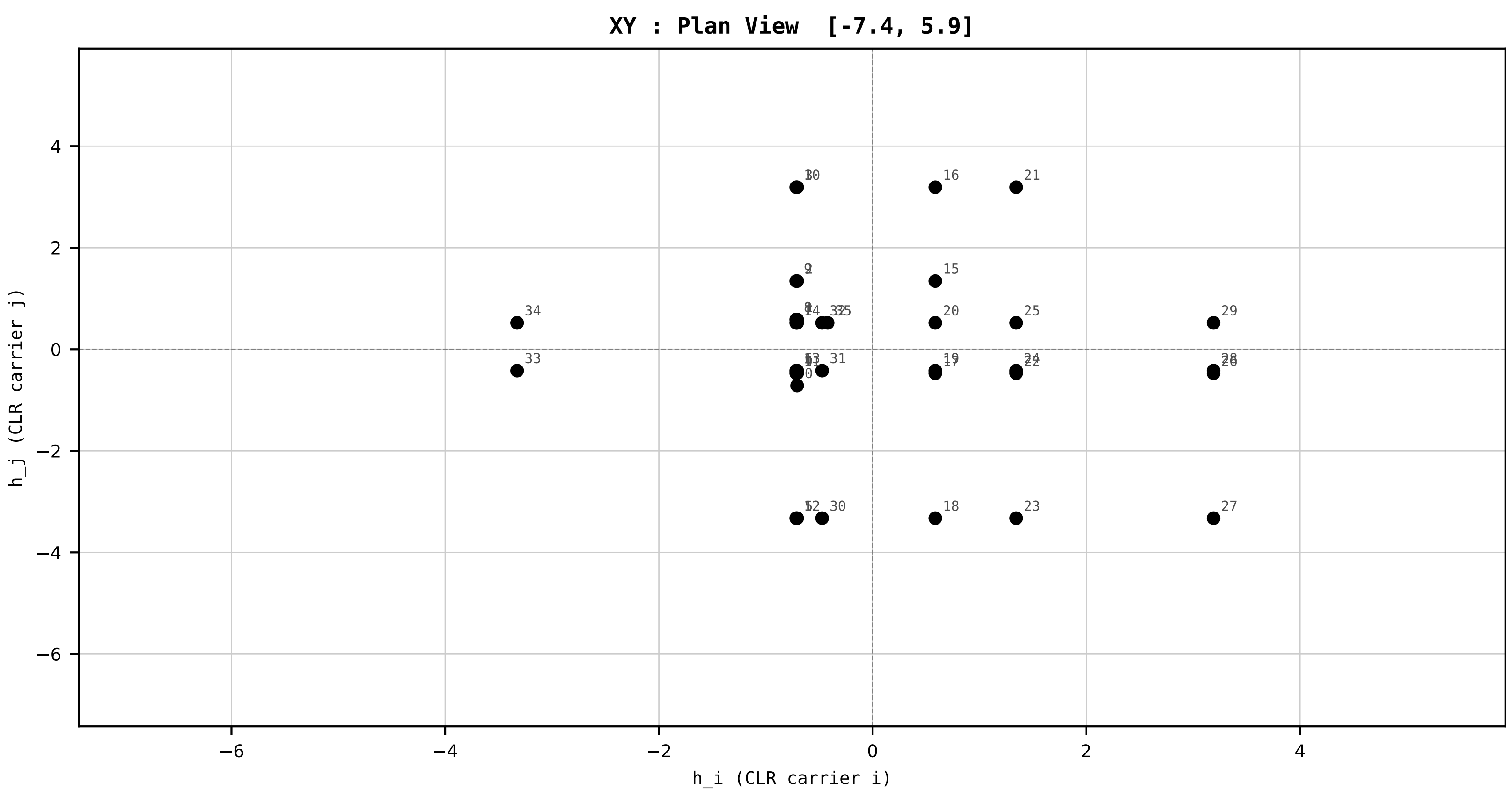


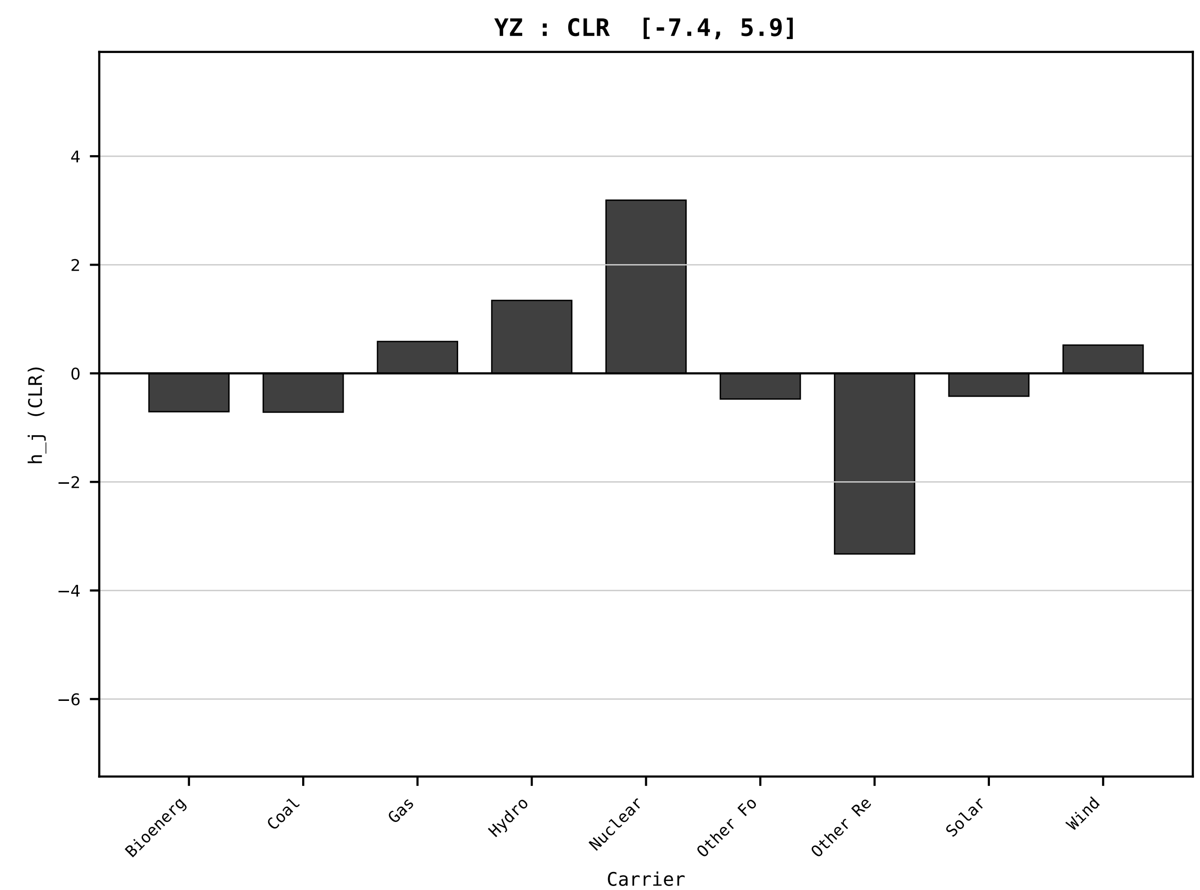
SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2018
t       : 18 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.513194
Ring    = Hs-4
E_metric= 25.0810
kappa_HS= 676.95
omega   = 7.5442 deg
d_A     = 0.662124
Helm    = Coal
Helm d  = -0.414639
DR      = 6.518 HLR
DR ratio= 677.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



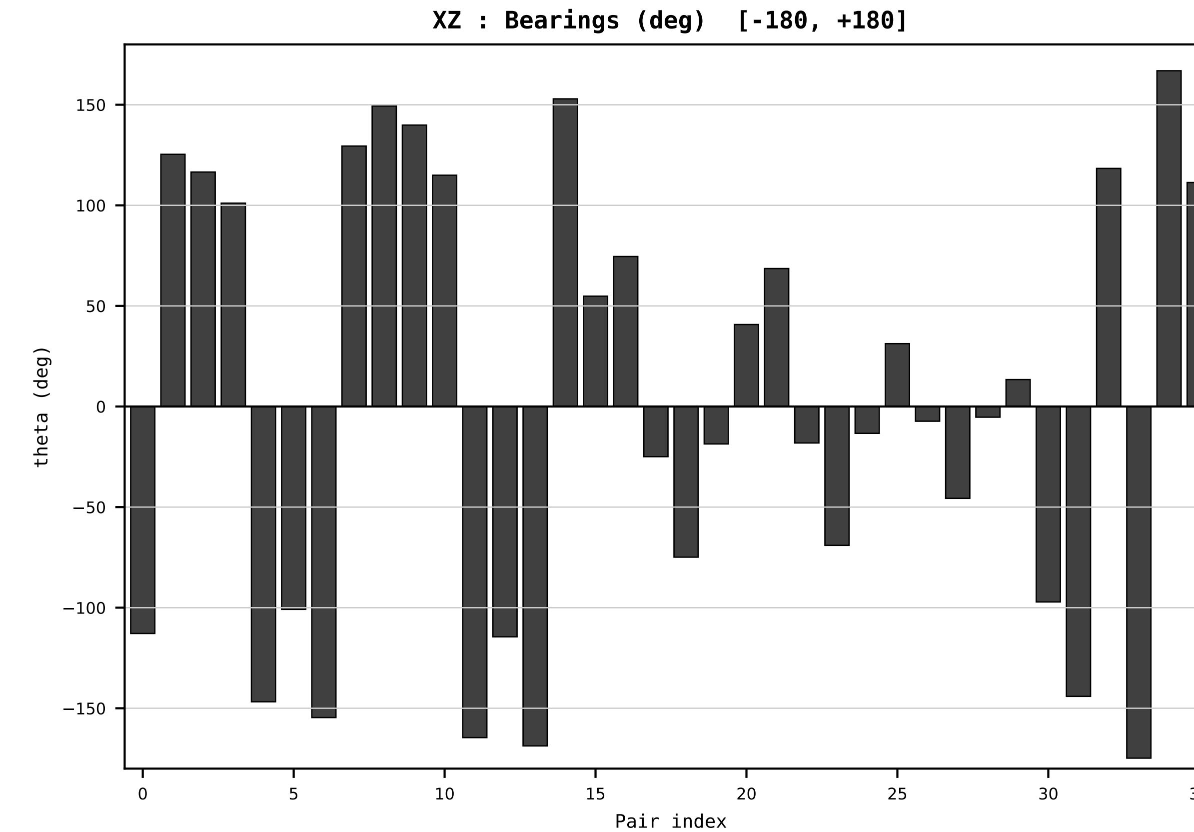


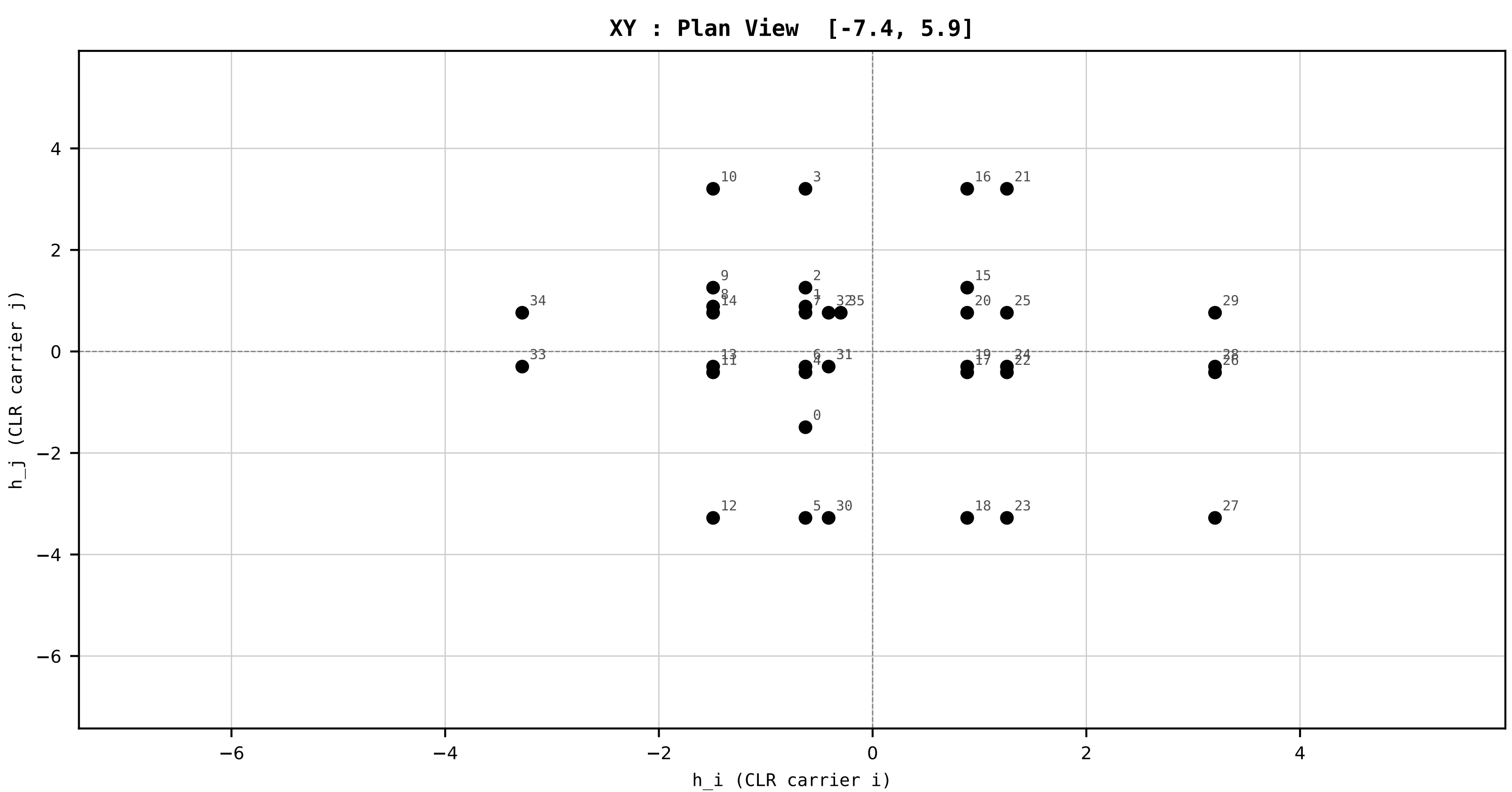


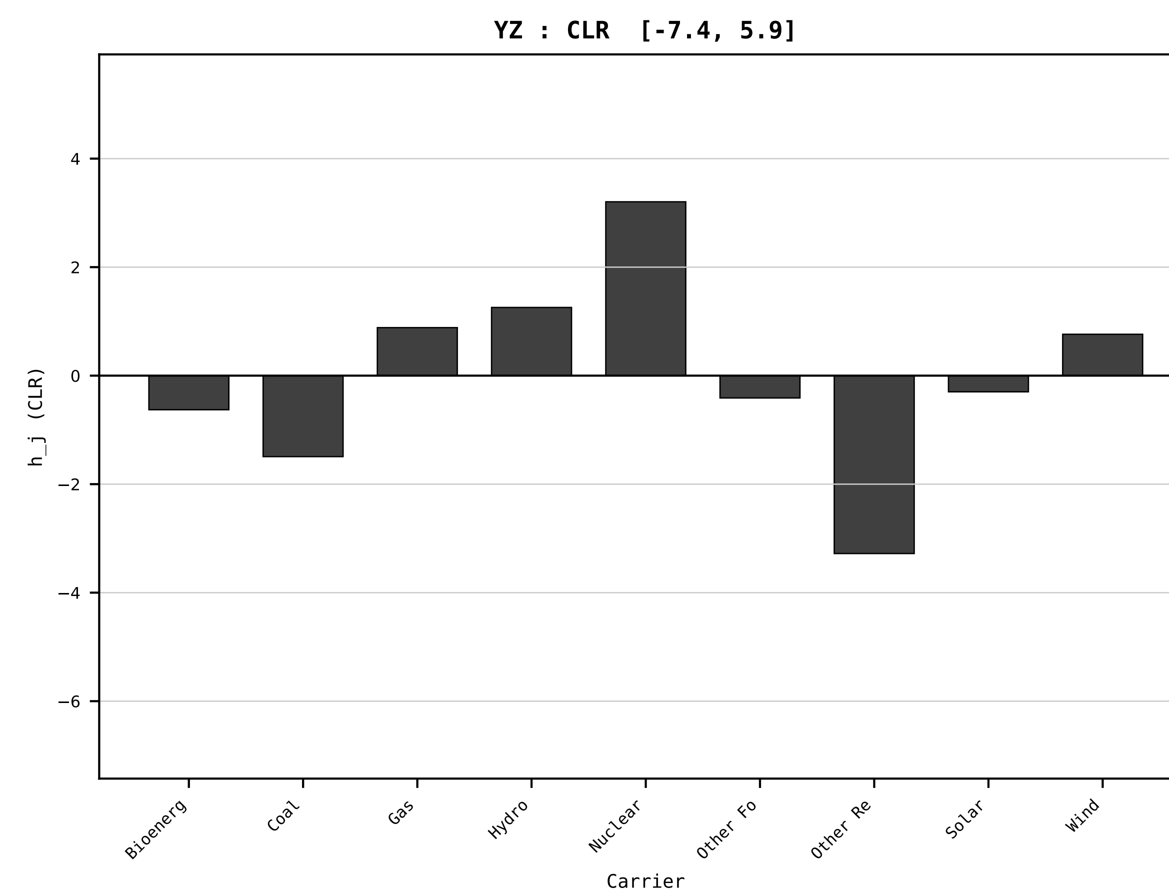
SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2019
t       : 19 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.501913
Ring    = Hs-4
E_metric= 26.8448
kappa_HS= 654.11
omega   = 9.7988 deg
d_A     = 0.887160
Helm    = Coal
Helm d  = -0.777550
DR      = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



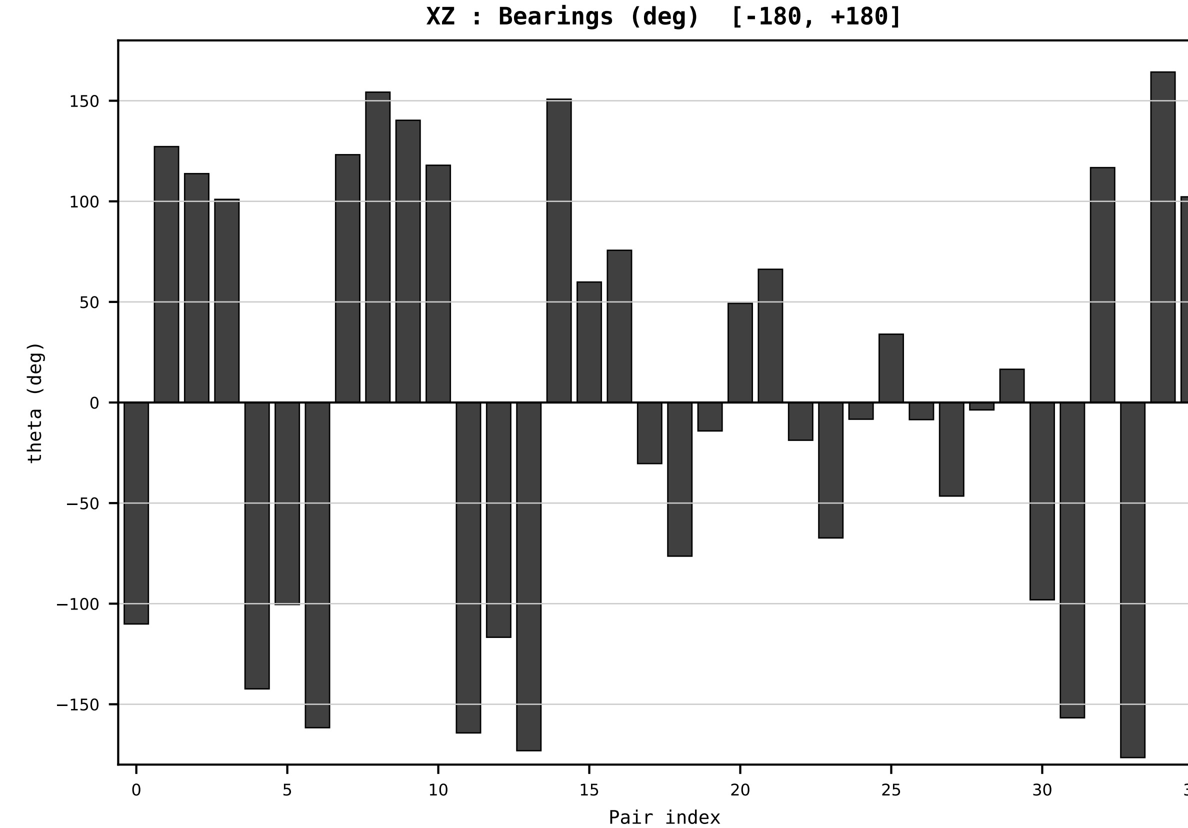


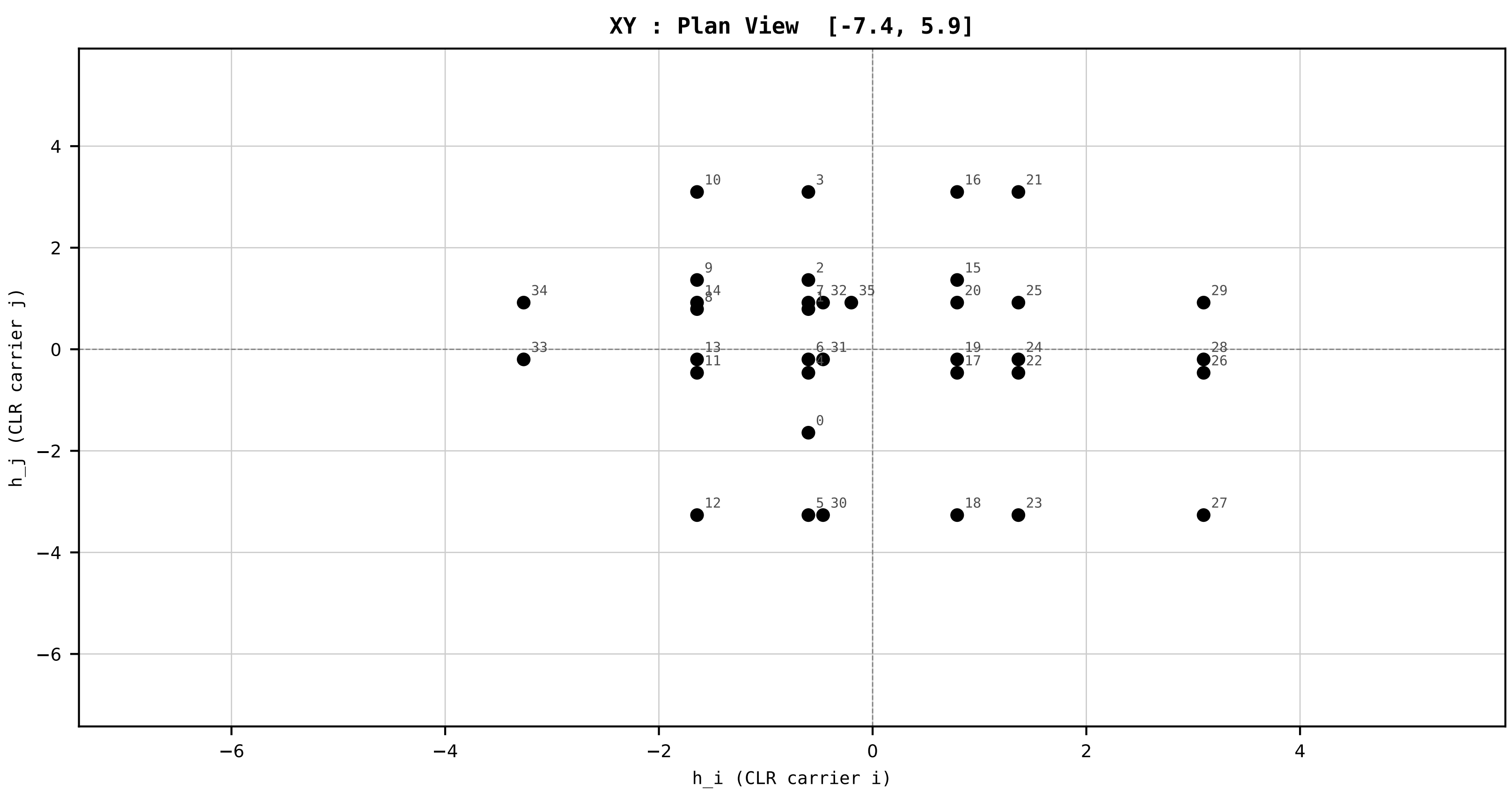


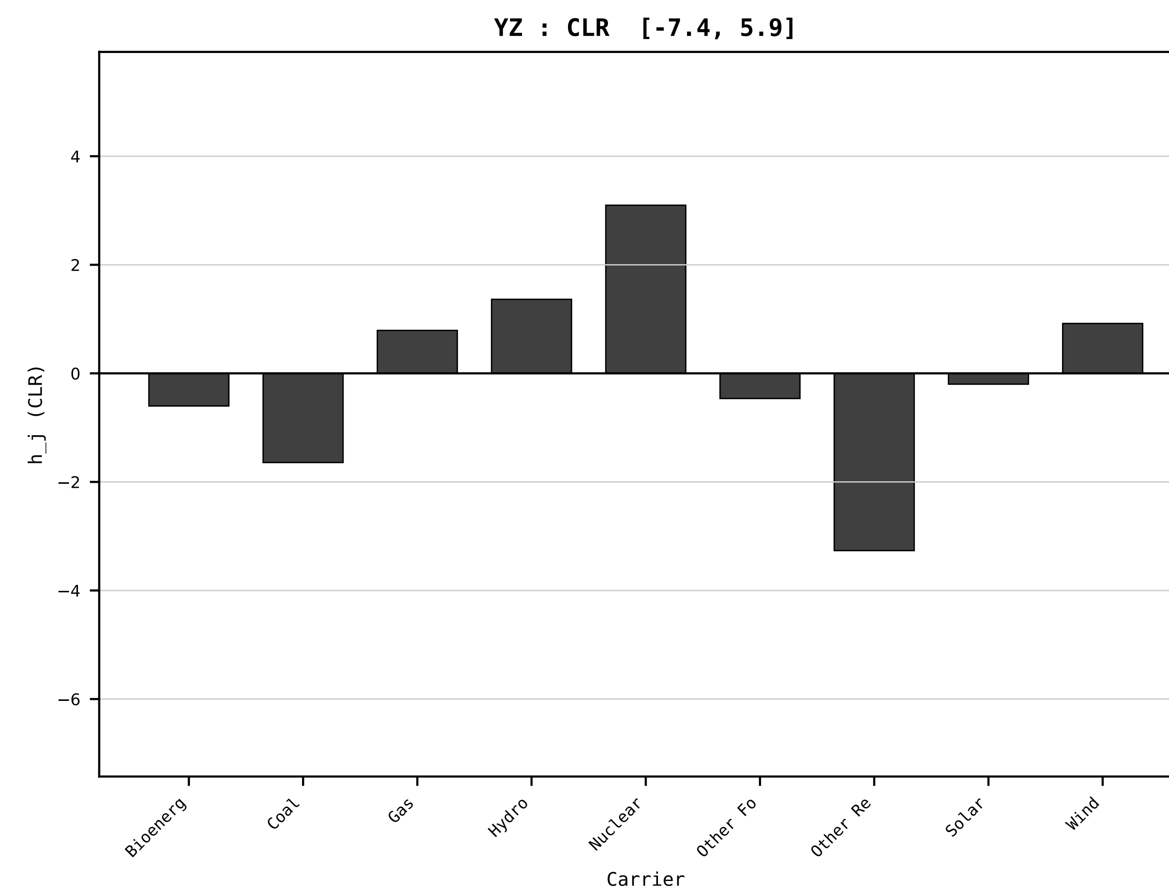
SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2020
t       : 20 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.467323
Ring    = Hs-3
E_metric= 26.9059
kappa_HS= 580.05
omega   = 3.3546 deg
d_A     = 0.303539
Helm    = Wind
Helm d  = +0.156233
DR      = 6.363 HLR
DR ratio= 580.0x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



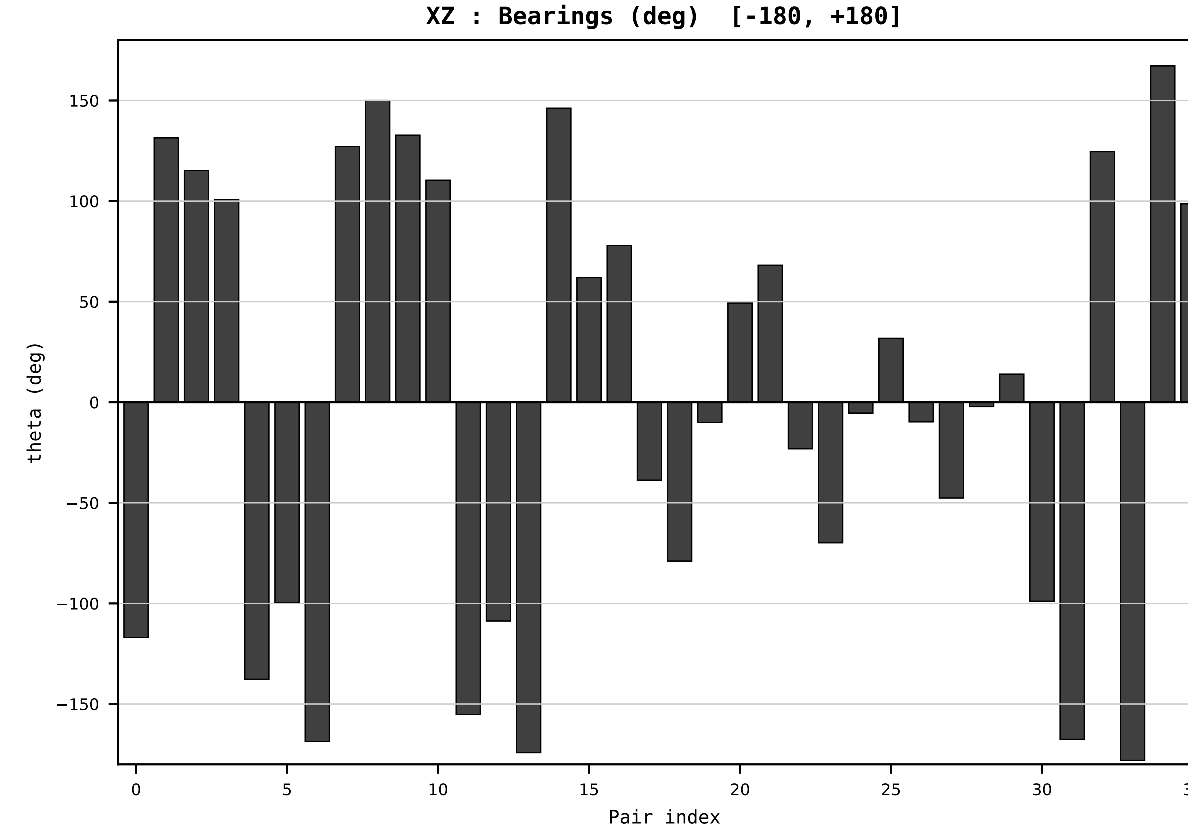


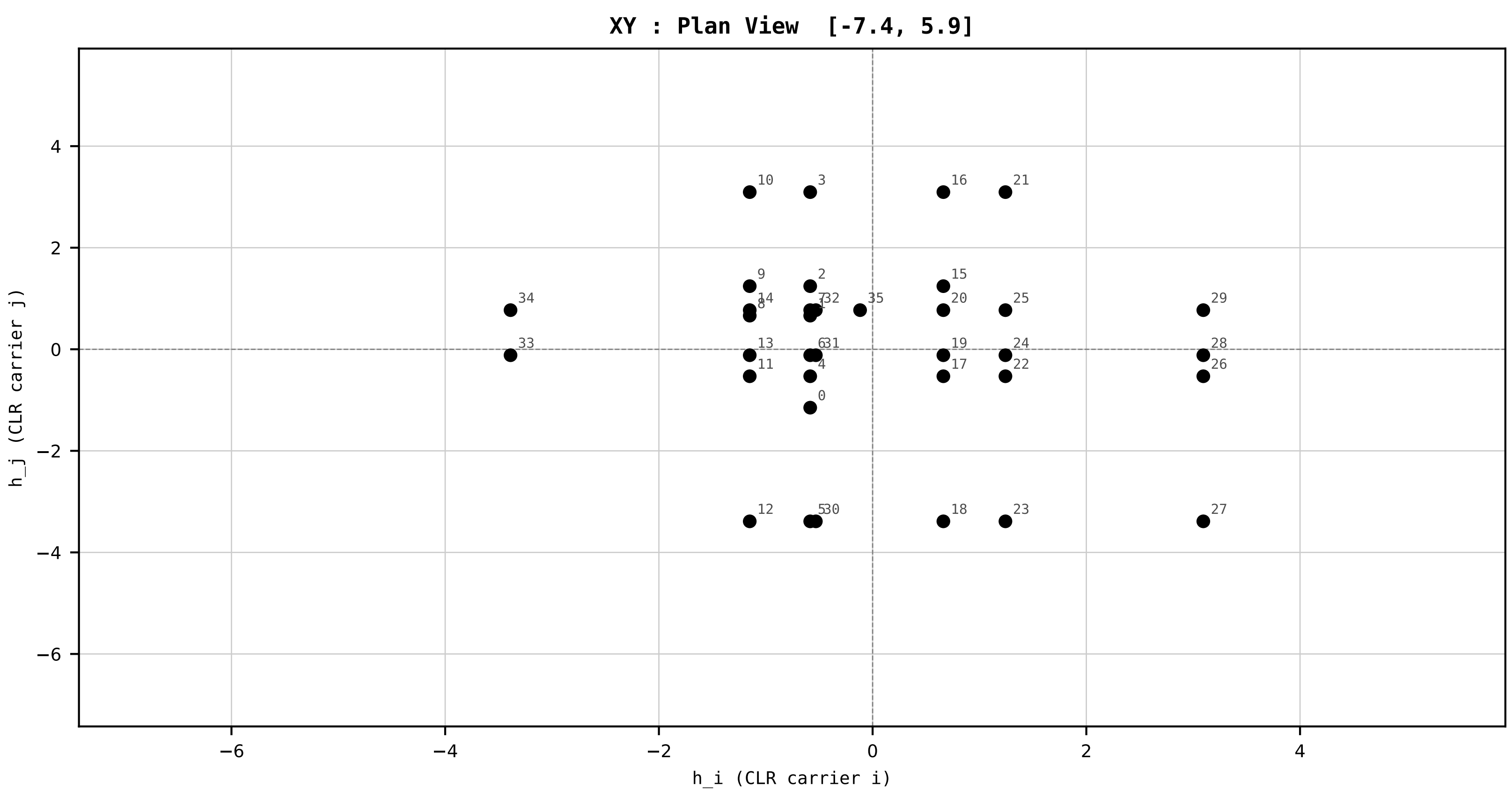


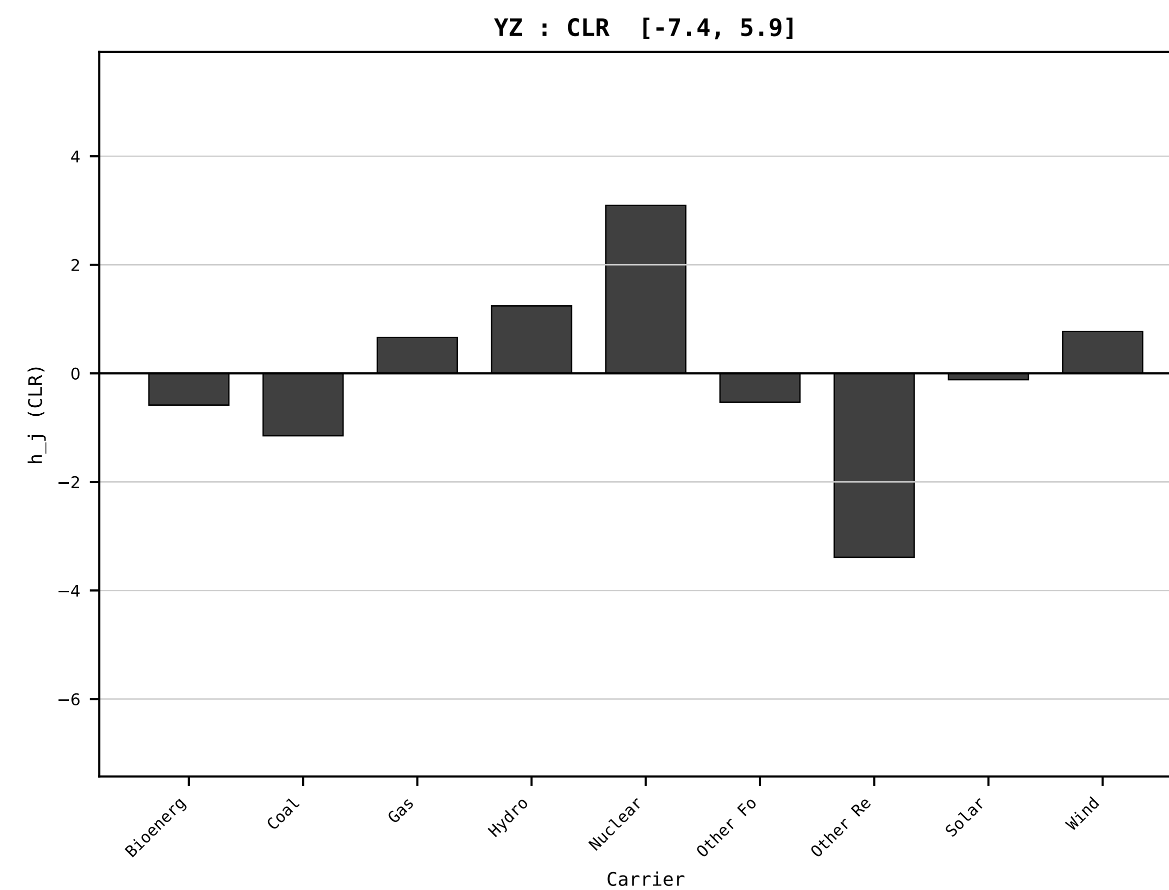
SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2021
t       : 21 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.478912
Ring    = Hs-3
E_metric= 25.5954
kappa_HS= 654.07
omega   = 6.1997 deg
d_A     = 0.568610
Helm    = Coal
Helm d  = +0.493100
DR      = 6.483 HLR
DR ratio= 654.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal        18=Gas-Other Renewables
 1=Bioenergy-Gas         19=Gas-Solar
 2=Bioenergy-Hydro       20=Gas-Wind
 3=Bioenergy-Nuclear     21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar       24=Hydro-Solar
 7=Bioenergy-Wind        25=Hydro-Wind
 8=Coal-Gas              26=Nuclear-Other Fossil
 9=Coal-Hydro            27=Nuclear-Other Renewables
10=Coal-Nuclear          28=Nuclear-Solar
11=Coal-Other Fossil     29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar            31=Other Fossil-Solar
14=Coal-Wind             32=Other Fossil-Wind
15=Gas-Hydro             33=Other Renewables-Solar
16=Gas-Nuclear           34=Other Renewables-Wind
17=Gas-Other Fossil      35=Solar-Wind
```







SECTION PLATE t=22 | Year 2022 | D=9 N=26 pairs=36

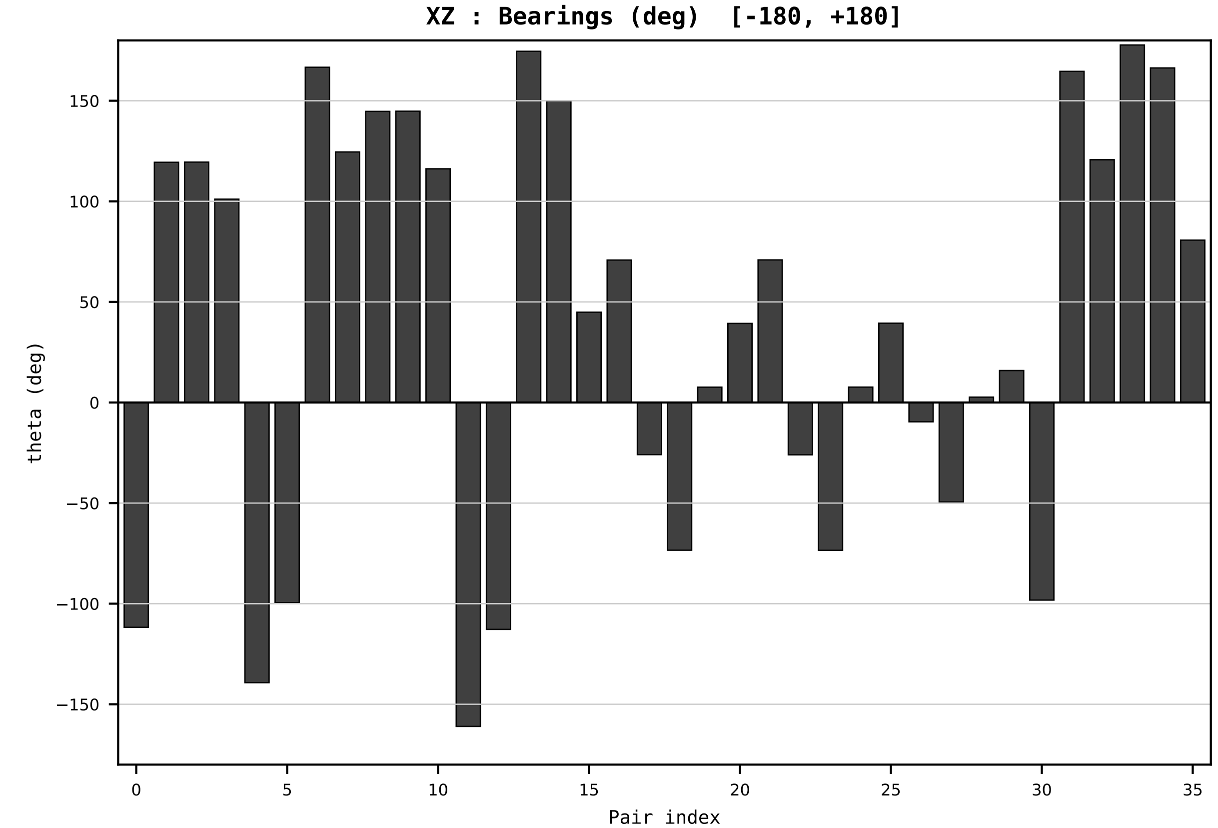
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

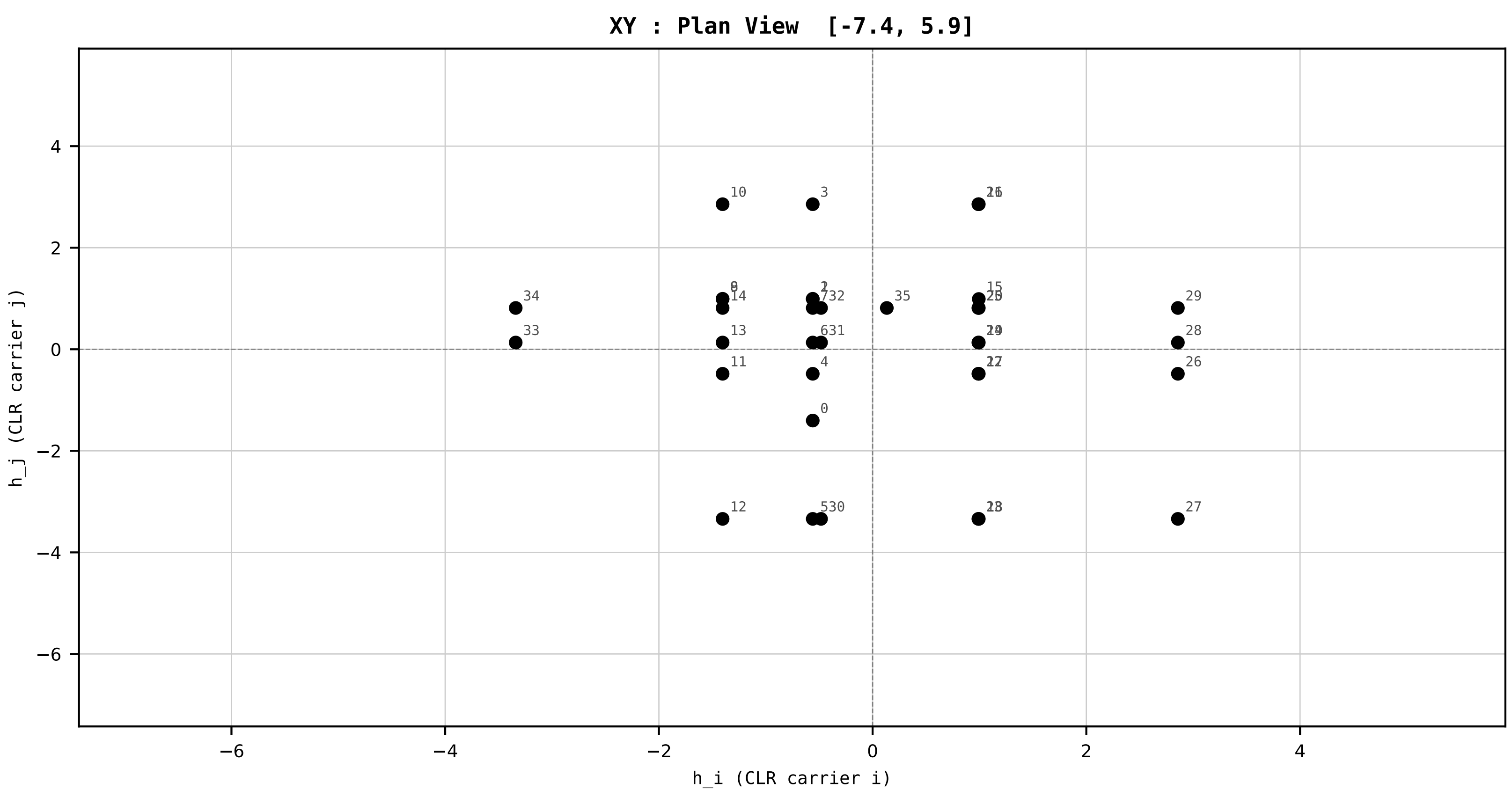
Dataset : ember_FRA_France_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

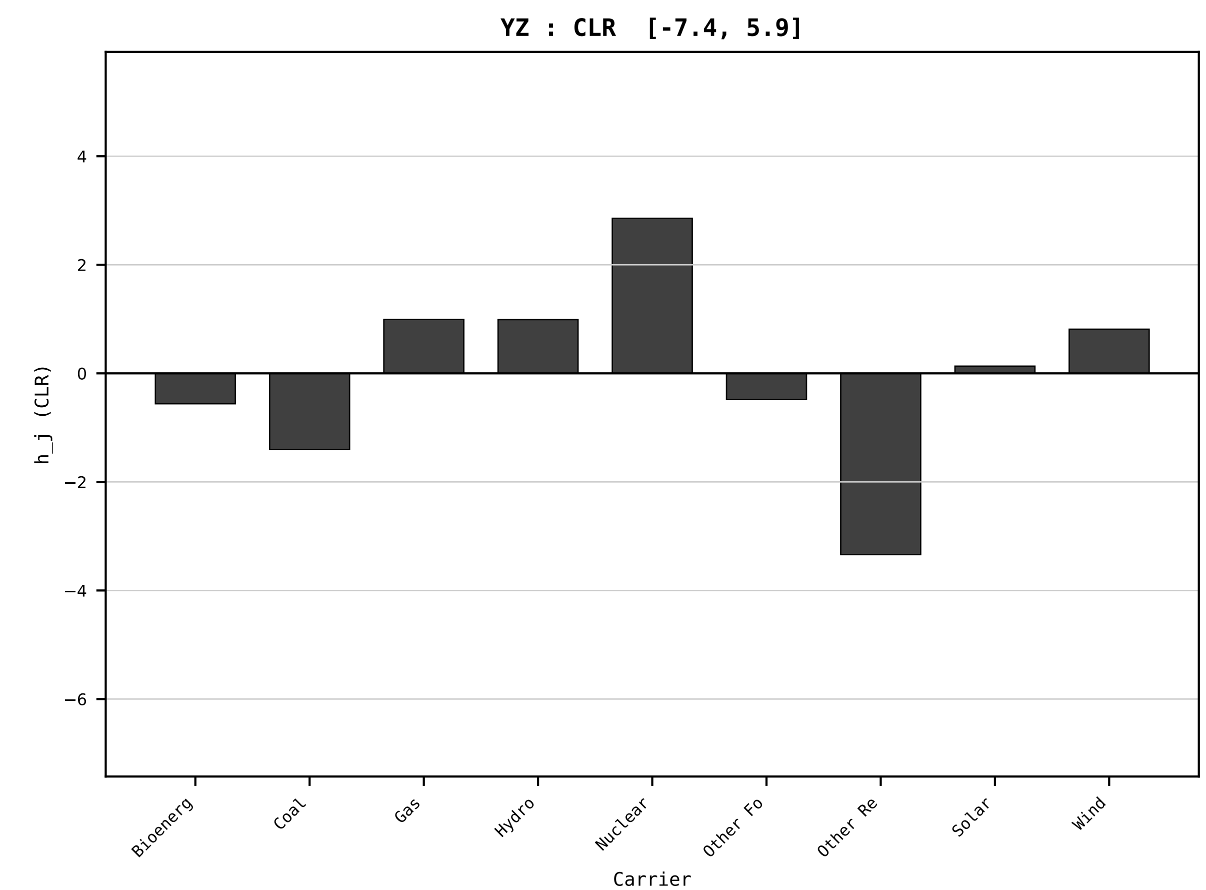
Hs = 0.409828
Ring = Hs-3
E_metric= 24.4815
kappa_HS= 491.22
omega = 6.8017 deg
d_A = 0.603939
Helm = Gas
Helm d = +0.331511
DR = 6.197 HLR
DR ratio= 491.2x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]

PAIR INDEX:
0=Bioenergy-Coal
1=Bioenergy-Gas
2=Bioenergy-Hydro
3=Bioenergy-Nuclear
4=Bioenergy-Other Fossil
5=Bioenergy-Other Renewables
6=Bioenergy-Solar
7=Bioenergy-Wind
8=Coal-Gas
9=Coal-Hydro
10=Coal-Nuclear
11=Coal-Other Fossil
12=Coal-Other Renewables
13=Coal-Solar
14=Coal-Wind
15=Gas-Hydro
16=Gas-Nuclear
17=Gas-Other Fossil
18=Gas-Other Renewables
19=Gas-Solar
20=Gas-Wind
21=Hydro-Nuclear
22=Hydro-Other Fossil
23=Hydro-Other Renewables
24=Hydro-Solar
25=Hydro-Wind
26=Nuclear-Other Fossil
27=Nuclear-Other Renewables
28=Nuclear-Solar
29=Nuclear-Wind
30=Other Fossil-Other Renewables
31=Other Fossil-Solar
32=Other Fossil-Wind
33=Other Renewables-Solar
34=Other Renewables-Wind
35=Solar-Wind



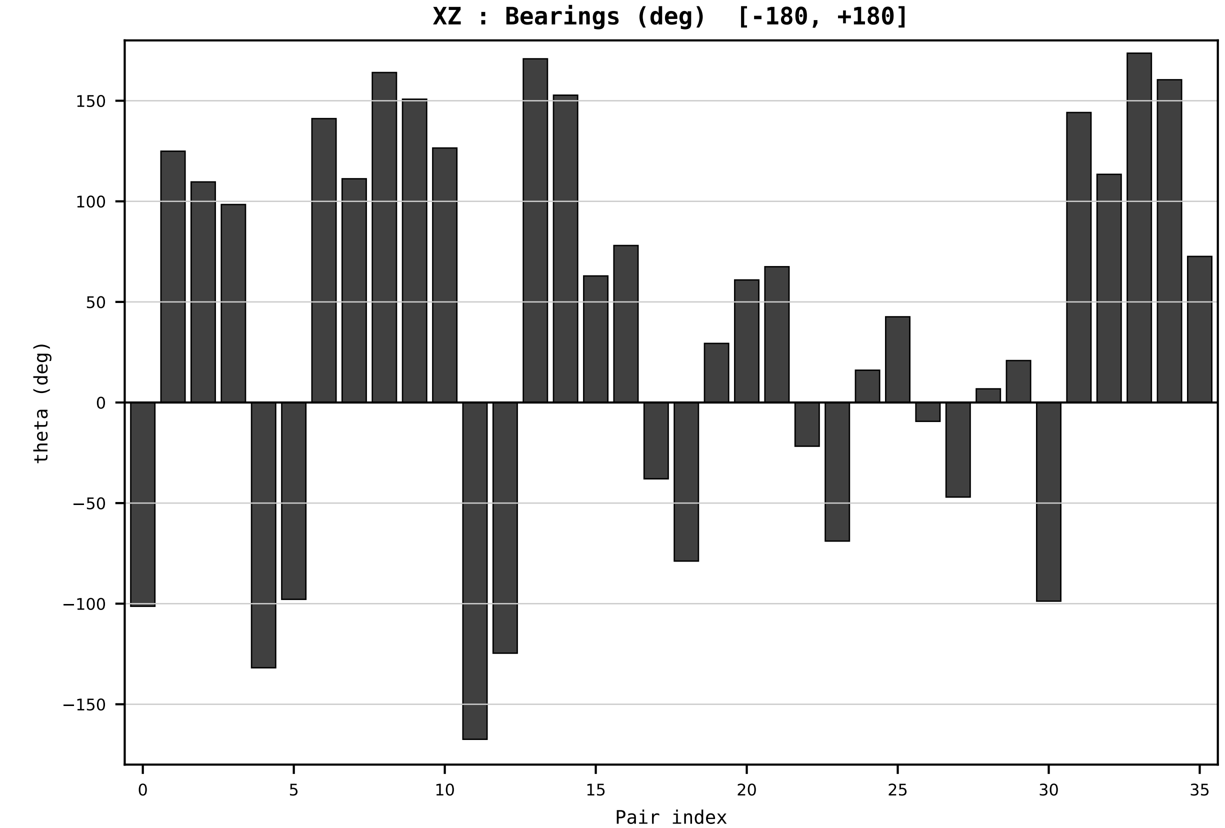


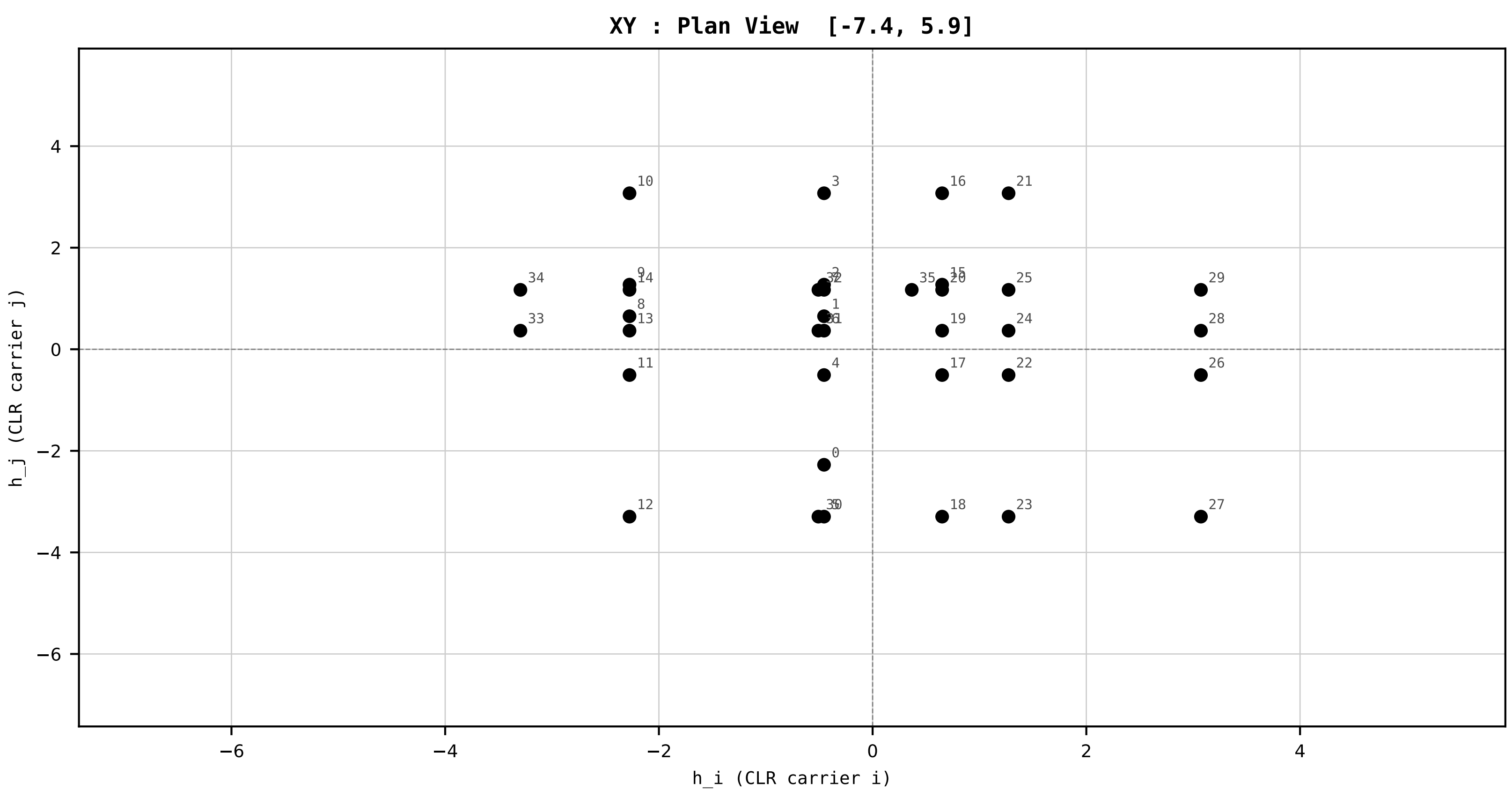


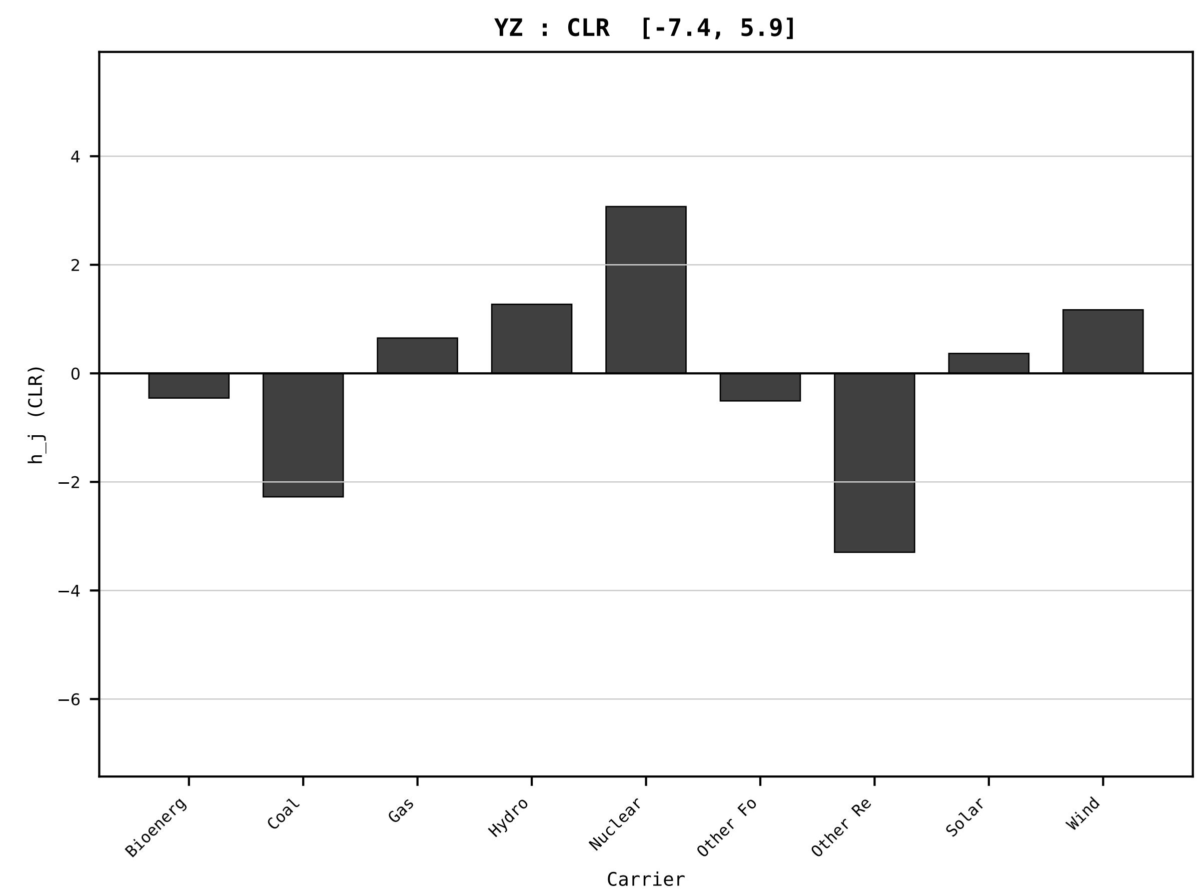
SECTION PLATE t=23 | Year 2023 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2023
t       : 23 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.444201
Ring    = Hs-3
E_metric= 29.4888
kappa_HS= 583.10
omega   = 10.8790 deg
d_A     = 1.094786
Helm    = Coal
Helm d  = -0.871139
DR      = 6.368 HLR
DR ratio= 583.1x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



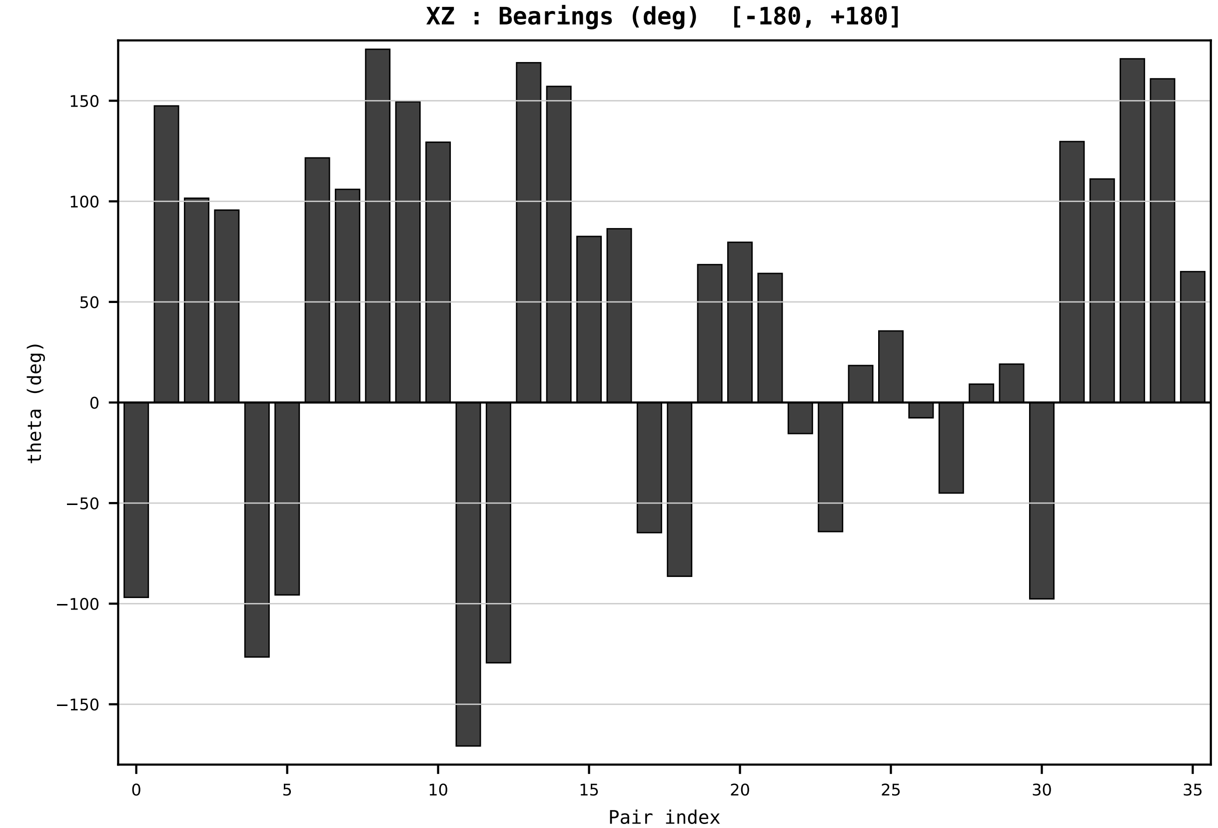


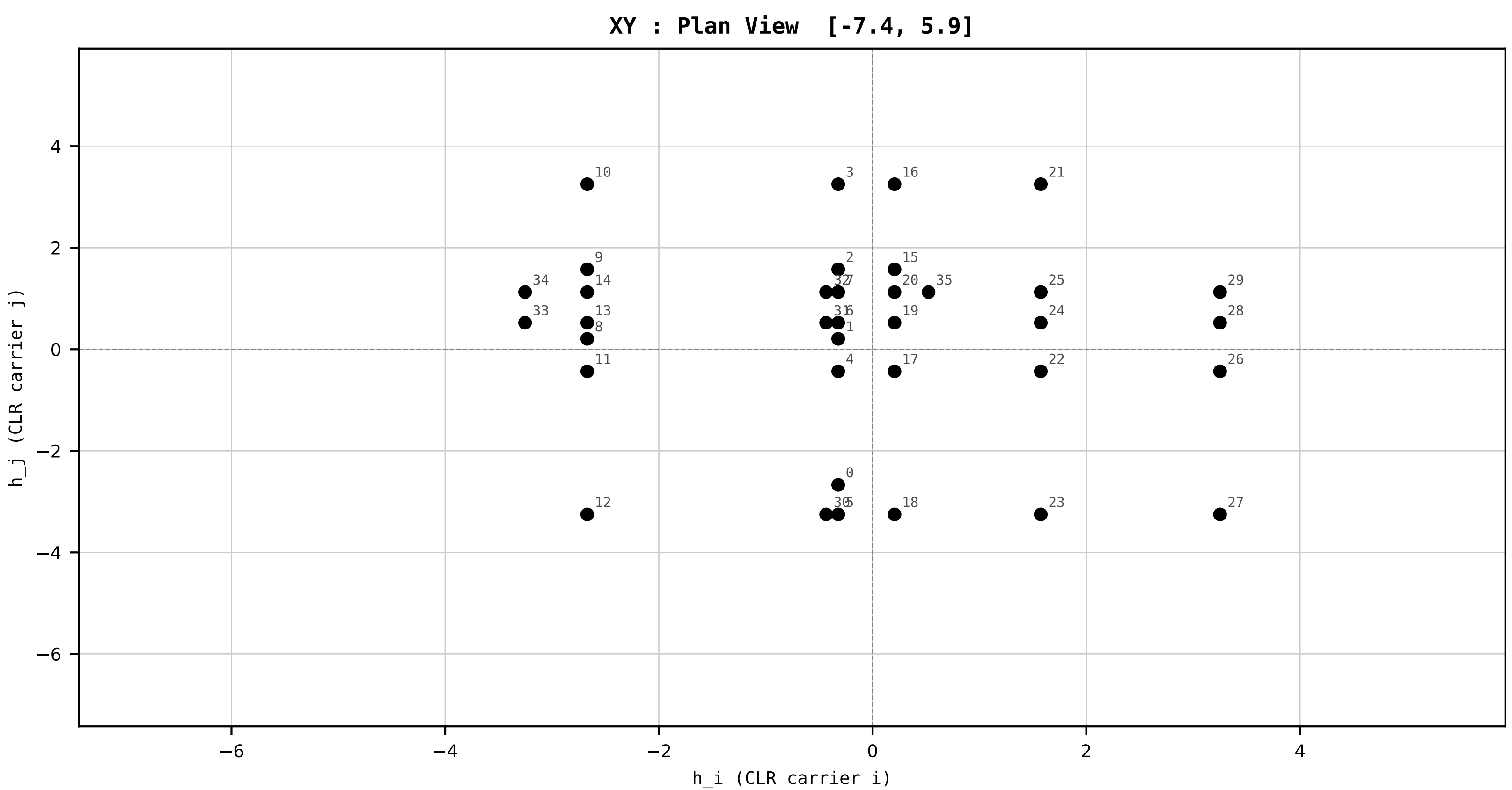


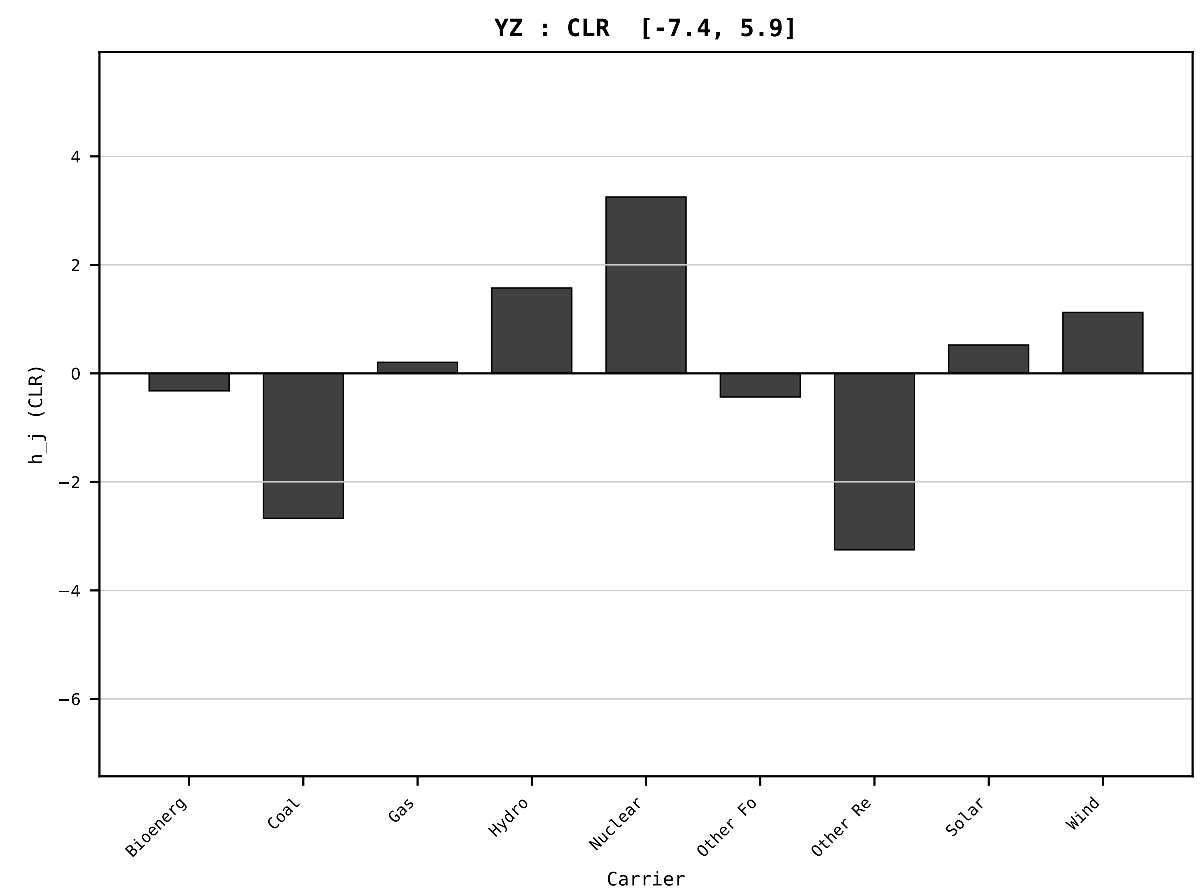
SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2024
t       : 24 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.480863
Ring    = Hs-3
E_metric= 32.6333
kappa_HS= 667.46
omega   = 6.8989 deg
d_A     = 0.727222
Helm    = Gas
Helm d  = -0.445163
DR      = 6.503 HLR
DR ratio= 667.5x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```



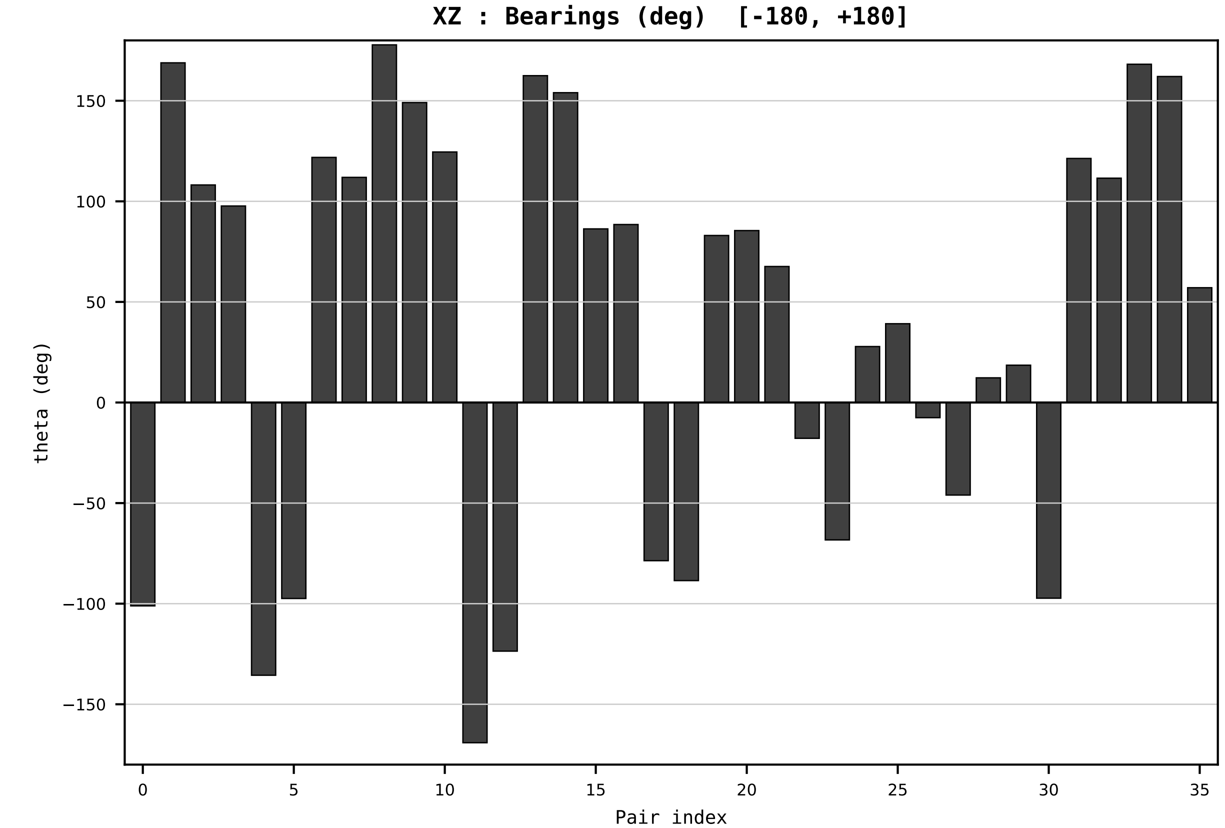


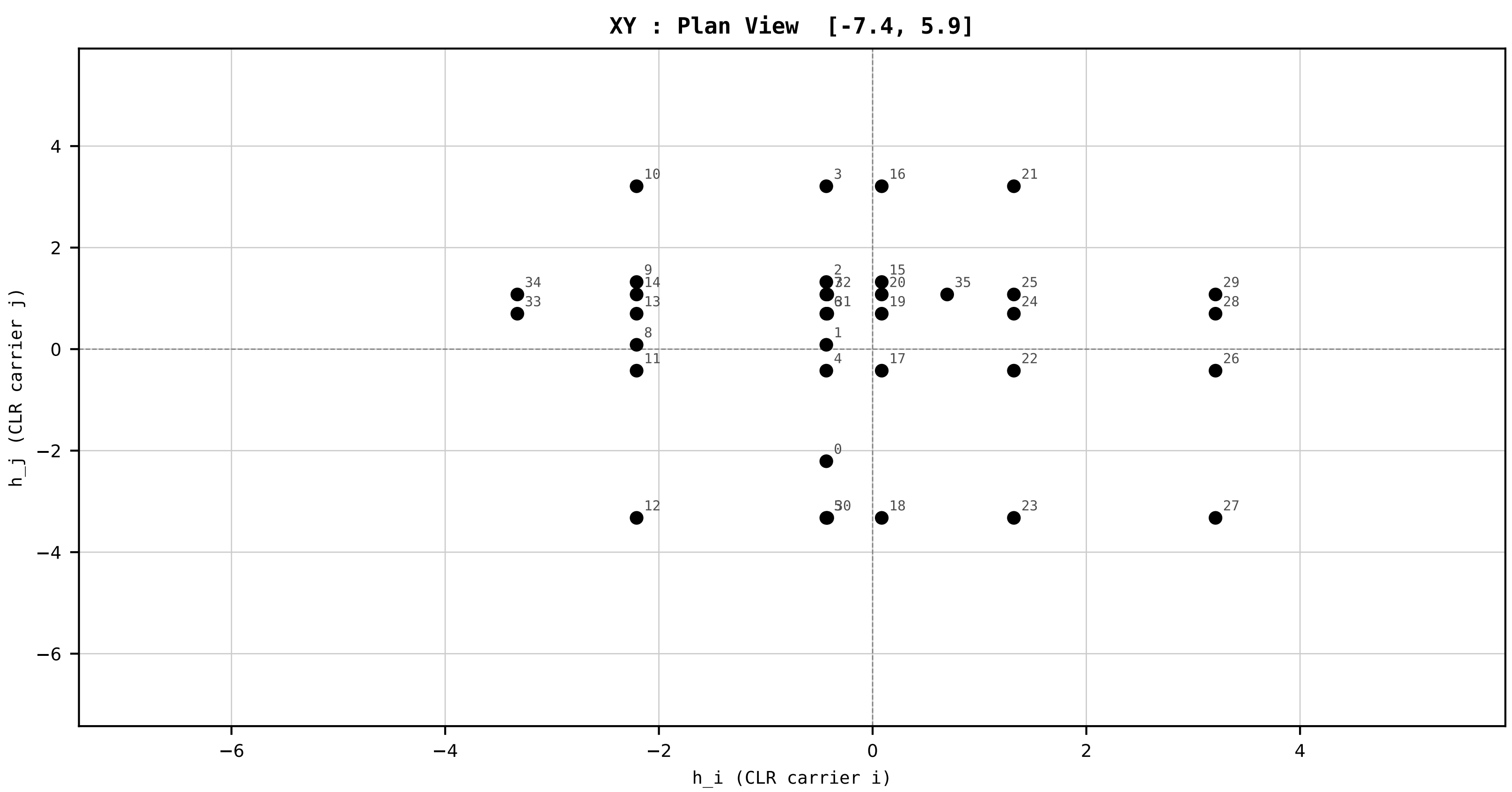


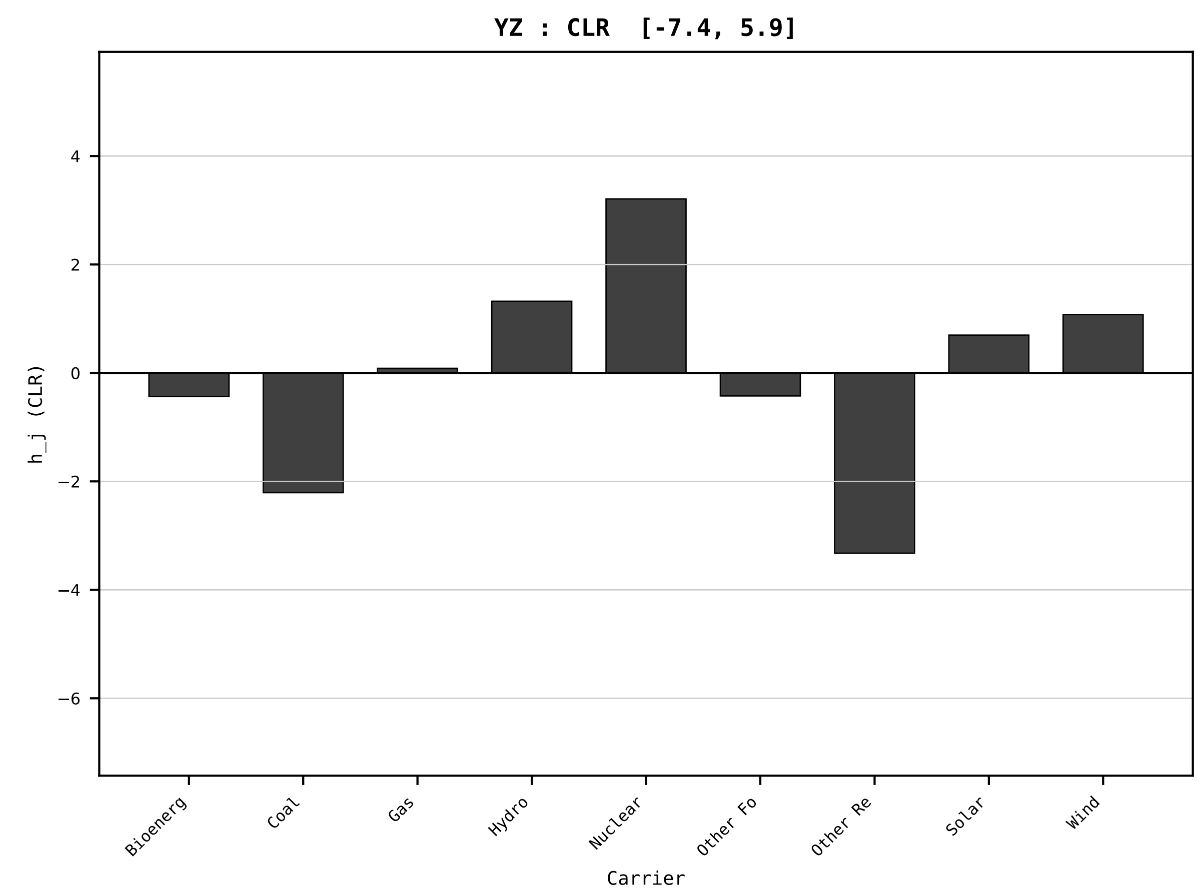
SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

```
HCI Stage 1  |  FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_FRA_France_generation_TWh.csv
Year    : 2025
t       : 25 / 25
D       : 9  carriers
Pairs   : 36  channels

Hs      = 0.483790
Ring    = Hs-3
E_metric= 29.9972
kappa_HS= 687.84
omega   = 5.5004 deg
d_A     = 0.586196
Helm    = Coal
Helm d  = +0.461974
DR      = 6.534 HLR
DR ratio= 687.8x

XZ [-180,180]
CLR[-7.4,5.9]
Hs [0, 1]
```

```
PAIR INDEX:
 0=Bioenergy-Coal       18=Gas-Other Renewables
 1=Bioenergy-Gas        19=Gas-Solar
 2=Bioenergy-Hydro      20=Gas-Wind
 3=Bioenergy-Nuclear    21=Hydro-Nuclear
 4=Bioenergy-Other Fossil22=Hydro-Other Fossil
 5=Bioenergy-Other Renewables23=Hydro-Other Renewables
 6=Bioenergy-Solar      24=Hydro-Solar
 7=Bioenergy-Wind       25=Hydro-Wind
 8=Coal-Gas             26=Nuclear-Other Fossil
 9=Coal-Hydro           27=Nuclear-Other Renewables
10=Coal-Nuclear         28=Nuclear-Solar
11=Coal-Other Fossil    29=Nuclear-Wind
12=Coal-Other Renewables30=Other Fossil-Other Renewables
13=Coal-Solar           31=Other Fossil-Solar
14=Coal-Wind            32=Other Fossil-Wind
15=Gas-Hydro            33=Other Renewables-Solar
16=Gas-Nuclear          34=Other Renewables-Wind
17=Gas-Other Fossil     35=Solar-Wind
```

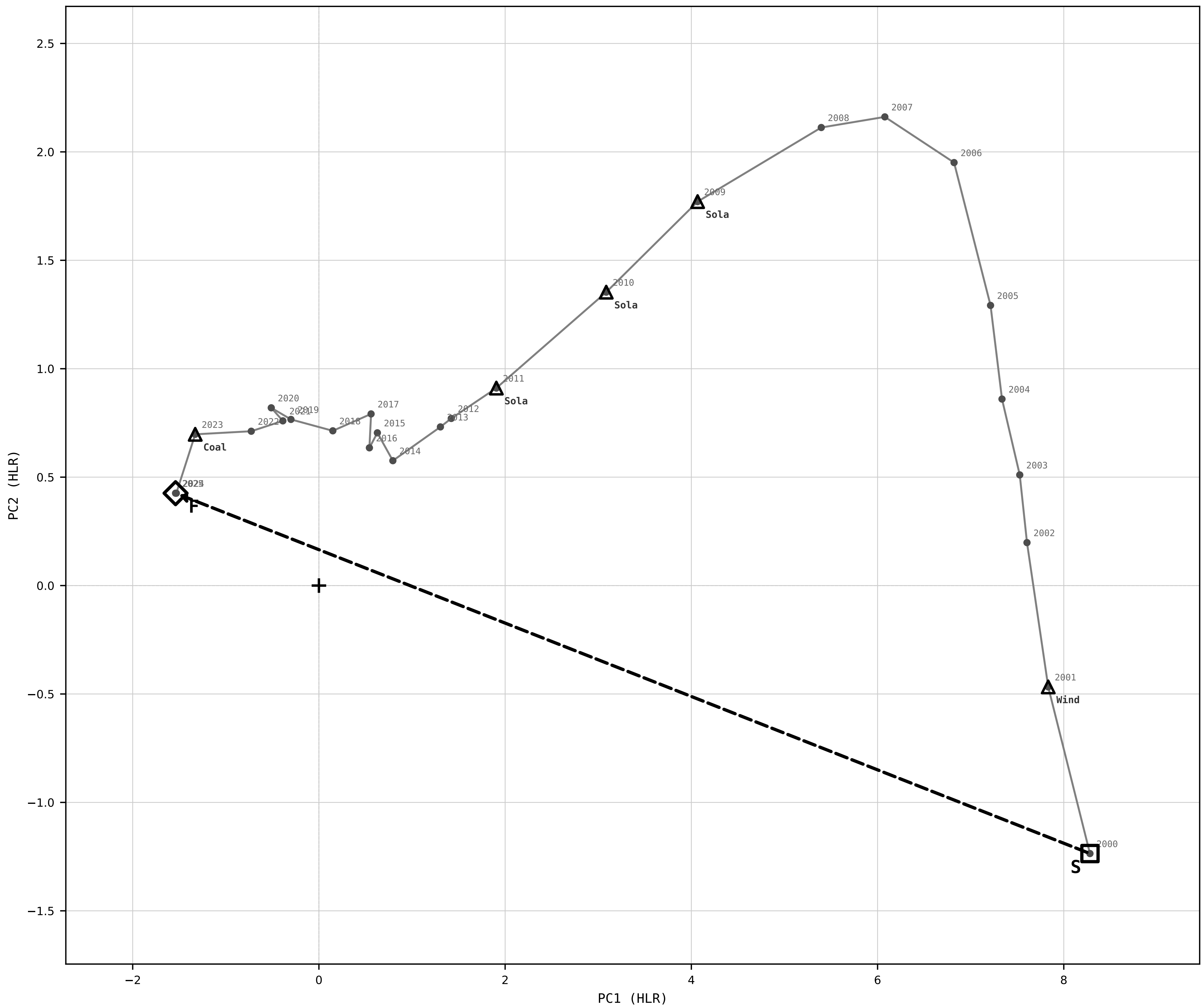






SYSTEM COURSE PLOT | ember_FRA_France_generation_TWh.csv | D=9 N=26 | Years 2000–2025

Compositional Course (PC1 vs PC2 of CLR)




```
COURSE METRICS
══════════════════════════════

Scale Provenance:
  Y_j → x_j=Y_j/T → h_j=CLR
  → PC projection
  Unit: HLR (Higgins Log-Ratio)

Net distance   = 10.0035 HLR
Path length    = 17.9314 HLR
Directness     = 0.5579

DR start       = 10.634 HLR
  ratio        = 41516.0x
DR final       = 6.534 HLR
  ratio        = 687.8x

E_metric start = 94.4242
E_metric final = 29.9972

PC1 loadings:
  Solar    -0.768
  Coal     +0.387
  Wind     -0.385

PC2 loadings:
  Wind     +0.841
  Solar    -0.448
  Hydro    -0.160

Top displacement events:
  2009 Solar    d=1.437
  2011 Solar    d=1.337
  2010 Solar    d=1.129
  2023 Coal     d=1.095
  2001 Wind     d=0.998

══════════════════════════════
The instrument reads.
The expert decides.
The loop stays open.
```